FOUR EXPONENTIALS AT PERIODS OF CM ABELIAN VARIETIES

DRAFT — NOT FOR CIRCULATION

Abstract. Let P be a nonzero eigen-period of an abelian variety with complex multiplication defined over $\overline{\mathbb{Q}}$ (for an elliptic curve: a period Ω, or a conjugate-eigen quasi-period). We prove the four exponentials conjecture at $t \in \overline{\mathbb{Q}}^{\times} P^{\pm 1}$ and at $t \in \overline{\mathbb{Q}}^{\times} (P/\pi)^{\pm 1}$: if $\log\alpha, \log\beta$ are linearly independent over $\mathbb{Q}$, then α^t and β^t are not both algebraic. For instance, 2^{ϖ} and 3^{ϖ} are not both algebraic, nor are $2^{\varpi/\pi}$ and $3^{\varpi/\pi}$, where ϖ is the real period of $y^2 = 4x^3 - 4x$; and 2^t, 3^t are not both algebraic for $t = \Gamma(1/5)^2/\Gamma(2/5)$. For CM elliptic periods we also prove the strong four exponentials conjecture at $t \in \overline{\mathbb{Q}}^{\times}\{\Omega^{\pm 1}, (\Omega/\pi)^{\pm 1}\}$ (e.g. $(2e)^{\varpi}$ and 3^{ϖ} are not both algebraic), and that one of e^{π^2}, $e^{\pi\Omega}$, $e^{\pi^2/\Omega}$ is transcendental. For two non-isogenous CM elliptic curves the same mechanism gives a bilinear statement: if u is a non-torsion elliptic logarithm on $y^2 = 4x^3 - 4x$, Ω_ρ a period of $y^2 = 4x^3 - 4$, and a, t are algebraic, then $\varpi\,\Omega_\rho\,(\zeta(u) + t)/(\pi(u + a))$ is transcendental; it also gives that $\pi, \varpi, \Omega_\rho, \varpi\Omega_\rho$ are linearly independent over $\overline{\mathbb{Q}}$, and so are $1, \varpi, \Omega_\rho, \varpi\Omega_\rho/\pi$. A homogeneous quadratic relation among first-kind CM periods of four different eigen-slots (Deligne–Koblitz–Ogus) is a further source of the period term; on the Jacobian of $y^7 = x(1-x)$ it gives that 1, $B(\frac{2}{7}, \frac{2}{7})$, $B(\frac{3}{7}, \frac{3}{7})$, $B(\frac{3}{7}, \frac{3}{7})/B(\frac{1}{7}, \frac{1}{7})$ are linearly independent over $\overline{\mathbb{Q}}$. We also record a single-curve theorem ("Theorem M") with a second proof by Laurent's identity and Wüstholz's analytic subgroup theorem.

Every class $\overline{\mathbb{Q}}^{\times} t$ above contains a real number. So, by the six exponentials theorem, Roy's strong six exponentials theorem and complex conjugation (Diaz, Waldschmidt), the four exponentials statements were already known at every $t \notin \mathbb{R} \cup i\mathbb{R}$, and the four exponentials and strong four exponentials statements were already known, at every t, for pairs whose span is not stable under complex conjugation. The new cases are pairs with a conjugation-stable span, such as $(\log 2, \log 3)$, at real or purely imaginary t (such as ϖ, ϖ/π and the Beta values themselves; for the strong statement, which depends only on the class $\overline{\mathbb{Q}}^{\times} t$, at every t in our classes); they contain all our examples.

The proofs apply Waldschmidt's algebraic subgroup theorem with period term (1988), in the K-version of Roy and Waldschmidt, to explicit quotient groups whose connected algebraic subgroups we list; the period term comes from a period lying on the relevant line, and in the two-curve case from the Legendre relations of both curves. We state precisely what the proofs use from the literature, and give a sketch of the one step of Waldschmidt's auxiliary-function construction (multiplicities along a subspace) that is only indicated in print. The main results on the curve $y^2 = 4x^3 - 4x$ (Theorems 1.1, 1.5 and 1.9, Theorem MX, Corollary 1.7, both degree-two statements, and, in the forms they need, the dimension bounds behind them) are checked in Lean 4 with Mathlib (standard axioms only), relative to named classical inputs where indicated, by a route through Waldschmidt's Theorem 1.1 in its period-refined form; the results on $y^2 = 4x^3 - 4$ in addition assume that g_3 of its period lattice is algebraic. The other results, including those on other CM curves and on CM abelian varieties of dimension ≥ 2, are proved modulo published theorems and that step; for the twisted two-curve statement we give a proof sketch.

1. Introduction

Let $\overline{\mathbb{Q}}$ be the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$, $\mathcal{L} = \{z \in \mathbb{C} : e^z \in \overline{\mathbb{Q}}^{\times}\}$ and $\widetilde{\mathcal{L}} = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}$. For an elliptic curve $E\colon y^2 = 4x^3 - g_2 x - g_3$ with $g_2, g_3 \in \overline{\mathbb{Q}}$ we write $\wp, \zeta$ for its Weierstrass functions, Λ_E for its period lattice, $\eta(\lambda) = \zeta(z + \lambda) - \zeta(z)$ for the quasi-periods, and $\mathcal{L}_E = \{u \in \mathbb{C} : u \in$

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Λ_E or $\wp(u) \in \overline{\mathbb{Q}}\}$. We use two curves throughout:

$$E_i\colon y^2 = 4x^3 - 4x, \qquad \Lambda = \varpi\,\mathbb{Z}[i], \qquad \varpi = \tfrac{\Gamma(1/4)^2}{2\sqrt{2\pi}}, \qquad \eta(\varpi) = \tfrac{\pi}{\varpi};$$
$$E_\rho\colon y^2 = 4x^3 - 4, \qquad \Lambda = \Omega_\rho\mathbb{Z}[\rho], \qquad \Omega_\rho = \tfrac{\Gamma(1/3)^3}{2^{4/3}\pi}, \qquad \eta(\Omega_\rho) = \tfrac{2\pi}{\sqrt{3}\,\Omega_\rho}.$$

(The quasi-periods follow from the Legendre relation and $\zeta(iz) = -i\zeta(z)$, $\zeta(\rho z) = \bar\rho\zeta(z)$. The real periods are Beta integrals, $\varpi = \int_1^\infty (x^3-x)^{-1/2}dx = \frac{1}{2}B(\frac{1}{4},\frac{1}{2})$ and $\Omega_\rho = \int_1^\infty (x^3-1)^{-1/2}dx = \frac{1}{3}B(\frac{1}{6},\frac{1}{2})$, and the Γ-forms follow from the reflection and duplication formulas; these are instances of the Chowla–Selberg formula [15].)

For a CM curve E let $E^\natural$ be its universal vector extension and $\Lambda(E^\natural)$ the logarithms of its algebraic points, written in coordinates (z, w) (base, and conjugate-eigen fibre; §2.3). For E_i, E_ρ the fibre coordinate can be normalised so that $X = (u, \zeta(u)) \in \Lambda(E^\natural)$ for every $u \in \mathcal{L}_E \setminus \Lambda_E$, and in general $w(X) - \zeta^*(z(X)) \in \overline{\mathbb{Q}}$ for $X \in \Lambda(E^\natural)$ with $z(X) \notin \Lambda_E$, where $\zeta^* = \zeta - \kappa_1\,\mathrm{id}$ and $\eta^* = \eta - \kappa_1\,\mathrm{id}$ are the conjugate-eigen versions of ζ and η ($\kappa_1 \in \overline{\mathbb{Q}}$, $\kappa_1 = 0$ for E_i, E_ρ; §2.3).

Four exponentials at CM periods. The *four exponentials conjecture* says: if $\log\alpha, \log\beta$ are linearly independent over $\mathbb{Q}$ and $t \notin \mathbb{Q}$, then $\alpha^t = e^{t\log\alpha}$ and $\beta^t = e^{t\log\beta}$ are not both algebraic. The *strong four exponentials conjecture* [38, Conj. 1.5] says: if y_1, y_2 are linearly independent over $\overline{\mathbb{Q}}$ and $t \notin \overline{\mathbb{Q}}$, then y_1, y_2, ty_1, ty_2 are not all in $\widetilde{\mathcal{L}}$. Roy's strong six exponentials theorem [24] is the analogous statement with three numbers y_j. The four exponentials conjecture is known for algebraic t (Gel'fond–Schneider) and in conjugation-generic situations (Diaz [11]; Waldschmidt [40]); see the paragraph after Corollary 1.4 and §8.1.

Theorem 1.1. *Let $E/\overline{\mathbb{Q}}$ have complex multiplication, let $\Omega \neq 0$ be a period of E, $c \in \overline{\mathbb{Q}}^\times$ and $\alpha, \beta \in \overline{\mathbb{Q}}^\times$. Fix logarithms $\log\alpha$, $\log\beta$ and put $\alpha^z = e^{z\log\alpha}$, $\beta^z = e^{z\log\beta}$.*

(a) *If $\log\alpha, \log\beta$ are linearly independent over $\mathbb{Q}$, then $\alpha^{c\Omega}$ and $\beta^{c\Omega}$ are not both algebraic.*
(b) *If $\log\alpha, \log\beta$ are linearly independent over $\mathbb{Q}$, then $\alpha^{c\Omega/\pi}$ and $\beta^{c\Omega/\pi}$ are not both algebraic.*

Status: [Proved; formally verified, standard axioms] for $\Omega \in \overline{\mathbb{Q}}^\times\varpi$ (every $c \in \overline{\mathbb{Q}}^\times$); [Proved modulo the algebraicity of $g_3(\Omega_\rho\mathbb{Z}[\rho])$; formally verified otherwise] for $\Omega \in \overline{\mathbb{Q}}^\times\Omega_\rho$; [Proved modulo published theorems and Lemma A.1] for every other CM period (§9.3).

For example (with the real logarithms), 2^ϖ and 3^ϖ are not both algebraic; nor are $2^{\varpi/\pi}$ and $3^{\varpi/\pi}$, nor 2^{Ω_ρ} and 3^{Ω_ρ}. Since the four exponentials statement is symmetric under $t \leftrightarrow 1/t$ (if $\alpha^{1/t}$, $\beta^{1/t}$ are algebraic, apply the statement at t to them), the same holds at $t \in \overline{\mathbb{Q}}^\times/\Omega$ and $t \in \overline{\mathbb{Q}}^\times\pi/\Omega$. Theorem 1.1 consists of Corollaries 3.5 and 3.2 of Theorems 3.4 and 3.1 (§3.1). These are dimension bounds that also allow elliptic logarithms or fibre values in the second row. Theorem 3.1 also gives an elliptic statement, Corollary 1.3 below (stated after Theorem 1.2, next to its abelian analogue).

Four exponentials at CM abelian periods. Let $A/\overline{\mathbb{Q}}$ be a simple abelian variety with complex multiplication by a CM field F of degree $2g$, with CM type Φ. For an embedding σ of F, an *eigen-differential* is a class $\omega_\sigma \in H^1_{\mathrm{dR}}(A/\overline{\mathbb{Q}})$ with $[x]^*\omega_\sigma = \sigma(x)\omega_\sigma$; it is of the first kind (holomorphic) iff $\sigma \in \Phi$. An *eigen-period* is a nonzero number $P_\sigma = \int_\gamma \omega_\sigma$, $\gamma \in H_1(A, \mathbb{Z})$. By Riemann's bilinear relations $P_\sigma P_{\bar\sigma} \in \pi\overline{\mathbb{Q}}^\times$. For a CM abelian variety that is not simple, every nonzero eigen-period is an algebraic multiple of an eigen-period of a simple factor, of the same kind (§2.2). For $N \geq 3$ and $0 < a, b < N$ with $a + b \neq N$, the Beta value $B(a/N, b/N) = \Gamma(a/N)\Gamma(b/N)/\Gamma((a+b)/N)$ is, up to $\overline{\mathbb{Q}}^\times$, an eigen-period of a CM abelian variety (a factor of the Jacobian of the Fermat curve of degree N), of the first kind iff $a + b < N$ (§2.2).

Theorem 1.2 (Theorem A)**.** *Let P be a nonzero eigen-period of a CM abelian variety over $\overline{\mathbb{Q}}$, let $c \in \overline{\mathbb{Q}}^\times$ and $t \in \{cP,\ c/P,\ cP/\pi,\ c\pi/P\}$. Let $\alpha, \beta \in \overline{\mathbb{Q}}^\times$, fix logarithms $\log\alpha$, $\log\beta$ and put $\alpha^t = e^{t\log\alpha}$, $\beta^t = e^{t\log\beta}$. If $\log\alpha, \log\beta$ are linearly independent over $\mathbb{Q}$, then α^t and β^t are not both algebraic.*

Status: [Proved modulo published theorems and Lemma A.1] for $g \geq 2$ (§9.3 lists the instances checked in Lean conditionally on Theorem 2.5 for CM abelian varieties of dimension 2 and 3). For $g = 1$ it consists of Theorem 1.1, which has the status stated there, and its $t \leftrightarrow 1/t$ extension to $t \in \overline{\mathbb{Q}}^\times/\Omega \cup \overline{\mathbb{Q}}^\times\pi/\Omega$, which follows in one line and is not formalised ([Proved modulo published theorems and Lemma A.1]).

In particular Theorem 1.2 holds at $t = B(a/N, b/N)^{\pm 1}$ and $t = (B(a/N, b/N)/\pi)^{\pm 1}$ for $0 < a, b < N$, $a + b \neq N$. For example, 2^t and 3^t are not both algebraic (for any choice of $\log 2$, $\log 3$; any two such choices are $\mathbb{Q}$-independent) for

$$t = \frac{\Gamma(1/5)^2}{\Gamma(2/5)}, \qquad t = \frac{\Gamma(1/5)^2}{\pi\,\Gamma(2/5)}, \qquad t = \frac{\Gamma(4/5)^2}{\Gamma(3/5)}, \qquad t = \frac{\Gamma(1/7)^2}{\Gamma(2/7)}.$$

Theorem 1.2 consists of Theorems 3.7 and 3.9 (§3.3), which are dimension bounds of the shape of Theorems 3.1 and 3.4, with the CM elliptic curve replaced by one eigen-coordinate of a CM abelian variety. Theorem 3.1 (for a CM elliptic curve) gives Corollary 1.3, and Theorem 3.7 its abelian analogue, Corollary 1.4:

Corollary 1.3. *For every CM curve with period Ω, at least one of e^{π^2}, $e^{\pi\Omega}$, $e^{\pi^2/\Omega}$ is transcendental. On $y^2 = 4x^3 - 4x$: one of $e^{\pi^2} \approx 19333.69$, $e^{\pi\varpi} \approx 3779.84$, $e^{\pi^2/\varpi} \approx 43.12$ is transcendental.*

Status: on $y^2 = 4x^3 - 4x$: [Proved modulo Wüstholz's theorem in the form $\varpi, \eta(\varpi) \notin \widetilde{\mathcal{L}}$; formally verified otherwise] (Corollary 7.1); [Proved modulo published theorems and Lemma A.1] for every other CM period.

Corollary 1.4. *For every first-kind CM eigen-period P, at least one of e^{π^2}, $e^{\pi P}$, $e^{\pi^2/P}$ is transcendental. For instance, one of e^{π^2}, $e^{\pi\Gamma(1/5)^2/\Gamma(2/5)}$, $e^{\pi^2\Gamma(2/5)/\Gamma(1/5)^2}$ is transcendental.*

Status: [Proved modulo published theorems and Lemma A.1].
The proof (Corollary 7.3) does not use the algebraic independence of π and P (which is not known for abelian periods); for $g = 1$ it is the proof of the elliptic case, Corollary 7.1, which therefore does not need Chudnovsky's theorem either (Remark 7.4).

Part of Theorems 1.1 and 1.2 also follows from the six exponentials theorem, Roy's strong six exponentials theorem and complex conjugation (Diaz [11]; Waldschmidt [40, Cor. 2.7, Thm. 4.1]). Every class $\overline{\mathbb{Q}}^\times t$ occurring in them contains a real number, and these arguments together cover every pair at every $t \notin \mathbb{R} \cup i\mathbb{R}$, and, at every t, every pair whose logarithms span a space that is not stable under complex conjugation (§8.1). They do not apply to real or purely imaginary t — such as $t = \varpi$, ϖ/π, Ω_ρ, or the Beta values themselves — with pairs whose logarithms span a conjugation-stable space, such as $(2, 3)$, which is where our examples lie, nor to Corollaries 1.3 and 1.4.

Strong four exponentials. For a CM elliptic curve $E/\overline{\mathbb{Q}}$ with period $\Omega \neq 0$ let Fib be the set of fibre values (§2.3): $\zeta^*(u) + \overline{\mathbb{Q}}$ for $u \in \mathcal{L}_E \setminus \Lambda_E$ and $\eta^*(\lambda) + \overline{\mathbb{Q}}$ for $\lambda \in \Lambda_E$; they are the conjugate-eigen fibre coordinates of logarithms of algebraic points of the universal vector extension $E^\natural$.

Theorem 1.5 (Theorem S)**.** *Let $E/\overline{\mathbb{Q}}$ have complex multiplication, $\Omega \neq 0$ a period of E, and $t \in \overline{\mathbb{Q}}^\times\Omega \cup \overline{\mathbb{Q}}^\times\Omega^{-1} \cup \overline{\mathbb{Q}}^\times\Omega/\pi \cup \overline{\mathbb{Q}}^\times\pi/\Omega$. If $y_1, y_2 \in \mathbb{C}$ are linearly independent over $\overline{\mathbb{Q}}$, then y_1, y_2, ty_1, ty_2 are not all in $\widetilde{\mathcal{L}}$.*

Status: [Proved modulo Wüstholz's theorem in the form $\Omega, \eta \notin \widetilde{\mathcal{L}}$ (Lemma 4.1); formally verified otherwise] for $\Omega \in \overline{\mathbb{Q}}^{\times}\varpi$; [Proved modulo that form of Wüstholz's theorem and the algebraicity of $g_3(\Omega_\rho \mathbb{Z}[\rho])$; formally verified otherwise] for $\Omega \in \overline{\mathbb{Q}}^{\times}\Omega_\rho$; [Proved modulo published theorems and Lemma A.1] for every other CM period. For the paper proof (trust point T), two internal referees graded the first write-up a proof sketch modulo this trust point, and a third, after the subgroup list was written out in full (§4), found no case missing.

It follows from the affine dimension bounds SMG and SMG$^\natural$ (Theorem 4.2), whose second rows may also contain elliptic logarithms, resp. fibre values. For instance (Corollary 4.5, which has label (c)):

- $\varpi \log 2$ and $\varpi \log 3$ are not both in $\widetilde{\mathcal{L}}$; $(2e)^{\varpi}$ and 3^{ϖ} are not both algebraic; $e^{\varpi/\pi}$ and $2^{\varpi/\pi}$ are not both algebraic (this recovers Corollary 7.7(ii), which §7.4 derives from a proof sketch); $e^{1/\varpi}$ and $2^{1/\varpi}$ are not both algebraic. Here $\varpi = \Gamma(1/4)^2/(2\sqrt{2\pi}) = 2.62205755429211981046\ldots$ is the real period of $y^2 = 4x^3 - 4x$.
- For every CM period Ω, one of e^{π^2}, $e^{\pi\Omega}$, $e^{\Omega/\pi}$ is transcendental (this recovers Corollary 7.7(i), which §7.4 derives from a proof sketch), and one of e^{π^2}, $e^{\pi^2/\Omega}$, $e^{1/\Omega}$ is transcendental.

Two CM curves. The same mechanism, applied to two non-isogenous CM curves, gives a statement with no logarithm of $\mathbb{G}_{\mathrm{m}}$ at all (and a twisted variant, Theorem 5.6 in §5.6, of which we give a proof sketch). Applied to periods, it also gives two statements of degree two about periods of the two curves (Corollary 7.5): $\pi, \varpi, \Omega_\rho, \varpi\Omega_\rho$ are linearly independent over $\overline{\mathbb{Q}}$, and so are $1, \varpi, \Omega_\rho, \varpi\Omega_\rho/\pi$. These are not, as far as we can see, consequences of the analytic subgroup theorem ([Heuristic]: period conjecture), and we flag them, with Corollary 1.11(b), as the most delicate claims of the paper.

Theorem 1.6 (Theorem MX)**.** *Let $X \in \Lambda(E_i^\natural)$, $Y \in \Lambda(E_\rho^\natural)$, $a, t \in \overline{\mathbb{Q}}$ and $\beta \in \overline{\mathbb{Q}}^{\times}$. If*

$$\big(z(X) + z(Y) + a\big) \cdot \beta\, \eta(\Omega_\rho) \;=\; \big(w(X) + w(Y) + t\big) \cdot \varpi,$$

then $a = 0$, $z(X) \in \mathbb{Q}(i)\varpi$ and $z(Y) \in \mathbb{Q}(\rho)\Omega_\rho$; in particular $\exp_{E_i}(z(X))$ and $\exp_{E_\rho}(z(Y))$ are torsion points. In particular, for $u \in \mathcal{L}_{E_i} \setminus \Lambda$ and $u' \in \mathcal{L}_{E_\rho} \setminus \Lambda_\rho$ not both torsion, $(u + u' + a)\,\beta\,\eta(\Omega_\rho) \neq (\zeta_i(u) + \zeta_\rho(u') + t)\,\varpi$. An analogous statement holds for any two non-isogenous CM elliptic curves (Remark 5.3).

Status: [Proved modulo Wüstholz's theorem (in the forms $\varpi\Omega_\rho/\pi \notin \overline{\mathbb{Q}}$, $\Omega_\rho/\varpi \notin \overline{\mathbb{Q}}$, and the linear independence of 1 and elliptic logarithms of E_i, E_ρ), Masser's theorem, and the algebraicity of $g_3(\Omega_\rho\mathbb{Z}[\rho])$; formally verified otherwise]. The analogue for other pairs of curves (Remark 5.3) is [Proved modulo published theorems and Lemma A.1]. (The dimension bound behind it, Proposition 5.1, is formally verified only in its single-point form; see its status line.)

Corollary 1.7. *For $u \in \mathcal{L}_{E_i} \setminus \mathbb{Q}(i)\varpi$ and $a, t \in \overline{\mathbb{Q}}$,*

$$\frac{\varpi\,\Omega_\rho\,(\zeta_i(u) + t)}{\pi\,(u + a)} \notin \overline{\mathbb{Q}}, \qquad \textit{equivalently} \qquad \frac{\zeta_i(u) + t}{u + a} \notin \overline{\mathbb{Q}} \cdot \frac{\pi^{5/2}}{\Gamma(1/4)^2\,\Gamma(1/3)^3}.$$

Symmetrically for non-torsion $u' \in \mathcal{L}_{E_\rho}$ with ζ_ρ.

Status: for $u \in \mathcal{L}_{E_i}$: [Proved modulo the inputs of Theorem 1.6 and Laurent's theorem that $1, u, \zeta(u)$ are $\overline{\mathbb{Q}}$-linearly independent [18]; formally verified otherwise]; the symmetric statement on E_ρ is [Proved modulo published theorems and Lemma A.1].

For example, for the point with $\wp(u) = 2$ one has $u = 0.72694593546890819853\ldots$ and $\varpi\Omega_\rho\zeta(u)/(\pi u) = 3.76379062156897182773\ldots$ (computed independently twice, by theta functions and by the Laurent series of ζ; an illustration only). The case $X = Y = 0$ of Theorem 1.6 gives:

Corollary 1.8. $\Gamma(1/4)^2\,\Gamma(1/3)^3/\pi^{5/2}$ *is transcendental.*

Status: [Proved modulo published theorems and Lemma A.1] (only Wüstholz's theorem is used, Remark 5.4; in the Lean certificate it is a named hypothesis).

This is immediate from Wüstholz's theorem (Remark 5.4), and probably also from the Wolfart–Wüstholz theorem on Beta values [44] (not checked); we found it stated nowhere, but do not claim it as new. Why Corollary 1.7 does not follow from known algebraic independence results is said in §8.3.

The single-curve version of the same count gives the following theorem, which turns out to be classical in substance.

Theorem 1.9 (Theorem M)**.** *Let $E/\overline{\mathbb{Q}}$ have complex multiplication by k, and let $\Omega \neq 0$ be a period. If $u \in \overline{\mathbb{Q}}\mathcal{L}_E$ and $\ell \in \widetilde{\mathcal{L}}$ satisfy $i\pi\, u = \Omega\,\ell$, then $u = s\Omega$ and $\ell = s\, i\pi$ for some $s \in \overline{\mathbb{Q}}$; if $u \in \mathcal{L}_E$ then $s \in k$.*

Status: for $\Omega \in \overline{\mathbb{Q}}^\times\varpi$: [Proved; formally verified, standard axioms] for the first conclusion, and [Proved modulo Masser's theorem; formally verified otherwise] for $s \in k$ when $u \in \mathcal{L}_E$; for $\Omega \in \overline{\mathbb{Q}}^\times\Omega_\rho$: [Proved modulo the algebraicity of $g_3(\Omega_\rho\mathbb{Z}[\rho])$; formally verified otherwise] for the first conclusion, and [Proved modulo published theorems and Lemma A.1] for $s \in k$; [Proved modulo published theorems and Lemma A.1] for every other CM period.

Corollary 1.10. *On E_i (and analogously on every CM curve): (M1) $\wp(c\varpi/\pi) \notin \overline{\mathbb{Q}}$ for $c \in \overline{\mathbb{Q}}^\times$; (M2) $\wp(\varpi \log\alpha/\pi) \notin \overline{\mathbb{Q}}$ for $\alpha \in \overline{\mathbb{Q}}^\times$ not a root of unity; (M3) at least one of e^{π^2}, $\wp(\pi\varpi)$ is transcendental; more precisely, if $e^{\pi^2} \in \overline{\mathbb{Q}}$ then $\wp(c\pi\Omega) \notin \overline{\mathbb{Q}}$ for every $c \in \overline{\mathbb{Q}}^\times$ and every period Ω of every CM curve; (M3′) for $\alpha \in \overline{\mathbb{Q}} \setminus \{0,1\}$ and any branch, one of $e^{i\pi\log\alpha}$, $\wp(\varpi\log\alpha)$ is transcendental; (Mη) $\eta(\varpi)\,u \notin \widetilde{\mathcal{L}}$ for $u \in \mathcal{L}_{E_i} \setminus \mathbb{Q}(i)\varpi$.*

Status: (M1) and (M3) on E_i (with $\Omega \in \overline{\mathbb{Q}}^\times\varpi$ in (M3)): [Proved; formally verified, standard axioms]; (M3′): [Proved modulo the Hermite–Lindemann theorem; formally verified otherwise]; (Mη): [Proved modulo Masser's theorem; formally verified otherwise]; (M2), (M3) for other CM periods, and the analogues on other CM curves: [Proved modulo published theorems and Lemma A.1].

Why MX is different. After multiplying by π, every relation in Theorem 1.9 with $u \in \mathcal{L}_E$ becomes a $\overline{\mathbb{Q}}$-linear relation among periods of 1-motives, because of Laurent's identity (4); this is why the analytic subgroup theorem proves it too. The hypothesis of Theorem 1.6 instead contains the products $\Omega_\rho\,\zeta_i(u)$ and $\eta(\varpi)\,u'$, each pairing a period of one curve with a logarithm of the other. No CM identity linearises these; by the period conjecture they are genuine degree-two periods (§5.8). The extra strength comes from a second true degree-two period relation (the Legendre relation of the second curve), which the count sees as a second period on the line.

First-kind period relations. In Theorem 1.6, the period term $\kappa = 2$ on a line came from two Legendre relations, i.e. from two eigen-pairs $\omega\,\eta^*(\omega) \in \overline{\mathbb{Q}}^\times\pi$; this forces universal vector extensions into the group. Products of CM abelian varieties involving a simple factor of dimension ≥ 3 can have homogeneous quadratic relations among first-kind periods of four different

eigen-slots (Remark 6.9). On the Jacobian A of $C\colon y^7 = x(1-x)$ (genus 3, CM by $\mathbb{Q}(\zeta_7)$), with E an elliptic curve with CM by $\mathbb{Q}(\sqrt{-7})$, the Beta cocycle and one reflection give

$$B(\tfrac{1}{7},\tfrac{1}{7})\,B(\tfrac{2}{7},\tfrac{2}{7}) = \frac{\sin(3\pi/7)}{\sin(\pi/7)}\,B(\tfrac{3}{7},\tfrac{3}{7})\,B(\tfrac{1}{7},\tfrac{2}{7}),$$

where $B(j/7, j/7)$ $(j = 1, 2, 3)$ are the periods of the holomorphic eigen-differentials dx/y^{7-j} on C and $B(1/7, 2/7) \in \overline{\mathbb{Q}}^{\times}\Omega_E$ (Chowla–Selberg [15]). Theorem 6.2 (§6) uses such a relation as the period term. On $y^7 = x(1-x)$ it gives:

Corollary 1.11 (Corollary X7)**.** (a) *For $O, Q \in C(\overline{\mathbb{Q}})$ with $Q - O$ of infinite order in A,*

$$\int_O^Q \frac{dx}{y^5} \Big/ \int_O^Q \frac{dx}{y^6} \;\notin\; \overline{\mathbb{Q}} \cdot \frac{B(3/7,3/7)}{B(1/7,1/7)}.$$

(b) *The numbers 1, $B(\frac{2}{7},\frac{2}{7})$, $B(\frac{3}{7},\frac{3}{7})$, $B(\frac{3}{7},\frac{3}{7})/B(\frac{1}{7},\frac{1}{7})$ are linearly independent over $\overline{\mathbb{Q}}$; equivalently, so are π, $\Gamma(\frac{2}{7})^2\Gamma(\frac{3}{7})$, $\Gamma(\frac{1}{7})\Gamma(\frac{3}{7})^2$, $\Gamma(\frac{2}{7})\Gamma(\frac{3}{7})^2/\Gamma(\frac{1}{7})$.*

Status: [Proved modulo published theorems and Lemma A.1], for the dimension statement (Proposition 6.1), for Theorem 6.2 (whose deduction from the proposition is the standard analytic-subgroup step, written out in §6) and for (a) and (b). That (a) and (b) are not linearisations, i.e. not consequences of the analytic subgroup theorem, is [Heuristic] (period conjecture).

How the parts fit together. Theorem 1.2 contains Theorem 1.1 ($g = 1$), and its proof is that of Theorems 3.1 and 3.4 with new numbers: the point is that the quotient group in the count has abelian part of dimension $n/2$ for every g (Lemma 2.2). Theorem 6.2 is a second source of the period term: Theorems 1.6 and 5.6 (and the abelian-surface sketch of §5.7) obtain $\kappa = 2$ on a line only from eigen-pair (Legendre) relations, which force vector extensions into the group; in Theorem 6.2 the group is an abelian variety times $\mathbb{G}_{\mathrm{a}}^2$ and the margin of the count is n instead of $n-2$. Theorem 1.5 at $t \in \overline{\mathbb{Q}}^{\times}(\Omega/\pi)^{\pm 1}$ contains the strong statement of Corollary 7.7(iii), which §7.4 derives from an affine version of Theorem 3.1 stated as a proof sketch with the subgroup list not written out (Theorem 7.6); §4 writes that list out (Theorem SMG) and adds the fibre version SMG$^\natural$, which gives the classes $\overline{\mathbb{Q}}^{\times}\Omega^{\pm 1}$.

Organisation. §2 states the tool (Waldschmidt's Theorem 4.1, completed by Lemma A.1 of Appendix A), the linear algebra of CM abelian varieties of any dimension and of their universal vector extensions, and the Roy–Waldschmidt count. §3 proves Theorems 1.1 and 1.2, §4 Theorem 1.5, §5 Theorems 1.9 and 1.6 and Proposition 5.5 (with a proof sketch of Theorem 5.6), and §6 Theorem 6.2. §7 collects the corollaries MG1, MG3, A2, the degree-two statements and the affine version. §8 discusses the overlap with known results, and §9 the consistency checks, the dependencies and the formal verification.

Conventions on labels. Each main result carries a status line with one of the following labels (or the label [Proof sketch], described after them); where the label depends on the period (typically when a result is formally verified for $\Omega \in \overline{\mathbb{Q}}^{\times}\varpi$ only), the status line says which label applies to which case.

(a) [Proved; formally verified, standard axioms]: a deduction of the statement (possibly by another route than the proof given here; see §9.3), including the instances of Waldschmidt's theorem (Theorem 2.1, or its consequence Theorem 2.5, with the step of Lemma A.1) on which that deduction rests, is checked in the Lean certificate described in §9.3, from Lean's standard axioms only. These are not always the instances used in

our paper proofs. The group models used there were refereed, not derived in Lean from algebraic geometry.

(b) [Proved modulo X; formally verified otherwise]: as (a), except that the named classical input X (for example Wüstholz's theorem in a stated form, Baker's theorem, the Hermite–Lindemann theorem, Masser's theorem, Laurent's theorem on ζ, or the algebraicity of $g_3(\Omega_\rho\mathbb{Z}[\rho])$) enters the Lean proof as an explicit hypothesis.

(c) [Proved modulo published theorems and Lemma A.1]: the proof is complete given Waldschmidt's Theorem 4.1 [33] as printed (Theorem 2.1), the other published results cited, and Lemma A.1, which is itself a proof sketch of the one step of [33, Prop. 6.1] that is only indicated in print (checked by two internal referees, not peer-reviewed). A result that uses fewer inputs (for instance only Wüstholz's theorem) also carries this label when it is not formally verified.

A main result one step of whose proof is only sketched carries the status label [Proof sketch] instead, and its status line names that step; this applies to Theorem 5.6 and Corollary 5.7 only. (Other proof sketches, such as Theorem 7.6, are not main results and say so in their headings.) We call the results with label (b) "formally verified modulo named inputs": the named classical theorems X are hypotheses of their Lean proofs, so these results are not unconditional.

For steps and remarks we use: [Proved]; [Lit] (a published result, cited); [Proof sketch] (the argument is written at the level of the case check, with some routine steps compressed, or the internal referees graded it so); [Heuristic] (depends on the period conjecture or the Rohrlich–Lang conjecture); [Numerical] (computed, illustrates only). "Internal referee" means an independent check within this project, not peer review.

2. The method

All the main results are obtained by applying one published theorem, Waldschmidt's algebraic subgroup theorem with period term (Theorem 2.1), to explicit quotient groups G'' whose connected algebraic subgroups we list. This section states that theorem and what its proof uses (§2.1), the linear algebra of CM abelian varieties of any dimension g and of their universal vector extensions (§§2.2–2.3), and the Roy–Waldschmidt count, which predicts the results and gives the first proof of Theorem 1.9 (§2.4).

2.1. The subgroup theorem with period term.

For a commutative algebraic group G over $\overline{\mathbb{Q}}$ we write T_G for its tangent space at the origin with its $\overline{\mathbb{Q}}$-structure, $\Omega(G) = \ker \exp_G$ and $\Lambda(G) = \exp_G^{-1}(G(\overline{\mathbb{Q}}))$; for a number field K, $\Omega_K(G), \Lambda_K(G)$ are their K-spans.

Theorem 2.1 ([33, Thm. 4.1, pp. 385–386])**.** *Let $G = \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times G_2$ be a connected commutative algebraic group over $\overline{\mathbb{Q}}$ of dimension $d = d_0 + d_1 + d_2 \geq 1$, with projections π_0, π_1 onto the first two factors (d_0, d_1 need not be maximal). Let $V \subset T_G(\mathbb{C})$ be a subspace of dimension $N < d$, $W \subset V$ a subspace defined over $\overline{\mathbb{Q}}$, and $Y \subset V$ a finitely generated subgroup with $\Gamma = \exp_G Y \subset G(\overline{\mathbb{Q}})$; put $\kappa = \mathrm{rk}_{\mathbb{Z}}(Y \cap \ker \exp_G)$. Then there is a connected algebraic subgroup $G' \neq G$ defined over $\overline{\mathbb{Q}}$ such that, with $\delta = \dim G/G'$, $\delta_0 = \dim \mathbb{G}_{\mathrm{a}}^{d_0}/\pi_0(G')$, $\delta_1 = \dim \mathbb{G}_{\mathrm{m}}^{d_1}/\pi_1(G')$, $\delta_2 = \delta - \delta_0 - \delta_1$, $\lambda = \mathrm{rk}_{\mathbb{Z}}\,\Gamma/(\Gamma \cap G')$ and $\tau = \dim W/(W \cap T_{G'})$, one has $\delta > \tau$ and*

$$(\lambda + \delta_1 + 2\delta_2)(d - N) \leq (\delta - \tau)(d_1 + 2d_2 - \kappa). \tag{1}$$

($G' = 0$ is allowed.) The proof [33, §§6–7] combines an auxiliary function [33, Prop. 6.1] with Philippon's zero estimate [23]. Prop. 6.1 asks for a polynomial that vanishes with multiplicity $\geq T$ along W at the points of $\exp_G(Y(S))$, with the saving κ in the parameters. The κ-saving is proved in [32, Prop. 2.1 and 2.4] (an auxiliary function constructed with periods), for $W = 0$ and $b = 1$ [33, p. 396]; for the rest W88 refers to [32] and to Bertrand [30, App. I, Lemme 7

and its Corollaire, with Lemme 1] (cited in [33, p. 396] as "Lemma 7"). All our applications use both $W \neq 0$ and $\kappa > 0$. We write out the multiplicity step in Appendix A (Lemma A.1). It holds under the condition

$$2N \geq d_1 + 2d_0 + \kappa, \tag{2}$$

which is satisfied in all our applications: for Theorem 3.1: $5n - 2 \geq 3n$; Theorem 3.4: $5n \geq 3n$; Theorem 1.6: $4n + 2 \geq 2n + 2$; Proposition 5.5: $4n + 2 \geq 2n$; Theorem 7.6: $5n + 2 \geq 3n + 4$; Theorems 3.7 and 3.9: $5n - 2 \geq 3n$ and $5n \geq 3n$; Proposition 6.1: $4n + 2 \geq 2n + 4$; Theorem 4.2: $5n + 2 \geq 3n + 4$ and $5n + 2 \geq 3n + 2$. When $d_0 \geq 1$, the proof of Theorem 2.1 uses an exponent $b > 1$ in the case $\delta = \tau$; the last paragraph of the proof of Lemma A.1 covers this for every d_0. Proposition 6.1 and Theorem 4.2(a) use $d_0 = 2$, whereas the other proved results use $d_0 \leq 1$. This is legitimate, for three reasons.

- Theorem 2.1 is stated for any d_0 ("we do not assume that d_0 and d_1 are maximal", p. 385), and d_0 enters (1) only through $\delta_2 = \delta - \delta_0 - \delta_1$.
- Waldschmidt applies his main theorem with $d_0 = 2$ himself: in the proof of [33, Cor. 2.1] (p. 377, $G = \mathbb{G}_{\mathrm{a}}^2 \times \mathbb{G}_{\mathrm{m}}^2$, with the same two-row affine mixing as in Theorem 4.2(a), without E). He also applies it with $d_0 = n$ and $W \neq 0$ in the second proof of [33, Cor. 3.3] (p. 384, $G = \mathbb{G}_{\mathrm{a}}^n \times \mathbb{G}_{\mathrm{m}}$, $W = \mathbb{C}(1, 0, \ldots, 0, 1)$). Both of these are applications of Theorem 1.1 there, which is the case $\kappa = 0$.
- Lemma A.1 is stated for general d_0. Its only d_0-dependent hypothesis is (2), which we check in each application; its other hypothesis $d_2 \geq 1$ holds in every application below ($d_2 \geq n/2$). The compensation of the exponent b by the factor $(\log S)^{bd_0}$ in its last paragraph is uniform in d_0. (This last point is a two-sentence step of Lemma A.1, so it is part of trust point T; Theorem 7.6, a proof sketch, uses the same bookkeeping with $d_0 = 2$, which is consistent with it but is not independent support.)

So $d_0 = 2$ adds no risk beyond trust point T itself. We call "trust point T" the use of Theorem 2.1 with $W \neq 0$ and $\kappa > 0$, completed by Lemma A.1; the main proofs of Theorems 3.1, 3.4, 1.6, 3.7, 3.9, 4.2 and of Proposition 6.1 use exactly this. In all our applications V'' is a hyperplane, $N = d - 1$, and $W'' = \ker \Psi''$, so that $\tau = \delta + r' - 2$ where $r' = \dim \Psi''(T_{G'})$; thus only G' with $r' \leq 1$ can satisfy $\delta > \tau$, and then $\delta - \tau = 2 - r'$.

2.2. **CM abelian varieties.** Let $A/\overline{\mathbb{Q}}$ be a simple abelian variety of dimension g with CM by the CM field F, $[F : \mathbb{Q}] = 2g$, with (primitive) CM type Φ. For $\sigma \in \Phi$ let $z_\sigma \colon \mathrm{Lie}(A) \to \mathbb{C}$ be the $\overline{\mathbb{Q}}$-rational eigen-coordinate dual to a holomorphic eigen-differential ω_σ, so that $z_\sigma([x]v) = \sigma(x) z_\sigma(v)$ for $x \in \mathcal{O}_F$. Since $H_1(A, \mathbb{Q}) = F\gamma_0$ for any $0 \neq \gamma_0 \in \Omega(A)$, we have $z_\sigma(x\gamma_0) = \sigma(x) P_\sigma$ with $P_\sigma = z_\sigma(\gamma_0)$, so $z_\sigma(\Omega(A)) \subset \sigma(F) P_\sigma$.

Reduction to the simple case. A CM abelian variety over $\overline{\mathbb{Q}}$ is isogenous to $\prod_i B_i^{m_i}$ with B_i simple CM abelian varieties of primitive type, with CM fields F_i. By its *CM algebra* we mean any commutative semisimple $\mathbb{Q}$-subalgebra $\mathcal{E} \subset \mathrm{End}^0(A)$ of degree $2 \dim A$ (so $\mathcal{E} = \prod_i \mathcal{E}_i$ with $\mathcal{E}_i \supseteq F_i$ maximal commutative in $M_{m_i}(F_i)$), and eigen-differentials of A are taken with respect to $\mathcal{E}$; the argument below does not depend on this choice. Being maximal commutative, $\mathcal{E}$ contains the centre $\prod_i F_i$ of $\mathrm{End}^0(A)$ and its central idempotents. An eigen-character $\mathcal{E} \to \overline{\mathbb{Q}}$ is nonzero on exactly one of these, so the eigen-differential lives on one isotypic part $B_i^{m_i}$, and the character restricts to an embedding σ_0 of F_i, which acts on $B_i^{m_i}$ diagonally. An eigen-differential is therefore, on that part, a σ_0-eigen class for F_i, i.e. a $\overline{\mathbb{Q}}$-combination of pull-backs $\mathrm{pr}_k^* \omega_{\sigma_0}$ of one σ_0-eigen differential of B_i, of the same kind. So every nonzero eigen-period lies in $\overline{\mathbb{Q}}\, \sigma_0(F_i) P_{\sigma_0}(B_i) \setminus \{0\} = \overline{\mathbb{Q}}^\times P_{\sigma_0}(B_i)$, and it suffices to prove Theorem 1.2 for simple A of primitive type.

Lemma 2.2 (Lemma K′)**.** *Let $F \subset K$, where K is a number field containing the Galois closure of F, $n = [K : \mathbb{Q}]$, and let $\theta_1, \dots, \theta_n$ be a basis of $K/\mathbb{Q}$. For an embedding $\sigma\colon F \to K$ put $\psi_\sigma(m) = \sum_j \theta_j \sigma(m_j)$ on F^n. Then $(\psi_\sigma)_\sigma\colon F^n \to K^{2g}$ is a $\mathbb{Q}$-linear bijection, each ψ_σ is σ-semilinear and onto K, and for every set Σ of embeddings the common kernel of $(\psi_\sigma)_{\sigma\in\Sigma}$ is an F-subspace of F-dimension $n - |\Sigma|\, n/(2g)$.*

Proof. $F^n \cong F \otimes_{\mathbb{Q}} K$ via $m \mapsto \sum m_j \otimes \theta_j$, and $F \otimes_{\mathbb{Q}} K \cong K^{2g}$ via $x \otimes y \mapsto (\sigma(x)y)_\sigma$, since K contains the Galois closure; the composite is $(\psi_\sigma)_\sigma$. Semilinearity is clear. The common kernel is the preimage of $\{0\}^\Sigma \times K^{2g-|\Sigma|}$, of $\mathbb{Q}$-dimension $(2g - |\Sigma|)n$, i.e. of F-dimension $(2g - |\Sigma|)n/(2g)$. □

For $g = 1$, $F = k$ is an imaginary quadratic field, ψ_{id} and ψ_- are the maps ψ, ϕ below, and Lemma 2.2 becomes the following lemma, which we use throughout for CM elliptic curves (we keep its direct proof).

Lemma 2.3. *Let $k \subset K$ be an imaginary quadratic field inside a number field of degree n, $\theta_1, \dots, \theta_n$ a basis of $K/\mathbb{Q}$, and on k^n put $\psi(m) = \sum \theta_j m_j$, $\phi(m) = \sum \theta_j \bar{m}_j$. Then $(\psi, \phi)\colon k^n \to K \times K$ is a $\mathbb{Q}$-linear bijection;* $\ker\psi$ *and* $\ker\phi$ *are k-subspaces of dimension $n/2$, and* $\ker\psi \cap \ker\phi = 0$.

Proof. $k^n \cong K \otimes_{\mathbb{Q}} k$ via $m \mapsto \sum \theta_j \otimes m_j$, and $K \otimes_{\mathbb{Q}} k \cong K \times K$ via $x \otimes y \mapsto (xy, x\bar{y})$; the composite is (ψ, ϕ). ψ is k-linear and ϕ is k-semilinear, and both are onto K, which has k-dimension $n/2$. □

An *eigen-slot* is a pair (A, σ) with $A/\overline{\mathbb{Q}}$ a simple CM abelian variety and σ an embedding of its CM field; its periods are the eigen-periods P_σ of ω_σ (on $A^\natural$ if $\sigma \notin \Phi$). Two eigen-slots are equal iff the abelian varieties are isogenous and the embeddings correspond under the isogeny. An eigen-slot is of the first kind iff $\sigma \in \Phi$, and of the second kind otherwise. We use the following standard facts.

- (S1) [Lit] (standard; see Shimura [28]). $\mathrm{End}^0(A^n) = M_n(F)$, and Hom between non-isogenous simple abelian varieties is 0. Abelian subvarieties of A^n are the B_S, $S \subset F^n$ an F-subspace, with $\dim B_S = g \dim_F S$ and $T(B_S) = \{(\sigma(M)u_\sigma)_{\sigma\in\Phi} : u_\sigma \in \mathbb{C}^{\dim S}\}$ in the eigen-coordinates (M a matrix whose columns span S). On $T(B_S)$ the form $\theta^{\mathsf{T}} z_\sigma$ vanishes iff $\psi_\sigma(S) = 0$, and different $\sigma \in \Phi$ carry independent u_σ.
- (P) [Proved] Every connected algebraic subgroup of $\mathbb{G}_{\mathrm{a}}^a \times \mathbb{G}_{\mathrm{m}}^b \times B$, B an abelian variety, is a product $(H' \cap \mathbb{G}_{\mathrm{a}}^a) \times (H' \cap \mathbb{G}_{\mathrm{m}}^b) \times B'$; subgroups of $\mathbb{G}_{\mathrm{a}}^a$ are $\overline{\mathbb{Q}}$-subspaces and of $\mathbb{G}_{\mathrm{m}}^b$ rational subtori. (The image of H' in B is an abelian subvariety B', and $H'/(H' \cap M)$, M the linear part, maps to the linear group $M/(H' \cap M)$ trivially; so $H' = (H' \cap M)^0 \times B'$ by dimension, and $H' \cap M$ splits since $\mathrm{Hom}(\mathbb{G}_{\mathrm{a}}, \mathbb{G}_{\mathrm{m}}) = \mathrm{Hom}(\mathbb{G}_{\mathrm{m}}, \mathbb{G}_{\mathrm{a}}) = 0$.)
- (Ch) By Dedekind's independence of characters, the vectors $(\sigma(x))_{\sigma\in\Phi}$, $x \in F$, span K^g over K. Hence $\Lambda_K(A)$, read in the eigen-coordinates, is $\prod_{\sigma\in\Phi} K\, z_\sigma(\Lambda(A))$, and likewise for $\Omega_K(A)$: the slot coordinates of K-combinations of logarithms (or periods) can be prescribed independently.
- (W) [Lit] (Wüstholz [45]). The numbers 1, π and nonzero periods of pairwise distinct eigen-slots (first or second kind, on one or several simple CM abelian varieties) are linearly independent over $\overline{\mathbb{Q}}$. In particular $P_\sigma \notin \overline{\mathbb{Q}}$ and $P_\sigma/\pi \notin \overline{\mathbb{Q}}$. A nonzero eigen-period of A (resp. a second-kind eigen-period of $A^\natural$) is not in $\overline{\mathbb{Q}}\mathcal{L}$.

All of these follow from Wüstholz's theorem on $\mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{m}} \times \prod_i A_i^\natural$ (resp. with $\mathbb{G}_{\mathrm{m}}^r$ added), applied to the point with coordinates 1, $2\pi i$ (resp. the logarithms) and the cycles, as in the following

example. For example, $P/\pi \notin \overline{\mathbb{Q}}$: otherwise the point $(\gamma_0, 2\pi i)$ of $\mathrm{Lie}(A \times \mathbb{G}_{\mathrm{m}})$ lies in a $\overline{\mathbb{Q}}$-hyperplane $z_\sigma = a\,x$; the smallest algebraic subgroup whose Lie algebra contains it is $A \times \mathbb{G}_{\mathrm{m}}$ (A is simple and the subgroup is a product by (P)), whose Lie algebra is not in the hyperplane.

Universal vector extension. For $\sigma \notin \Phi$ the eigen-coordinate w_σ lives on $\mathrm{Lie}(A^\natural)$. We use the following facts. The input from the literature is [Lit] (Mazur–Messing [22]): $\mathrm{Lie}(A^\natural) = H^1_{\mathrm{dR}}(A)^\vee$ functorially, and $A^\natural$ is universal among extensions of A by vector groups (every such extension of B by U is the push-out of $B^\natural$ along a unique linear map $W_B \to U$, where W_B is the fibre of $B^\natural$). The rest is [Proved] by the one-line arguments given.

- (F1) $\mathrm{Lie}(A^\natural) = H^1_{\mathrm{dR}}(A)^\vee$ with functorial $\mathcal{O}_F$-action. In the dual eigenbasis (e_τ) the fibre Lie algebra is $\bigoplus_{\tau\notin\Phi} \mathbb{C}e_\tau$, and w_σ ($\sigma \notin \Phi$) is nonzero exactly on e_σ, which lies in the fibre. *Proof:* the Lie algebra sequence of $0 \to V \to A^\natural \to A \to 0$ is the dual of $0 \to H^0(A,\Omega^1) \to H^1_{\mathrm{dR}}(A) \to H^1(A,\mathcal{O}) \to 0$, so the fibre is the annihilator of $H^0(A,\Omega^1) = \bigoplus_{\tau\in\Phi} \overline{\mathbb{Q}}\omega_\tau$.
- (F2) For $B \subset A^n$, $B^\natural \to (A^n)^\natural$ is injective on fibres and $\pi^{-1}(B) = B^\natural + V^n$ (V the fibre). *Proof:* on fibres the map is dual to the restriction $H^1(A^n,\mathcal{O}) \to H^1(B,\mathcal{O})$, which is onto because B is a direct factor of A^n up to isogeny (Poincaré reducibility); and $B^\natural$ maps onto B.
- (F3) $\mathrm{Ext}(B,\mathbb{G}_{\mathrm{a}}) \cong H^1(B,\mathcal{O})$ and $\mathrm{Hom}(B,\mathbb{G}_{\mathrm{a}}) = 0$: a connected H over B with given intersection W' with the fibre exists iff $W' \supseteq W_B$, and then $H = B^\natural + W'$ is unique. *Proof:* such an H is a splitting over B of $\pi^{-1}(B)/W'$, which by (F2) and universality is the push-out of $B^\natural$ along $W_B \to V^n/W'$; it splits iff this map is 0, i.e. $W_B \subseteq W'$, and the splitting is unique since $\mathrm{Hom}(B, V^n/W') = 0$.
- (F4) $T(B^\natural_S) = \bigoplus_\tau \tau(S) \otimes \mathbb{C}$ and $W_{B_S} = \bigoplus_{\tau\notin\Phi} \tau(S) \otimes \mathbb{C}$. *Proof:* under the comparison isomorphism $H^1_{\mathrm{dR}}(B_S)^\vee \cong H_1(B_S,\mathbb{Q}) \otimes \mathbb{C}$, with $H_1(B_S,\mathbb{Q}) = S\gamma_0 \subset F^n\gamma_0$, the τ-eigen component is $\tau(S) \otimes \mathbb{C}$; the fibre part is $\tau \notin \Phi$ by (F1).

The periods of $A^\natural$ satisfy $\Omega(A^\natural) \otimes \mathbb{Q} = F\tilde{\gamma}_0$ with $w_\sigma(x\tilde{\gamma}_0) = \sigma(x)P_\sigma$, and the values $w_\sigma(\Lambda(A^\natural))$ ("σ-fibre values") are second-kind abelian integrals $\int \omega_\sigma + \overline{\mathbb{Q}}$.

Beta values [Lit] (Weil [43]; Rohrlich's appendix to [15]; Koblitz–Ogus [16]). For $N \geq 3$ and $0 < a, b < N$ with $a+b \not\equiv 0 \bmod N$, $B(a/N, b/N)$ is, up to $\overline{\mathbb{Q}}^\times$, the period of an eigen-differential on the Fermat curve of degree N, with Koblitz–Ogus label $\Phi_{a,b} = \{u \in (\mathbb{Z}/N)^\times : \langle ua/N\rangle + \langle ub/N\rangle + \langle u(N-a-b)/N\rangle = 1\}$ (after reducing $a/N, b/N$ to lowest common denominator); the corresponding isotypic factor of the Jacobian is a power of a simple CM abelian variety, and the differential is of the first kind iff $a + b < N$. Deligne's theorem (stated by Koblitz–Ogus [16]; it rests on Deligne's theorem that Hodge cycles on abelian varieties are absolutely Hodge [10]) gives: if $\mathbf{1}_{\Phi_a} + \mathbf{1}_{\Phi_b} = \mathbf{1}_{\Phi_c} + \mathbf{1}_{\Phi_d}$ on $(\mathbb{Z}/N)^\times$, then $P_aP_b \in \overline{\mathbb{Q}}^\times P_cP_d$. Together with the rule that π corresponds to the constant function 1, and $\mathbf{1}_\Phi + \mathbf{1}_{-\Phi} \equiv 1$, the same dictionary contains $P_\sigma P_{\bar\sigma} \in \pi\overline{\mathbb{Q}}^\times$. That these are all multiplicative relations is the Rohrlich–Lang conjecture, used below only in remarks labelled [Heuristic].

2.3. **CM elliptic curves and their universal vector extensions.** For a CM elliptic curve E let $E^\natural$ be its universal vector extension, with base coordinate z and conjugate-eigen fibre coordinate w: $\mathcal{O} = \mathrm{End}(E)$ acts on $T(E^\natural) = H^1_{\mathrm{dR}}(E)^\vee$ by its identity character on z and by the conjugate character on w (the eigenline of H^1_{dR} for $\bar a$ is spanned by $x\,dx/y + c\,dx/y$ with $c \in \overline{\mathbb{Q}}$, so these coordinates are rational; this is (F1) for $g = 1$). In these coordinates the periods are $(\lambda, \eta^*(\lambda))$, $\lambda \in \Lambda_E$, where $\eta^*(\lambda) = \eta(\lambda) - \kappa_1\lambda$ satisfies $\eta^*(a\lambda) = \bar a\,\eta^*(\lambda)$; the logarithms of points of $E^\natural(\overline{\mathbb{Q}})$ are $(u, \zeta^*(u) + t)$ with $u \in \mathcal{L}_E$, $t \in \overline{\mathbb{Q}}$ and $\zeta^* = \zeta - \kappa_1\,\mathrm{id}$. On E_i and E_ρ, $\kappa_1 = 0$. Since

the characters $a \mapsto a, \bar{a}$ of k are K-independent for $K \supseteq k$,

$$\Omega_K(E^\natural) = K\lambda_0 \times K\eta^*(\lambda_0) \qquad (\Lambda_E \otimes \mathbb{Q} = k\lambda_0). \tag{3}$$

We call *fibre values* the w-coordinates of $\Lambda(E^\natural)$, that is, $\zeta^*(u)+\overline{\mathbb{Q}}$ for $u \in \mathcal{L}_E \backslash \Lambda_E$ and $\eta^*(\lambda)+\overline{\mathbb{Q}}$ at lattice points, and write Fib for the set of fibre values. By Legendre and $\eta^*(a\lambda) = \bar{a}\eta^*(\lambda)$, $\Omega\eta^*(\Omega) \in \overline{\mathbb{Q}}^\times \pi$ for every CM period Ω.

2.4. **The Roy–Waldschmidt count, and the hyperplane case.** For G as in §2.1, let $d_0(G)$ (resp. $d_1(G)$) be the largest d with a surjection $G \to \mathbb{G}_{\mathrm{a}}^d$ (resp. $\mathbb{G}_{\mathrm{m}}^d$) over $\overline{\mathbb{Q}}$, and $\alpha(G) = 2\dim G - 2d_0(G) - d_1(G)$. Fix a basis $\theta_1, \dots, \theta_n$ of $K/\mathbb{Q}$ and $p_\theta \colon T_{G^n} = T_G^n \to T_G$, $(z_j) \mapsto \sum \theta_j z_j$. Let H be the largest connected algebraic subgroup of G^n defined over $\overline{\mathbb{Q}}$ with $p_\theta(T_H) = 0$ and put $\alpha_K(G) = \alpha(G^n/H)/n$ [26, p. 360]. $S_K(G)$ is the set of subspaces $p_\theta(T_{G'})$, $G' \subset G^n$ an algebraic subgroup defined over $\overline{\mathbb{Q}}$; they are $\overline{\mathbb{Q}}$-rational.

Theorem 2.4 ([26, Théorème, p. 363])**.** *Let $\mathcal{E}$ be a finite-dimensional $\mathbb{C}$-space with a $\overline{\mathbb{Q}}$-structure, $\Psi \colon T_G(\mathbb{C}) \to \mathcal{E}$ a $\overline{\mathbb{Q}}$-rational linear map, $W \subset V \subset \mathcal{E}$ with W rational, and $\kappa = \dim_K(V \cap \Psi(\Omega_K(G)))$. If (i) V contains no nonzero $\Psi(S)$ with $S \in S_K(G)$, then*

$$\dim_K\big(V \cap \Psi(\Lambda_K(G))\big) \le (\alpha_K(G) - \kappa)\dim_{\mathbb{C}}(V/W).$$

By [26, Remarque 1], (i) holds when V has no nonzero $\overline{\mathbb{Q}}$-rational point (and then $W = 0$), since every $\Psi(S)$ is rational. We use only this case. Theorem 2.4 is used below only for the first proof of Theorem 1.9, in the sketch of §5.7, and to explain the counts (Remarks 5.2 and 3.8). The main proofs of Theorems 3.1, 3.4, 1.6, 3.7, 3.9, 4.2 and of Proposition 6.1 apply Theorem 2.1 directly; the proof of Proposition 5.5 (on which the sketch of Theorem 5.6 rests) and the second derivations of Remarks 3.3, 3.6, 5.2 and 6.5 use its hyperplane consequence, Theorem 2.5 below.

Where Theorem 2.4 is proved, and the dependence. The proof [26, pp. 363–364] has two steps.

(b) (Reduction, fully written.) With $\Psi_\theta = \Psi \circ p_\theta$, let H be the largest connected subgroup with $T_H \subset \ker \Psi_\theta$, $G' = G^n/H$, $\Psi' \colon T_{G'} \to \mathcal{E}$ the induced map, $V' = \Psi'^{-1}(V)$, $W' = \Psi'^{-1}(W)$. Then $\dim_{\mathbb{Q}}(V' \cap \Lambda(G')) = n\dim_K(V \cap \Psi(\Lambda_K))$ and $\dim_{\mathbb{Q}}(V' \cap \Omega(G')) = n\kappa$, and the theorem follows from case (a) for G' and $\alpha(G') \le n\alpha_K(G)$.

(a) (Case $K = \mathbb{Q}$, $\Psi = \mathrm{id}$.) *Statement R*: if $V \subset T_G(\mathbb{C})$ contains no nonzero Lie algebra of an algebraic subgroup of G defined over $\overline{\mathbb{Q}}$, and $W \subset V$ is $\overline{\mathbb{Q}}$-rational, then

$$\dim_{\mathbb{Q}}(V \cap \Lambda(G)) \le \big(\alpha(G) - \dim_{\mathbb{Q}}(V \cap \Omega(G))\big)\dim_{\mathbb{C}}(V/W).$$

RW93 derive (i)⇒(iii) here "de manière purement formelle" from Theorem 4.1 of Waldschmidt [33] (the algebraic subgroup theorem with the period rank $\kappa = \mathrm{rk}(Y \cap \ker \exp_G)$; we checked its statement on pp. 385–386), by applying Roy's categorical Proposition 2.1 of [25] to the category $\mathcal{G}_w$ and the functions a', b, c, d, r of [25, §6]. Their Remark 2 says that Statement R with the κ-term was known when $\dim V = \dim G - 1$ (citing [33, Thm. 1.1]) or with κ replaced by 0 ([25, §6, Remark (i)]), and is proved in general "en raffinant les arguments du §6 de [R1] (en travaillant avec la fonction a' au lieu de a'')".

Step (b) also uses case (a) for $\ker \Psi'$, to see that Ψ' is injective on $\Lambda(G')$; this also follows from Wüstholz's analytic subgroup theorem [45].

The dependence, stated precisely. When $V' = \Psi'^{-1}(V)$ is a **hyperplane** of $T_{G'}$ (as in all our applications), Statement R with the κ-term is in substance the following refinement of [33, Thm. 1.1], proved in [33, §5d, p. 394].

Theorem 2.5 ([33, Thm. 1.1, pp. 375–376, under (5.4) of §5d, p. 394])**.** *Let $G = \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times G_2$ be a connected commutative algebraic group over $\overline{\mathbb{Q}}$ of dimension $d \geq 1$, with d_0, d_1 maximal and $d_2 = d - d_0 - d_1$. Let V be a hyperplane of $T_G(\mathbb{C})$, $W \subset V$ a subspace of dimension $t \geq 0$ defined over $\overline{\mathbb{Q}}$, and $Y = \mathbb{Z}y_1 + \cdots + \mathbb{Z}y_m \subset V$ a subgroup of rank m with $\exp_G Y \subset G(\overline{\mathbb{Q}})$. Put $\kappa = \mathrm{rk}_{\mathbb{Z}}(Y \cap \ker \exp_G)$. If*

$$m > (d_1 + 2d_2 - \kappa)(d - 1 - t), \tag{5.4}$$

then V contains a nonzero algebraic Lie subalgebra of $T_G(\mathbb{C})$ defined over $\overline{\mathbb{Q}}$, that is, $T_{G'} \subset V$ for some connected algebraic subgroup $G' \neq 0$ of G defined over $\overline{\mathbb{Q}}$.

The printed Theorem 1.1 has the hypothesis $m > (d_1 + 2d_2)(d - 1 - t)$, without κ; [33, §5d] proves it under the weaker hypothesis (5.4) (p. 394 prints $\kappa = \ell - m$, a misprint for $m - \ell$, cf. Lemma 5.2 there). The case $W = 0$ ($t = 0$) of Theorem 2.5 is also printed as [38, Thm. 5.1, p. 352], where it is attributed to [33, Thm. 1.1] and to [34, Thm. 4.1]. The proof in [33, §5d] applies Theorem 2.1 (in the form [33, Cor. 5.3]) to G with $N = d - 1$, and then again to an auxiliary subgroup G' given by [33, Lemma 5.1]. The condition of Lemma A.1 below covers the first application in the cases listed above; we have not checked it for the second. So every use of Theorem 2.5 rests on [33, §5d] as published.

For Theorems 3.1, 3.4, 1.6, 3.7, 3.9, 4.2 and Proposition 6.1 the main proofs bypass Statement R: §§3.1–3.3, 4, 5.4 and 6 apply Theorem 2.1 to explicit quotient groups and exclude every connected algebraic subgroup. Theorems 3.1, 3.4, 1.6 and Proposition 6.1 also have a second derivation from Theorem 2.5, with a short subgroup exclusion (Remarks 3.3, 3.6, 5.2 and 6.5). The proof of Proposition 5.5 uses Theorem 2.5 (a full case check against Theorem 2.1 is not written out). For Theorem 1.9, the cases (M1), (M3) were checked against Theorem 2.1 in the same way, and the second proof (§5.2) avoids [26] for $u \in \mathcal{L}_E$; only $u \in \overline{\mathbb{Q}}\mathcal{L}_E$ uses Theorem 2.4 as stated.

Weights used. $\alpha_K(\mathbb{G}_{\mathrm{a}}) = 0$, $\alpha_K(\mathbb{G}_{\mathrm{m}}) = 1$, and for an elliptic curve $\alpha_K(E) = 1$ if E has CM by k and $K \supseteq k$, else 2 [26, Lemme, p. 361]; α_K is additive on products [26, (2)]. For the universal vector extension $E^\natural$ of a CM curve and $K \supseteq k$ we need:

Lemma 2.6. *$\alpha_K(E^\natural) = 2 + 2/n$.*

Proof. In the coordinates (z, w) of §2.3, let $H \subset (E^\natural)^n$ be connected with $p_\theta(T_H) = 0$, with abelian image $B \subset E^n$. Then $B = B_S$ for a k-subspace $S \subset k^n$ on which both ψ (base) and ϕ (fibre) of Lemma 2.3 vanish; since $\ker\psi \cap \ker\phi = 0$, $B = 0$. Hence H lies in the fibre $\mathbb{G}_{\mathrm{a}}^n$, and the largest such H is the rational hyperplane $\sum \theta_j t_j = 0$, of dimension $n - 1$. The quotient has dimension $n + 1$ and admits no surjection to $\mathbb{G}_{\mathrm{a}}$ or $\mathbb{G}_{\mathrm{m}}$ (it is a non-split extension of E^n by $\mathbb{G}_{\mathrm{a}}$), so $\alpha = 2(n + 1)$ and $\alpha_K = 2 + 2/n$. □

3. Four exponentials at CM eigen-periods

This section proves Theorems 1.1 and 1.2. §§3.1–3.2 treat a CM elliptic curve (Theorems MG and MG$^\natural$, the K-forms behind Theorem 1.1); §3.3 treats one eigen-coordinate of a CM abelian variety of any dimension (Theorems MG-A and MG$^\natural$-A, behind Theorem 1.2). The abelian case checks are those of the elliptic ones, because the quotient group has abelian part of dimension $n/2$ for every g.

3.1. **Theorem MG.** Throughout this section $E/\overline{\mathbb{Q}}$ has complex multiplication by k, $\Omega \neq 0$ is a period of E, and for a number field K we write $K\mathcal{L}$, $K\mathcal{L}_E$ for the K-spans of $\mathcal{L}$ and $\mathcal{L}_E$. By Baker's theorem $K\mathcal{L} \cong K \otimes_{\mathbb{Q}} \mathcal{L}$, so elements of $\mathcal{L}$ that are $\mathbb{Q}$-linearly independent are K-linearly independent.

Theorem 3.1 (Theorem MG). *Let $c \in \overline{\mathbb{Q}}^\times$ and $s = c\,\Omega/\pi$. For every number field $K \supseteq k$,*

$$\mathcal{S}_K := \{x \in K\mathcal{L} :\ s\,x \in K\mathcal{L} + K\mathcal{L}_E\} \quad \textit{satisfies} \quad \dim_K \mathcal{S}_K \le 2,$$

and $2\pi i \in \mathcal{S}_K$ when $ic \in K$ (since then $s \cdot 2\pi i = 2ic\,\Omega \in K\mathcal{L}_E$).

Status: [Proved; formally verified, standard axioms] for $\Omega \in \overline{\mathbb{Q}}^\times\varpi$ and $K \ni i, c$ (the passage to an arbitrary $K \supseteq k$ uses Baker's theorem and is not formalised); [Proved modulo published theorems and Lemma A.1] in general.

Proof. Enlarging K does not change the conclusion (by Baker, K-independence in $K\mathcal{L}$ is preserved under base change), so assume $K \supseteq k(i,c)$; $n = [K:\mathbb{Q}]$ is even (the count below closes for every such n). Let $G = \mathbb{G}_{\mathrm{m}} \times \mathbb{G}_{\mathrm{m}} \times E$ with coordinates (x,y,z), $\Psi(x,y,z) = (x, y+z)$, $V = \mathbb{C}(1,s)$. Fix a basis $\theta_1, \dots, \theta_n$ of $K/\mathbb{Q}$, put $p_\theta \colon T_{G^n} \to T_G$, $(x_j) \mapsto \sum \theta_j x_j$, and $\Psi_\theta = \Psi \circ p_\theta$.

The group. Connected algebraic subgroups of $\mathbb{G}_{\mathrm{m}}^{2n} \times E^n$ are products $T_L \times B_S$ of a subtorus (given by a subspace $L \subset \mathbb{Q}^{2n}$) and an abelian subvariety (given by a k-subspace $S \subset k^n$). Such a subgroup has tangent space in $\ker \Psi_\theta$ iff $L = 0$ and $\psi(S) = 0$, where $\psi(m) = \sum \theta_j m_j$; by Lemma 2.3, $\ker \psi$ has k-dimension $n/2$. Put $U = B_{\ker \psi}$ and $G'' = G^n/U = \mathbb{G}_{\mathrm{m}}^{2n} \times A''$, $A'' = E^n/U$, of dimension $n/2$. Then $d_0 = 0$, $d_1 = 2n$, $d_2 = n/2$, $d = 5n/2$. The induced map Ψ'' is onto $\mathbb{C}^2$, $V'' = \Psi''^{-1}(V)$ is a hyperplane, and $W'' = \ker \Psi'' \subset V''$ is $\overline{\mathbb{Q}}$-rational of dimension $d-2$. For a connected $G' \subset G''$ put $r' = \dim \Psi''(T_{G'})$; then $\tau = \delta + r' - 2$, so only $r' \le 1$ can satisfy $\delta > \tau$, and $\delta - \tau = 2 - r'$.

The data. $w = (2\pi i, 2ic\Omega) \in \Psi(\Omega_K(G))$. Suppose $x_1, x_2 \in \mathcal{S}_K$ with $2\pi i, x_1, x_2$ K-independent, and put $v_j = (x_j, sx_j) \in V \cap \Psi(\Lambda_K(G))$. Let Y'' be generated by the images in $T_{G''}$ of elements of $\Lambda(G^n)$ with Ψ_θ-images $D\theta_h w$ (taken in $\Omega(G^n)$, D a common denominator), $\theta_h v_1$, $\theta_h v_2$ ($h = 1, \dots, n$). Then $\operatorname{rk} Y'' = 3n$, $\Gamma = \exp Y'' \subset G''(\overline{\mathbb{Q}})$, and $\kappa = n$: a period in Y'' has first coordinate in $2\pi i K$, so its v-parts vanish. The inequality of Theorem 2.1 reads $\lambda + \delta_1 + 2\delta_2 \le (2 - r') \cdot 2n$.

No obstruction.

- $G' = 0$: $\lambda = 2n$, so the left side is $2n + 2n + n = 5n > 4n$.
- $r' = 1$ with image $\mathbb{C}(0,1)$ (this covers every G' with an abelian part, since ψ does not vanish on any nonzero abelian subvariety of A''): an element of Y'' lying in $T_{G'} + \Omega(G'')$ has first coordinate in $2\pi i K$, so $\operatorname{rk}(Y'' \cap (T_{G'} + \Omega(G''))) \le n$ and $\lambda \ge 2n$; the left side is $\ge 3n > 2n$.
- $r' = 1$ with image a line $\mathbb{C}(p,q) \ne \mathbb{C}(0,1)$: then $G' = T_L$ with $\dim L \le n$. Two K-independent data in $T_{G'} + \Omega(G'')$ would put x_1, x_2 in a K-space of dimension ≤ 2 containing $2\pi i$ (namely $K \cdot 2\pi i + K\pi^2/\Omega$ if $q = 0$, and $K \cdot 2\pi i + K \cdot p\,\Omega/(sp-1)$ otherwise, using $s \notin K$), against the K-independence of $2\pi i, x_1, x_2$. So $\lambda \ge n$, $\delta_1 \ge n$, $\delta_2 = n/2$, and the left side is $\ge 3n > 2n$.
- $r' = 0$ forces $G' = 0$.

So no connected $G' \ne G''$ satisfies the conclusion of Theorem 2.1, a contradiction. □

Controls. If E has no CM then $U = 0$, $d_2 = n$, and $G' = 0$ satisfies the inequality ($6n \le 6n$); so does $G' = 0$ with κ deleted ($5n \le 6n$), with only one datum, or with $k \not\subset K$. If $s \in K$ (a rational line) the full subtorus over V satisfies it with equality; replacing E by $\mathbb{G}_{\mathrm{m}}$ makes s algebraic, and then the statement is false ($s = 2$, $x = \log 2$).

Corollary 3.2 (MG2: four exponentials at $t \in \overline{\mathbb{Q}}^\times\Omega/\pi$). *Let $t \in \overline{\mathbb{Q}}^\times \cdot \Omega/\pi$ and $\alpha, \beta \in \overline{\mathbb{Q}}^\times$ with $\log\alpha, \log\beta$ linearly independent over $\mathbb{Q}$. Then $\alpha^t = e^{t\log\alpha}$ and $\beta^t = e^{t\log\beta}$ are not both algebraic. If moreover $\log\alpha, \log\beta, 2\pi i$ are linearly independent over $\mathbb{Q}$, then $t\log\alpha$ and $t\log\beta$ are not both in $\overline{\mathbb{Q}}\mathcal{L} + \overline{\mathbb{Q}}\mathcal{L}_E$. For example, $2^{\varpi/\pi}$ and $3^{\varpi/\pi}$ are not both algebraic.*

Status: the first sentence is Theorem 1.1(b) and has its status. The second sentence is [Proved modulo published theorems and Lemma A.1] (for $\Omega \in \overline{\mathbb{Q}}^{\times}\varpi$ it is the formally verified K-form of Theorem 3.1 combined with Baker's theorem).

Proof. If $\log\alpha, \log\beta, 2\pi i$ are $\mathbb{Q}$-independent, take $x_1 = \log\alpha$, $x_2 = \log\beta$ in Theorem 3.1: with $2\pi i$ they are three $\mathbb{Q}$-independent logarithms, hence K-independent. Otherwise $m\log\beta = a\log\alpha + 2\pi i\, b$ with integers $m \neq 0$, $b \neq 0$. If α^t and β^t are algebraic, so is $e^{2\pi i b t} = (\beta^t)^m(\alpha^t)^{-a}$, i.e. $2\pi i b t = 2ib\,c\,\Omega \in \mathcal{L}$ (with $t = c\Omega/\pi$). Then $\Omega \in \overline{\mathbb{Q}}\mathcal{L}$, which the analytic subgroup theorem [45] excludes (on $\mathbb{G}_{\mathrm{m}} \times E$, the smallest algebraic subgroup whose tangent space contains (ℓ, Ω) is $\mathbb{G}_{\mathrm{m}} \times E$). (Remark 3.3 gives a proof that needs no such step.) □

The determinations matter: $\log\alpha = \log 2$ and $\log\beta = \log 2 + 2\pi i$ are $\mathbb{Q}$-independent, and for the determinations $e^{t\log 2}$ of both powers the statement would say $2^t \notin \overline{\mathbb{Q}}$, a one-exponential statement that we do not prove.

Remark 3.3 (Second derivation via §5d)**.** Theorem 2.5 gives Theorem 3.1 with a shorter subgroup analysis, and the first sentence of Corollary 3.2 without the case distinction. Keep $G'' = \mathbb{G}_{\mathrm{m}}^{2n} \times A''$, Ψ'', $V'' = \Psi''^{-1}(V)$ and $W'' = \ker\Psi''$ from the proof of Theorem 3.1: $d = 5n/2$, $d_0 = 0$, $d_1 = 2n$, $d_2 = n/2$, and W'' is $\overline{\mathbb{Q}}$-rational of dimension $t_W = d-2$ (the t of Theorem 2.5; we write t_W to keep t for the exponent of Corollary 3.2), so $d-1-t_W = 1$. Let Y'' be generated by the $3n$ lifts of $D\theta_h w$, $\theta_h v_1$, $\theta_h v_2$ ($h = 1, \dots, n$). The n period lifts lie in $Y'' \cap \ker\exp$, so $\kappa \geq n$ (in fact $\kappa = n$), and condition (5.4) of [33] reads $3n > (2n + n - \kappa)\cdot 1$, i.e. $3n > 2n$.

The rank $m = 3n$. For the data of Theorem 3.1 ($2\pi i, x_1, x_2$ K-independent) the first coordinates of the Ψ''-images $D\theta_h \cdot 2\pi i$, $\theta_h x_j$ are $\mathbb{Q}$-independent. For the data of Corollary 3.2 ($x_j = \log\alpha_j$, $sx_j = t\log\alpha_j \in \mathcal{L}$) only the $\mathbb{Q}$-independence of $\log\alpha, \log\beta$ is needed. Take the period lifts with toric part $(2\pi i D e_h, 0)$ and E^n-part $m^{(h)}\Omega \in \Lambda_E^n$, where $\psi(m^{(h)}) = 2icD\theta_h$ (possible by Lemma 2.3, as $ic \in K$; D a common denominator), and the data lifts toric, $(x_j e_h,\ sx_j e_h)$. The linear form induced by $\theta^{\mathsf{T}} z$ on $T_{A''}$ takes the $\mathbb{Q}$-independent values $2icD\theta_h\Omega$ on the period lifts and 0 on the data lifts, so a $\mathbb{Q}$-relation among the $3n$ lifts involves only the data lifts; their first-block coordinates $x_j e_h$ are $\mathbb{Q}$-independent.

No subgroup in V''. Theorem 2.5 gives a connected $G' \neq 0$ with $T_{G'} \subset V''$, and $G' = T_L \times B_S$. If $B_S \neq 0$, then $T_{G'}$ contains a vector $(0, b)$ with $\theta^{\mathsf{T}} b \neq 0$ (since ψ does not vanish on any nonzero abelian subvariety of A''), whose image $(0, \theta^{\mathsf{T}} b)$ is not on $V = \mathbb{C}(1, s)$. If $B_S = 0$, then $L \neq 0$, and for $0 \neq l = (l_1, l_2) \in L$ the image $(\theta^{\mathsf{T}} l_1, \theta^{\mathsf{T}} l_2) \in K^2 \setminus 0$ lies on $\mathbb{C}(1, s)$; so $\theta^{\mathsf{T}} l_1 \neq 0$ and $s \in K$, which is false ($\Omega/\pi \notin \overline{\mathbb{Q}}$ [20]).

For $K = \mathbb{Q}(i)$, $n = 2$, $\theta = (1, i)$: $A'' \cong E$ via $(z_1, z_2) \mapsto z_1 + iz_2$, so $G'' = \mathbb{G}_{\mathrm{m}}^4 \times E$, $d = 5$, $t_W = 3$, $m = 6$, $\kappa = 2$, and (5.4) is $6 > 6 - \kappa$. This route uses neither the K-independence of $2\pi i, \log\alpha, \log\beta$ nor Wüstholz's theorem: it proves the first sentence of Corollary 3.2 directly, also when $2\pi i$ lies in the $\mathbb{Q}$-span of $\log\alpha, \log\beta$. Its subgroup exclusion uses only $s \notin K$. It rests on Theorem 2.5 as published, where the main proof rests on Theorem 2.1 and Lemma A.1.

3.2. **The fibre twin: Theorem MG$^\natural$.** Let $E^\natural$ be read on its conjugate-eigen fibre coordinate w (§2.3), with $\eta^* = \eta^*(\Omega)$; recall that $\Omega\eta^* \in \overline{\mathbb{Q}}^{\times}\pi$ and that the fibre values are the w-coordinates of $\Lambda(E^\natural)$.

Theorem 3.4 (Theorem MG$^\natural$)**.** *Let $c \in \overline{\mathbb{Q}}^{\times}$ and $s = c\,\eta^*/\pi \in \overline{\mathbb{Q}}^{\times}/\Omega$. For every number field $K \supseteq k$,* $\dim_K\{x \in K\mathcal{L} :\ s\,x \in K\mathcal{L} + K \cdot (\textit{fibre values})\} \leq 2$.

Status: [Proved; formally verified, standard axioms] for $\Omega \in \overline{\mathbb{Q}}^{\times}\varpi$ and $K \ni c$ with $[K : \mathbb{Q}] \geq 4$ (the passage to an arbitrary $K \supseteq k$ uses Baker's theorem and is not formalised); [Proved modulo published theorems and Lemma A.1] in general.

Proof. As in the proof of Theorem 3.1 we may enlarge K; we take $K \supseteq k(i,c)$ with $n = [K:\mathbb{Q}] \geq 4$ (for $n = 2$ the count below does not close). Let $G = \mathbb{G}_{\mathrm{m}} \times \mathbb{G}_{\mathrm{m}} \times E^\natural$, $\Psi(x,y,X) = (x,\, y + w(X))$, $V = \mathbb{C}(1,s)$, and θ, p_θ, $\Psi_\theta = \Psi \circ p_\theta$ as before.

The group. There are no nonzero homomorphisms between tori and connected subgroups of $(E^\natural)^n$ in either direction, so connected subgroups of $\mathbb{G}_{\mathrm{m}}^{2n} \times (E^\natural)^n$ are products. A connected $H \subset (E^\natural)^n$ over B_S contains the canonical lift $B_S^\natural$ (since $\mathrm{Hom}(B_S^\natural, \mathbb{G}_{\mathrm{a}}) = 0$), whose tangent space is $\{(\sigma(M)z, \bar\sigma(M)w)\}$; so $\theta^{\mathsf{T}} w$ vanishes on T_H only if $\phi(S) = 0$. Hence the largest connected U with $T_U \subset \ker \Psi_\theta$ is $U = B^\natural_{\ker\phi} + \{\theta^{\mathsf{T}} w = 0\}$, and $G'' = \mathbb{G}_{\mathrm{m}}^{2n} \times Q''$ with $Q'' = (E^\natural)^n / U$, an extension of $A'' = E^n / B_{\ker\phi}$ (dimension $n/2$) by $\mathbb{G}_{\mathrm{a}}$ (coordinate $\varsigma = \theta^{\mathsf{T}} w$). Over an abelian subvariety $B' \neq 0$ of A'', coming from $S' \supsetneq \ker\phi$, we have $\phi(S') \neq 0$, so the image of $B^\natural_{S'}$ contains the fibre. Thus Q'' has no splitting over any $B' \neq 0$, and its connected subgroups are 0, the fibre F, and the $\pi^{-1}(B')$ (dimension $\leq n/2+1$). Q'' is a quotient of an anti-affine group, so $\mathrm{Hom}(Q'', \mathbb{G}_{\mathrm{a}}) = 0$: $d_0 = 0$, $d_1 = 2n$, $G_2 = Q''$, $d_2 = n/2+1$. The induced map is $\Psi''(t_1, t_2, q) = (\theta^{\mathsf{T}} t_1, \theta^{\mathsf{T}} t_2 + \varsigma(q))$, V'' is a hyperplane, $W'' = \ker \Psi''$, and $\tau = \delta + r' - 2$ as before.

The data. The period datum is $(2\pi i, 2ic\eta^*) \in \Psi(\Omega_K(G))$ (using $ic \in K$ and that ϕ is onto K). With x_1, x_2 as in Theorem 3.1, $\mathrm{rk}\, Y'' = 3n$ and $\kappa = n$ (a period of G'' has first coordinate in $2\pi i K$). The inequality reads $\lambda + \delta_1 + 2\delta_2 \leq (2 - r')(2n+2)$.

No obstruction. Every G' is $T_L \times H$ with $H \in \{0, F, \pi^{-1}(B')\}$, and $H \neq 0$ forces $\mathbb{C}(0,1) \subset \Psi''(T_{G'})$.

- $G' = 0$: $\lambda = 2n$, and the left side is $2n + 2n + (n+2) = 5n + 2 > 4n + 4$.
- Image $\mathbb{C}(0,1)$ ($H \neq 0$, or L in the second block; $l = \dim L \leq n$, $h = \dim H \leq n/2+1$): elements of Y'' in $T_{G'} + \Omega(G'')$ have first coordinate in $2\pi i K$, so $\lambda \geq 2n$, and the left side is $\geq 2n + (2n - l) + 2(n/2 + 1 - h) \geq 3n > 2n + 2$.
- $H = 0$ and image a line $\mathbb{C}(1, q)$, $q \in K$: a datum X in $T_{G'} + \Omega(G'')$ satisfies $X(s - q) \in K \cdot 2\pi i + K \cdot 2\pi i s$, so $X \in K \cdot 2\pi i + K \cdot 2\pi i/(s-q)$ (here $s \notin K$ because Ω is transcendental). So $\lambda \geq n$, and the left side is $\geq n + n + (n+2) = 3n + 2 > 2n + 2$.
- $r' = 0$ forces $G' = 0$; $r' = 2$ fails $\delta > \tau$.

The minimum margin is $n - 2$. □

Corollary 3.5 (MG$^\natural$2: four exponentials at $t \in \overline{\mathbb{Q}}^\times \Omega$)**.** *If $\log\alpha, \log\beta$ are linearly independent over $\mathbb{Q}$ and $c \in \overline{\mathbb{Q}}^\times$, then $\alpha^{c\Omega} = e^{c\Omega \log\alpha}$ and $\beta^{c\Omega} = e^{c\Omega\log\beta}$ are not both algebraic. For example, 2^{ϖ} and 3^{ϖ} (real logarithms) are not both algebraic.*

Status: as Theorem 1.1(a).

Proof. Put $x_j = c\Omega \log\alpha_j$ (with $\alpha_1 = \alpha$, $\alpha_2 = \beta$). If $\alpha^{c\Omega}, \beta^{c\Omega}$ are algebraic then $x_j \in \mathcal{L}$, and $s x_j \in \overline{\mathbb{Q}} \log\alpha_j \subset \overline{\mathbb{Q}}\mathcal{L}$. A K-relation among $2\pi i, x_1, x_2$ without the $2\pi i$ term contradicts Baker; one with it gives $0 \neq \ell \in \overline{\mathbb{Q}}\mathcal{L} \cap \overline{\mathbb{Q}}^\times \eta^*$, impossible by the analytic subgroup theorem on $\mathbb{G}_{\mathrm{m}}^r \times E^\natural$ [45]. (Remark 3.6 gives a proof that needs no such step.) □

Remark 3.6 (Second derivation via §5d)**.** As in Remark 3.3, keep $G'' = \mathbb{G}_{\mathrm{m}}^{2n} \times Q''$, Ψ'', V'' and $W'' = \ker\Psi''$ from the proof of Theorem 3.4: $d = 5n/2 + 1$, $d_0 = 0$, $d_1 = 2n$, $d_2 = n/2+1$, $t_W = \dim W'' = d - 2$. With the $3n$ lifts of $D\theta_h w$, $\theta_h v_1$, $\theta_h v_2$ one has $m = 3n$ and $\kappa \geq n$ (in fact $\kappa = n$), and condition (5.4) of [33] reads $3n > (2n + n + 2 - \kappa)\cdot 1$, i.e. $3n > 2n + 2$. It holds as soon as $\kappa \geq 3$, so for every admissible $n \geq 4$ (at $n = 2$ it would read $6 > 6$). For $K = \mathbb{Q}(i, \sqrt{2})$, $n = 4$: $G'' = \mathbb{G}_{\mathrm{m}}^8 \times Q''$, $d = 11$, $t_W = 9$, $m = 12$, $\kappa = 4$, and (5.4) is $12 > 14 - \kappa$.

The rank. For the data of Theorem 3.4 it follows from the first coordinates, as in Remark 3.3. For Corollary 3.5 take $s = 1/(c\Omega) \in \overline{\mathbb{Q}}^\times \eta^*/\pi$, so that $s x_j = \log\alpha_j \in \mathcal{L}$, and take the data lifts toric, $(x_j e_h,\ \log\alpha_j\, e_h)$. Take the period lifts with toric part $(2\pi i D e_h, 0)$; the second coordinate

$D\theta_h \cdot 2\pi i s$ of their image is then carried by the fibre coordinate ς of Q''. So ς separates the period lifts from the data lifts, and the first-block coordinates $x_j e_h$ of the data lifts are $\mathbb{Q}$-independent as soon as $\log\alpha, \log\beta$ are.

No subgroup in V''. A connected $G' \neq 0$ with $T_{G'} \subset V''$ is $T_L \times H$ with $H \in \{0, F, \pi^{-1}(B')\}$. If $H \neq 0$, then $F \subset H$, and T_F contains a vector with image $(0,1) \notin V$. If $H = 0$, then $L \neq 0$ and, as in Remark 3.3, a nonzero $l \in L$ gives $s \in K$, which is false since $s \in \overline{\mathbb{Q}}^\times/\Omega$ and Ω is transcendental.

So this route, too, needs only the $\mathbb{Q}$-independence of $\log\alpha, \log\beta$ and no Wüstholz step. It rests on Theorem 2.5 as published.

By the symmetry $t \leftrightarrow 1/t$, Corollaries 3.2 and 3.5 give the four exponentials conjecture at every $t \in \overline{\mathbb{Q}}^\times\Omega/\pi \cup \overline{\mathbb{Q}}^\times\pi/\Omega$ and at every $t \in \overline{\mathbb{Q}}^\times\Omega \cup \overline{\mathbb{Q}}^\times/\Omega$, for every CM period Ω.

Proof of Theorem 1.1. Part (a) is Corollary 3.5, and part (b) is Corollary 3.2. □

3.3. **CM abelian varieties: Theorems MG-A and MG$^\natural$-A.** Throughout this subsection A, F, Φ are as in §2.2, F^c is the Galois closure of F, and $K\mathcal{L}$ is the K-span of $\mathcal{L}$. By Baker's theorem $K\mathcal{L} \cong K \otimes_{\mathbb{Q}} \mathcal{L}$, so $\mathbb{Q}$-independent elements of $\mathcal{L}$ are K-independent, and enlarging K does not change the conclusions below.
By the reduction to the simple case in §2.2, we may and do assume A simple of primitive type.

Theorem 3.7 (Theorem MG-A)**.** *Let $\sigma \in \Phi$, $P = P_\sigma \neq 0$, $c \in \overline{\mathbb{Q}}^\times$ and $s = cP/\pi$. For every number field $K \supseteq F^c(i,c)$,*

$$\mathcal{S}_K := \{x \in K\mathcal{L} :\ s\,x \in K\mathcal{L} + K\,z_\sigma(\Lambda(A))\} \quad \textit{satisfies} \quad \dim_K \mathcal{S}_K \leq 2,$$

and $2\pi i \in \mathcal{S}_K$.

Status: [Proved modulo published theorems and Lemma A.1]; two internal referees (correct with fixes, applied; exact independent enumerations for $g = 1, \dots, 6$).

Proof. Let $n = [K : \mathbb{Q}]$ (a multiple of $2g$), $G = \mathbb{G}_{\mathrm{m}} \times \mathbb{G}_{\mathrm{m}} \times A$ with coordinates (x, y, z), $\Psi(x,y,z) = (x,\ y + z_\sigma(z))$, $V = \mathbb{C}(1, s)$, and θ, $p_\theta \colon T_{G^n} \to T_G$, $(x_j) \mapsto \sum \theta_j x_j$, $\Psi_\theta = \Psi \circ p_\theta$ as in the proof of Theorem 3.1 (Φ is the CM type).

The group. By (P) and (S1), connected algebraic subgroups of $\mathbb{G}_{\mathrm{m}}^{2n} \times A^n$ are $T_L \times B_S$, and $T_{T_L \times B_S} \subset \ker\Psi_\theta$ iff $L = 0$ and $\psi_\sigma(S) = 0$. By Lemma 2.2 with $\Sigma = \{\sigma\}$, $\ker\psi_\sigma$ has F-dimension $n - n/(2g)$. Put $U = B_{\ker\psi_\sigma}$, of dimension $gn - n/2$, and $G'' = G^n/U = \mathbb{G}_{\mathrm{m}}^{2n} \times A''$, $A'' = A^n/U$. Then $\dim A'' = n/2$ *for every* g, and $d_0 = 0$, $d_1 = 2n$, $d_2 = n/2$, $d = 5n/2$: the numbers of Theorem 3.1. An abelian subvariety $B \neq 0$ of A'' comes from some $S \supsetneq \ker\psi_\sigma$, so $\psi_\sigma(S) \neq 0$ and $\Psi''(T_B) = \mathbb{C}(0,1)$; and $\dim B \leq n/2$.

The data. Since ψ_σ is onto K, the θ-span of $z_\sigma(\Omega(A^n))$ is KP, so $w = (2\pi i, 2icP) \in \Psi(\Omega_K(G))$ (using $ic \in K$); in particular $2\pi i \in \mathcal{S}_K$. Suppose $x_1, x_2 \in \mathcal{S}_K$ with $2\pi i, x_1, x_2$ K-independent, and put $v_j = (x_j, s x_j) \in V \cap \Psi(\Lambda_K(G))$. Let Y'' be generated by the images in $T_{G''}$ of lifts of $D\theta_h w$ (in $\Omega(G^n)$, D a common denominator), $\theta_h v_1$, $\theta_h v_2$, $h = 1, \dots, n$. Then $\operatorname{rk} Y'' = 3n$, $\Gamma = \exp Y'' \subset G''(\overline{\mathbb{Q}})$ and $\kappa = n$: since U is connected, a period of G'' is the image of a period of G^n, so its Ψ''-image has first coordinate in $2\pi i K$, and its v-components vanish. Inequality (1) reads $\lambda + \delta_1 + 2\delta_2 \leq (2 - r') \cdot 2n$.

No obstruction. We need $s \notin K$, i.e. $P/\pi \notin \overline{\mathbb{Q}}$, which is (W). The case check is that of the proof of Theorem 3.1 verbatim, because the only possible dependence on g is through $\dim A''$ and the bound $\dim B \leq \dim A''$, both independent of g: $G' = 0$ gives $5n > 4n$; image $\mathbb{C}(0,1)$ gives $\lambda \geq 2n$ and left side $\geq 3n > 2n$ (here a subtorus $T_L \subset G'$ lies in the second $\mathbb{G}_{\mathrm{m}}^n$ block, because θ^{T} vanishes on no nonzero rational vector, so $\dim L \leq n$; and $\dim B \leq n/2$); a subtorus with image a line $\mathbb{C}(p,q) \neq \mathbb{C}(0,1)$ has $\dim L \leq n$, and two K-independent data in $T_{G'} + \Omega(G'')$

would put x_1, x_2 in a 2-dimensional K-space containing $2\pi i$ (using $s \cdot 2\pi i \in KP$ and $s \notin K$), so $\lambda \geq n$ and the left side is $\geq 3n > 2n$; and $r' = 0$ forces $G' = 0$ (maximality of U). The minimum margin is n. Condition (2) reads $5n - 2 \geq 3n$. □

Remark 3.8 (The Roy–Waldschmidt count)**.** Roy–Waldschmidt's theorem [26, Théorème (iii)] as stated uses $\alpha_K(G)$, and $\alpha_K(A) = g$ for CM A [26, Lemme, p. 361]; for $G = \mathbb{G}_{\mathrm{m}}^2 \times A$ it gives the bound $(2 + g - 1) \cdot 1 \geq 3$, no contradiction with three data once $g \geq 2$. The inequality inside its proof, step (b), has $\alpha(G^n/H)/n$ with H the largest subgroup killed by $\Psi \circ p_\theta$; here $\alpha(G'')/n = (2n + 2 \cdot n/2)/n = 3$, and $(3 - 1) \cdot 1 = 2 < 3$. The saving comes from Ψ reading a single eigen-slot.

Theorem 3.9 (Theorem MG$^\natural$-A)**.** *Let $\sigma \notin \Phi$, $P = P_\sigma \neq 0$ the corresponding (second-kind) eigen-period, $c \in \overline{\mathbb{Q}}^\times$ and $s = cP/\pi$, so $s \in \overline{\mathbb{Q}}^\times/P_{\bar\sigma}$. For every number field $K \supseteq F^c(i, c)$,*

$$\mathcal{S}_K^\natural := \{x \in K\mathcal{L} : \ s\,x \in K\mathcal{L} + K\,w_\sigma(\Lambda(A^\natural))\} \quad \textit{satisfies} \quad \dim_K \mathcal{S}_K^\natural \leq 2,$$

and $2\pi i \in \mathcal{S}_K^\natural$ (as $s \cdot 2\pi i = 2icP_\sigma \in K\,w_\sigma(\Omega(A^\natural))$).

Status: [Proved modulo published theorems and Lemma A.1], given (F1)–(F4); two internal referees, one of whom built U and Q'' explicitly for $g = 2$ ($n = 8, 16$) and $g = 3$ ($n = 12$).

Proof. Prove it over $K' \supseteq K$ with $[K' : \mathbb{Q}] \geq 4$ and descend by Baker, as in the proof of Theorem 3.4. Let $G = \mathbb{G}_{\mathrm{m}} \times \mathbb{G}_{\mathrm{m}} \times A^\natural$, $\Psi = (x,\ y + w_\sigma)$, $V = \mathbb{C}(1, s)$. By (F1)–(F4), the largest connected subgroup with tangent space in $\ker \Psi_\theta$ is $U = B^\natural_{\ker \psi_\sigma} + \{\theta^{\mathsf{T}} t_\sigma = 0\}$, where t_σ is the σ-fibre coordinate, of dimension $(gn - n/2) + (gn - 1)$. The quotient $Q'' = (A^\natural)^n/U$ is an extension of A'' (dimension $n/2$) by $\mathbb{G}_{\mathrm{a}}$; it is a quotient of an anti-affine group, so $\mathrm{Hom}(Q'', \mathbb{G}_{\mathrm{a}}) = 0$, and it does not split over any abelian subvariety $B \neq 0$ of A'' (the image of $B^\natural_S$, $S \supsetneq \ker \psi_\sigma$, contains the fibre since $\psi_\sigma(S) \neq 0$). Its connected subgroups are 0, the fibre, and the $\pi^{-1}(B)$, and every nonzero one maps onto $\mathbb{C}(0, 1)$. So $G'' = \mathbb{G}_{\mathrm{m}}^{2n} \times Q''$ with $d_0 = 0$, $d_1 = 2n$, $d_2 = n/2 + 1$, connected subgroups of G'' are $T_L \times H''$, the period datum is $w = (2\pi i, 2icP_\sigma)$, $\kappa = n$, $\mathrm{rk}\, Y'' = 3n$, and the subtorus case needs $s \notin K$, i.e. $P_{\bar\sigma} \notin \overline{\mathbb{Q}}$ (W). This is exactly the situation of Theorem 3.4 (same type of quotient, same dimensions, same subgroup list, same data), with $d_1 + 2d_2 - \kappa = 2n + 2$, so (1) reads $\lambda + \delta_1 + 2\delta_2 \leq (2 - r')(2n + 2)$. Its case table:

G'	r'	left side	right side
0 $(\lambda = 2n)$	0	$5n + 2$	$4n + 4$
$T_L \times H''$ with image $\mathbb{C}(0, 1)$ $(\lambda \geq 2n$, $\dim L \leq n$, $\dim H'' \leq n/2 + 1)$	1	$\geq 3n$	$2n + 2$
subtorus with image a line $\neq \mathbb{C}(0, 1)$ $(\lambda \geq n$, $\dim L \leq n$, $H'' = 0)$	1	$\geq 3n + 2$	$2n + 2$

and $r' = 0$ forces $G' = 0$. So there is no obstruction iff $n \geq 3$, hence for every admissible n (which is even and ≥ 4: n is a multiple of $2g$, and $[K' : \mathbb{Q}] \geq 4$); the minimum margin is $n - 2$. Condition (2) reads $5n \geq 3n$. □

Proof of Theorem 1.2. By the reduction in §2.2, P is an algebraic multiple of an eigen-period of the same kind of a simple CM abelian variety of primitive type, and the classes in the statement depend only on $\overline{\mathbb{Q}}^\times P$. Write P^+ for a first-kind and P^- for a second-kind eigen-period; $P_\sigma P_{\bar\sigma} \in \pi\overline{\mathbb{Q}}^\times$ gives $\pi/P^+ \in \overline{\mathbb{Q}}^\times P^-$ and $\pi/P^- \in \overline{\mathbb{Q}}^\times P^+$ (for the conjugate slots).

Direct families: $t = cP^+/\pi$ (Theorem 3.7 with $s = t$) and $t = cP^-/\pi$ (Theorem 3.9 with $s = t$). Take $x_1 = \log \alpha$, $x_2 = \log \beta$; if $\alpha^t, \beta^t \in \overline{\mathbb{Q}}$ then $sx_j \in \mathcal{L}$. If $2\pi i, \log \alpha, \log \beta$ are $\mathbb{Q}$-independent they are K-independent (Baker), contradicting the theorem (which also puts

$2\pi i$ in the set). Otherwise $m\log\beta = a\log\alpha + 2\pi ib$ with integers $m, b \neq 0$, so $e^{2\pi ibt} \in \overline{\mathbb{Q}}$ and $t\cdot 2\pi i \in \overline{\mathbb{Q}}\mathcal{L}$; but $t\cdot 2\pi i = 2ic\,P^+$ (resp. $2ic\,P^-$) is a nonzero algebraic multiple of an eigen-period, which is not in $\overline{\mathbb{Q}}\mathcal{L}$ by (W) (on $A\times\mathbb{G}_{\mathrm{m}}^r$, resp. $A^\natural\times\mathbb{G}_{\mathrm{m}}^r$).

Inverted families: the four exponentials statement at t for all $\mathbb{Q}$-independent pairs is equivalent to the statement at $1/t$ (if $\alpha^t, \beta^t \in \overline{\mathbb{Q}}$, apply the statement at $1/t$ to α^t, β^t, whose logarithms $t\log\alpha, t\log\beta$ are $\mathbb{Q}$-independent). The inverses of the direct classes are $\overline{\mathbb{Q}}^\times\pi/P^+ = \overline{\mathbb{Q}}^\times P^-$ and $\overline{\mathbb{Q}}^\times\pi/P^- = \overline{\mathbb{Q}}^\times P^+$. So all of $t \in \overline{\mathbb{Q}}^\times P^{\pm1}$, $\overline{\mathbb{Q}}^\times(P/\pi)^{\pm1}$ are covered, for P of either kind ($1/P^+ \in \overline{\mathbb{Q}}^\times P^-/\pi$, $1/P^- \in \overline{\mathbb{Q}}^\times P^+/\pi$).

Beta values. By §2.2, $B(a/N, b/N)$ is a first-kind eigen-period up to $\overline{\mathbb{Q}}^\times$ if $a+b<N$ and a second-kind one if $a+b>N$. Of the four examples, $\Gamma(1/5)^2/(\pi\Gamma(2/5)) = B(\frac15,\frac15)/\pi$ and $\Gamma(4/5)^2/\Gamma(3/5) \in \overline{\mathbb{Q}}^\times B(\frac45,\frac45)$ (in fact $B(\frac45,\frac45) = \Gamma(\frac45)^2/\Gamma(\frac85) = \frac53\Gamma(\frac45)^2/\Gamma(\frac35)$) are covered by Theorem 3.7 (the latter inverted; the two are inverse to each other up to $\overline{\mathbb{Q}}^\times$), and $B(\frac15,\frac15)$, $B(\frac17,\frac17)$ by Theorem 3.9 (inverted). □

Remark 3.10 (Are the examples new instances?)**.** [Heuristic] Theorem 1.2 for a given Beta value t would already follow from Theorem 1.1 if t lay in $\overline{\mathbb{Q}}^\times\Omega^{\pm1}$ or $\overline{\mathbb{Q}}^\times(\Omega/\pi)^{\pm1}$ for a CM elliptic period Ω. For $t \in \overline{\mathbb{Q}}^\times\{P^{\pm1}, (P/\pi)^{\pm1}\}$ this happens iff $P \in \overline{\mathbb{Q}}^\times\{\Omega,\ \pi/\Omega,\ 1/\Omega,\ \Omega/\pi,\ \pi\Omega,\ \pi^2/\Omega\}$. Wüstholz's theorem excludes the two linear coincidences $P \in \overline{\mathbb{Q}}^\times\Omega$ and $P \in \overline{\mathbb{Q}}^\times\pi/\Omega = \overline{\mathbb{Q}}^\times\eta^*$ (for A not isogenous to a power of the curve); the other four are quadratic period relations, excluded only by the Rohrlich–Lang conjecture. So each named t is a new instance only conditionally.

Remark 3.11 (Not a linearisation)**.** [Heuristic] The hypothesis of Theorem 3.7 on the data $x_j = \log\alpha_j$ reads $cP\cdot\log\alpha = \pi\cdot\log\gamma$: a period times a logarithm against π times a logarithm. The abelian analogue of Laurent's identity (4) (an internal note, not reproduced here) linearises products of a period with the logarithm of the conjugate slot of the same variety, not $P\log\alpha$. So we do not expect Theorem 1.2 to follow from the analytic subgroup theorem.

Remark 3.12 (Strong form, not checked)**.** The $\mathbb{G}_{\mathrm{a}}$-bookkeeping of Theorem 4.2(a) below should transfer to Theorem 3.7, giving the strong four exponentials conjecture at $t \in \overline{\mathbb{Q}}^\times(P/\pi)^{\pm1}$ for first-kind abelian P. We have not checked this, and claim nothing.

3.4. **The general form, and elliptic four exponentials.** Theorems 3.1 and 3.4 are the cases $s_1 = s_3 = \mathbb{G}_{\mathrm{m}}$ of the following count. Let s_1, s_2, s_3 be *factors*: $\mathbb{G}_{\mathrm{m}}$ (period $2\pi i$), the base of a CM curve (period Ω), or the fibre of the universal vector extension of a CM curve (period η^*). In the count of Theorem 2.1 the first two have weight 1 per factor and a fibre factor has weight $1+2/n$. Write $\Lambda_K(s)$ for the K-span of the corresponding logarithms ($\mathcal{L}$, $\mathcal{L}_E$, or fibre values). Let p_j be the period of σ_j, $r = p_2/p_1$, and take Ψ reading s_1 into the first coordinate and s_2+s_3 into the second.

Theorem 3.13 (General form; proof sketch)**.** *Assume:*

- *s_1, s_2 are not of the same kind on isogenous groups (so $r \notin \overline{\mathbb{Q}}$);*
- *if s_2 and s_3 are of the same kind on isogenous groups, they are merged into one factor;*
- *(allowing $r \in \overline{\mathbb{Q}}^\times p_2/p_1$) K contains this constant, the CM fields, and the Legendre and isogeny constants, and $[K:\mathbb{Q}] \geq 2f+2$, where f is the number of fibre factors;*
- *s_1, s_3 are not both fibre factors.*

Then

$$\dim_K\{\lambda \in \Lambda_K(s_1) :\ r\lambda \in \Lambda_K(s_2) + \Lambda_K(s_3)\} \leq 2.$$

The count is that of Theorem 3.1, with the threshold $n > 2f$ coming from $G' = 0$ ($5n+2f > 4n+4f$). An internal referee checked all 27 factor-kind triples by an exact integer case check.

The load-bearing step (a bound on the data in subtori with image a K-line) was done by hand, and the subgroup lists are not written out in this paper. The excluded case is "four exponentials for quasi-logarithms": s_1, s_3 fibre factors of isogenous curves (for non-isogenous curves we have not written it out). It has mixed subgroups whose fibre slope μ is algebraic but possibly outside K, and it needs the $K(\mu)$-linear independence of K-independent fibre values, which we have not proved. When $s_1 = s_3$ is the base of a CM curve E, Theorem 3.13 is an elliptic analogue of four exponentials:

Corollary 3.14 (Elliptic four exponentials at CM slopes; proof sketch)**.** *Let E have CM, with period Ω, and let t be one of π/Ω^2, Ω'/Ω, $\eta^*(\Omega')/\Omega$, where Ω' is a period of a CM curve E' not isogenous to E, all up to $\overline{\mathbb{Q}}^\times$. If $u_1, u_2 \in \mathcal{L}_E$ and Ω are linearly independent over k, then tu_1 and tu_2 do not both lie in $\overline{\mathbb{Q}}\mathcal{L}_E + \overline{\mathbb{Q}}\Lambda(s_2)$. Here $\Lambda(s_2)$ is the set of fibre values of E (for $t = \pi/\Omega^2$), $\mathcal{L}_{E'}$ (for $t = \Omega'/\Omega$), or the set of fibre values of E' (for $t = \eta^*(\Omega')/\Omega$). In particular tu_1, tu_2 are not both elliptic logarithms of algebraic points of E.*

4. Strong four exponentials at CM elliptic periods

Throughout, $E/\overline{\mathbb{Q}}$ has CM by the imaginary quadratic field k, $\Omega \neq 0$ is a period, and $\eta = \eta^*(\Omega)$ its conjugate-eigen quasi-period (§2.3); Legendre's relation gives $\eta \in \overline{\mathbb{Q}}^\times \pi/\Omega$ (for $y^2 = 4x^3 - 4x$: $\Omega = \varpi$, $\eta = \pi/\varpi$). $\mathcal{L}_E$ and Fib are as in §2.3; for a number field K, $K\mathcal{L}$, $K\mathcal{L}_E$, KFib are K-spans.

Lemma 4.1. *$\Omega \notin \widetilde{\mathcal{L}}$ and $\eta \notin \widetilde{\mathcal{L}}$.*

Proof. [Lit: Wüstholz [45]; Baker–Wüstholz [2, Ch. 6]] If $\Omega = \beta_0 + \sum \beta_k \log \alpha_k$, the point $(1, \log\alpha, \Omega)$ of $\mathrm{Lie}(\mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m}^r \times E)$ lies in a $\overline{\mathbb{Q}}$-hyperplane; by Wüstholz the smallest algebraic subgroup through it has tangent space in the hyperplane, but it is a product (P) whose E-factor is E, and $0 \times 0 \times T(E)$ is not in the hyperplane. For η use $(1, \log\alpha, (\Omega, \eta))$ on $\mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m}^r \times E^\natural$. The torus splits off; the smallest subgroup $R \subset \mathbb{G}_\mathrm{a} \times E^\natural$ maps onto E, so its image in $E^\natural$ is $E^\natural$ (as $\mathrm{Hom}(E^\natural, \mathbb{G}_\mathrm{a}) = 0$, $E^\natural$ has no proper subgroup onto E); if $R \cap (\mathbb{G}_\mathrm{a} \times 0) = 0$, R would be the graph of a homomorphism $E^\natural \to \mathbb{G}_\mathrm{a}$, which is 0, contradicting the $\mathbb{G}_\mathrm{a}$-coordinate 1. So $R = \mathbb{G}_\mathrm{a} \times E^\natural$, which contains the fibre direction, not in the hyperplane. □

Theorem 4.2. *Let $c \in \overline{\mathbb{Q}}^\times$, and let $y_1, y_2 \in \widetilde{\mathcal{L}}$ be such that $2\pi i, y_1, y_2$ are linearly independent over $\overline{\mathbb{Q}}$.*

(a) (Theorem SMG) *If $s = c\Omega/\pi$, then sy_1 and sy_2 are not both in $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathcal{L}_E$.*
(b) (Theorem SMG$^\natural$) *If $s = c\eta/\pi$ (so $s \in \overline{\mathbb{Q}}^\times/\Omega$), then sy_1 and sy_2 are not both in $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}$Fib.*

Status: [Proved; formally verified, standard axioms] for $\Omega \in \overline{\mathbb{Q}}^\times \varpi$ with $sy_1, sy_2 \in \widetilde{\mathcal{L}}$ (the case used for Theorem 1.5); [Proved modulo the algebraicity of $g_3(\Omega_\rho \mathbb{Z}[\rho])$; formally verified otherwise] for $\Omega \in \overline{\mathbb{Q}}^\times \Omega_\rho$ with $sy_1, sy_2 \in \widetilde{\mathcal{L}}$; [Proved modulo published theorems and Lemma A.1] for the mixed second rows ($\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathcal{L}_E$, $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}$Fib) and for every other CM period. The paper proof uses Lemma A.1 with its exponent $b > 1$, since $d_0 \geq 1$ (§2.1). Two internal referees (correct with fixes, applied; independent exact enumerations) graded the first write-up as a proof sketch modulo trust point T; a third found every $r' \leq 1$ subgroup listed and nothing missing. Falsity hunt negative (§9.1).

Proof. Field. Choose a number field $K \supseteq k(i, c)$ containing all coefficients needed to write $y_j \in K + K\mathcal{L}$ and sy_j in $K + K\mathcal{L} + K\mathcal{L}_E$, resp. $K + K\mathcal{L} + K$Fib, with $[K : \mathbb{Q}] \geq 4$ in (b) (at $n = 2$ the count of (b) fails, with equality, for types Z and F of the table below). The $\overline{\mathbb{Q}}$-independence of $2\pi i, y_1, y_2$ gives their K-independence; no base change is needed (and base

change would not obviously preserve K-independence of elements of $\overline{\mathbb{Q}} + K\mathcal{L}$). Let $n = [K : \mathbb{Q}]$ and θ a basis of $K/\mathbb{Q}$.

(a) The group. $G = \mathbb{G}_a \times \mathbb{G}_m \times \mathbb{G}_a \times \mathbb{G}_m \times E$ with coordinates (a_1, x, a_2, y, z), $\Psi(a_1, x, a_2, y, z) = (a_1 + x,\, a_2 + y + z)$, $V = \mathbb{C}(1, s)$, $\Psi_\theta = \Psi \circ p_\theta$. By (P) and the structure of E^n, connected subgroups of G^n are $H \times T_L \times B_S$, and $T \subset \ker \Psi_\theta$ iff $H \subset \{\theta \cdot a_1 = \theta \cdot a_2 = 0\}$, $L = 0$, $\psi(S) = 0$ ($\psi(m) = \sum \theta_j m_j$; $\dim_k \ker \psi = n/2$ by Lemma 2.3). So $G'' = \mathbb{G}_a^2 \times \mathbb{G}_m^{2n} \times A''$ with $\dim A'' = n/2$: $d_0 = 2$, $d_1 = 2n$, $d_2 = n/2$, $d = 5n/2 + 2$, $d_1 + 2d_2 = 3n$. V'' is a hyperplane and $W'' = \ker \Psi''$ is $\overline{\mathbb{Q}}$-rational of dimension $d - 2$.

Data. $w = (2\pi i, 2ic\Omega) \in \Psi(\Omega_K(G))$, and $v_j = (y_j, sy_j) \in V \cap \Psi(\Lambda_K(G))$, writing $y_j = b_j + x_j$ with $b_j \in \overline{\mathbb{Q}} = \Lambda(\mathbb{G}_a)$. Let Y'' be generated by lifts of $\theta_h w, \theta_h v_1, \theta_h v_2$; Ψ'' is injective on it since $2\pi i, y_1, y_2$ are K-independent, so $m = \operatorname{rk} Y'' = 3n$, and $\kappa \geq n$ (in fact $\kappa = n$). With $Y_{G'} = \{y \in Y'' : \exp y \in G'\}$, $\lambda = 3n - \operatorname{rk} Y_{G'}$, and an obstruction needs $r' \leq 1$ and

$$(\star) \qquad \lambda + \delta_1 + 2\delta_2 \leq (2 - r')(3n - \kappa) \leq (2 - r') \cdot 2n.$$

Case check. $r' = 0$ forces $G' = 0$: the preimage of G' in G^n is connected (as U is) with tangent space in $\ker \Psi_\theta$, so it lies in U. For $G' = 0$, $(\star)$ reads $6n - \kappa \leq 2(3n - \kappa)$, i.e. $\kappa \leq 0$: excluded, margin $\kappa \geq n$. Let $r' = 1$, $\ell = \Psi''(T_{G'})$, $G' = H \times T_L \times B$ with h, l, b the dimensions. Then $\Psi''(T_H) = T_H$ (so $h \leq 1$, and $h = 1$ forces $\ell = T_H$); $L \neq 0$ forces ℓ to be K-rational and $l \leq n$; $B \neq 0$ forces $\ell = \mathbb{C}(0, 1)$. With $\delta_1 + 2\delta_2 = 3n - l - 2b$, $(\star)$ becomes $\operatorname{rk} Y_{G'} + l + 2b \geq 4n$. Since $y \in Y_{G'}$ implies $\Psi''(y) \in \ell + \Omega_\Psi$, $\Omega_\Psi := \Psi(\Omega_K(G)) = K2\pi i \times (K2\pi i + K\Omega)$ (a K-space), we have $\operatorname{rk} Y_{G'} \leq n \dim_K \mathcal{N}$, $\mathcal{N} = \{\beta \in K^3 : \beta_0 w + \beta_1 v_1 + \beta_2 v_2 \in \ell + \Omega_\Psi\}$.

- $\ell = \mathbb{C}(0, 1)$: first coordinates give $\beta_1 y_1 + \beta_2 y_2 \in K2\pi i$, so $\dim \mathcal{N} \leq 1$ and $\mathrm{rk} + l + 2b \leq n + n + n = 3n < 4n$.
- ℓ K-rational, $\neq \mathbb{C}(0, 1)$, $L \neq 0$: $\dim \mathcal{N} \leq 2$ (Lemma 4.4), so $\mathrm{rk} + l \leq 3n < 4n$.
- $L = 0$, $B = 0$, $h = 1$ (ℓ any $\overline{\mathbb{Q}}$-line): $\mathrm{rk} \leq 3n$, $l = b = 0$: $3n < 4n$.

No other G' has $r' = 1$. Hence no obstruction, for every $n \geq 2$; the minimum margin is n. Condition (2): $5n + 2 \geq 3n + 4$.

(b) The group. $G = \mathbb{G}_a \times \mathbb{G}_m \times \mathbb{G}_m \times E^\natural$, $\Psi(a, x, y, (z, w)) = (a + x,\, y + w)$ (w the conjugate-eigen fibre coordinate), $V = \mathbb{C}(1, s)$, $s = c\eta/\pi$. As in the proof of Theorem 3.4, $G'' = \mathbb{G}_a \times \mathbb{G}_m^{2n} \times Q''$, where the $\mathbb{G}_a$-part of U is the hyperplane $\{\theta \cdot a = 0\}$ and Q'' is the anti-affine extension of A'' ($\dim n/2$) by its fibre $\mathbb{G}_a$, with no splitting over any abelian subvariety $B \neq 0$: $d_0 = 1$, $d_1 = 2n$, $d_2 = n/2 + 1$, $d = 5n/2 + 2$, $d_1 + 2d_2 = 3n + 2$. Although $\mathbb{G}_a^n \times (E^\natural)^n$ has non-product subgroups, U is as stated: with $U_0 = \{\theta \cdot a = 0\} \times U_E$, where $U_E = B^\natural_{\ker \phi} + \{\theta^\mathsf{T} w = 0\} \subset (E^\natural)^n$ is the group U of the proof of Theorem 3.4, U/U_0 is a connected subgroup of $\mathbb{G}_a \times \mathbb{G}_m^{2n} \times Q''$ with tangent space in $\ker \Psi''$, and by Lemma 4.3 it is 0. The data are $w = (2\pi i, 2ic\eta)$ and $v_j = (y_j, sy_j)$; $m = 3n$, $\kappa \geq n$.

Lemma 4.3. *A connected subgroup of G'' is $T_L \times R$ with $R \subset \mathbb{G}_a \times Q''$ connected and one of: 0; a $\overline{\mathbb{Q}}$-line $\{(\alpha u, \beta u)\}$ in the vector group $\mathbb{G}_a \times \mathbb{G}_a^{\mathrm{fib}}$; $\mathbb{G}_a \times \mathbb{G}_a^{\mathrm{fib}}$; $0 \times Q''_B$; $\mathbb{G}_a \times Q''_B$ ($B \neq 0$, Q''_B the preimage of B).*

Proof. There are no nonzero homomorphisms between tori and $\mathbb{G}_a \times Q''$ or its subgroups, so the torus splits off. A subgroup of Q'' mapping onto $B \neq 0$ contains the fibre (otherwise it would split Q'' over B), so subgroups of Q'' are 0, $\mathbb{G}_a^{\mathrm{fib}}$, Q''_B. If R maps onto $B \neq 0$, its image in Q'' is Q''_B and $R \cap (\mathbb{G}_a \times 0)$ is $\mathbb{G}_a$ or 0; in the latter case R is the graph of a homomorphism $Q''_B \to \mathbb{G}_a$, which is 0 since a nonzero one would split Q''_B over B. If $B = 0$, R is a $\overline{\mathbb{Q}}$-subspace of $\mathbb{G}_a \times \mathbb{G}_a^{\mathrm{fib}}$. □

The diagonal lines $\{(\alpha u, \beta u)\}$ are the new feature compared with (a). With $\delta_0 = 1 - \dim \pi_0(R)$ one has $\delta_1 + 2\delta_2 = 3n + 4 - l - 2\dim R - 2\delta_0$, and for $r' = 1$ an obstruction needs $\operatorname{rk} Y_{G'} + l + 2\dim R + 2\delta_0 \geq 4n + 2$ (using $\kappa \geq n$). The cases:

type	R	image	$\operatorname{rk} Y_{G'}$	l	$\dim R$	δ_0	total
Z	0 ($G' = 0$)	0					$(\star) \iff \kappa \leq 2$
F	$0 \times Q''_B$	$\mathbb{C}(0,1)$	$\leq n$	$\leq n$	$\leq n/2 + 1$	1	$\leq 3n + 4$
VF	fibre line	$\mathbb{C}(0,1)$	$\leq n$	$\leq n$	1	1	$\leq 2n + 4$
T0	0, $L \neq 0$	$\mathbb{C}(0,1)$	$\leq n$	$\leq n$	0	1	$\leq 2n + 2$
VK	line, $\alpha \neq 0$, ℓ K-rat.	ℓ	$\leq 2n$	$\leq n$	1	0	$\leq 3n + 2$
VQ	line, $\alpha \neq 0$, ℓ not K-rat.	ℓ	$\leq 3n$	0	1	0	$\leq 3n + 2$
TK	0, $L \neq 0$, ℓ K-rat., $\neq \mathbb{C}(0,1)$	ℓ	$\leq 2n$	$\leq n$	0	1	$\leq 3n + 2$

Types with $R \supseteq \mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{a}}^{\mathrm{fib}}$ or $R = \mathbb{G}_{\mathrm{a}} \times Q''_B$ have $r' = 2$. For VK and TK, Lemma 4.4 applies with $\Omega_\Psi = K2\pi i \times (K2\pi i + K\eta)$. So there is no obstruction for $n \geq 3$, hence for every even $n \geq 4$ (type Z needs $\kappa \geq 3$, type F needs $n \geq 3$); the minimum margin is $n - 2$, attained by types Z and F; for $n = 2$ both obstruct, with equality, which is why $[K : \mathbb{Q}] \geq 4$. Condition (2): $5n + 2 \geq 3n + 2$. □

Lemma 4.4. *In the proofs above, if ℓ is a K-rational line different from $\mathbb{C}(0,1)$, then $\dim_K \mathcal{N} \leq 2$.*

Proof. Otherwise $v_1, v_2 \in \ell + \Omega_\Psi$. If $\ell = \mathbb{C}(1,0)$: $sy_j \in K2\pi i + K\omega_0$ ($\omega_0 = \Omega$, resp. η), so $y_j \in K \cdot 2\pi i + K \cdot \pi^2/\omega_0$ (as $ic \in K$). If $\ell = \mathbb{C}(p,1)$, $p \in K$: $y_j = \lambda_j p + 2\pi i a_j$ and $sy_j = \lambda_j + 2\pi i b_j + \omega_0 c_j$, so $y_j(1 - ps) = 2\pi i(a_j - pb_j) - p\omega_0 c_j$; since $\omega_0 = 2\pi i\, s/(2ic) \in K \cdot 2\pi i s$ (the period datum), $y_j \in 2\pi i(K + Ks)/(1 - ps)$, where $1 - ps \neq 0$ because $s \notin K$. In both cases y_1, y_2 lie in a 2-dimensional K-space containing $2\pi i$, against the K-independence of $2\pi i, y_1, y_2$. Here $s \notin K$: for (a), $s \in \overline{\mathbb{Q}}$ would give $\Omega \in \overline{\mathbb{Q}} \cdot 2\pi i \subset \widetilde{\mathcal{L}}$, against Lemma 4.1; for (b), $s \in \overline{\mathbb{Q}}^\times/\Omega$ is transcendental (Schneider). □

Proof of Theorem 1.5. The statement is invariant under $t \mapsto \gamma t$ ($\gamma \in \overline{\mathbb{Q}}^\times$) and under $t \mapsto 1/t$ (replace y_j by ty_j), and $1/(c\Omega) \in \overline{\mathbb{Q}}^\times \eta/\pi$; so $t = c\Omega/\pi$ and $t = c\eta/\pi$ suffice. Let $y_1, y_2 \in \widetilde{\mathcal{L}}$ be $\overline{\mathbb{Q}}$-independent with $ty_1, ty_2 \in \widetilde{\mathcal{L}}$. If $2\pi i \in \overline{\mathbb{Q}}y_1 + \overline{\mathbb{Q}}y_2$, then $t \cdot 2\pi i \in \widetilde{\mathcal{L}}$, i.e. $\Omega \in \widetilde{\mathcal{L}}$, resp. $\eta \in \widetilde{\mathcal{L}}$, contradicting Lemma 4.1. Otherwise $2\pi i, y_1, y_2$ are $\overline{\mathbb{Q}}$-independent and Theorem 4.2 applies, since $\widetilde{\mathcal{L}} \subset \widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathcal{L}_E$, resp. $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathrm{Fib}$. □

Corollary 4.5. (i) *$\varpi \log 2$ and $\varpi \log 3$ are not both in $\widetilde{\mathcal{L}}$; $(2e)^\varpi$ and 3^ϖ are not both algebraic; ϖ/π and $(\varpi/\pi)\log 2$ are not both in $\widetilde{\mathcal{L}}$ (so $e^{\varpi/\pi}$ and $2^{\varpi/\pi}$ are not both algebraic); $e^{1/\varpi}$ and $2^{1/\varpi}$ are not both algebraic. The same holds with $\Omega_\rho = 2^{-4/3}\Gamma(1/3)^3/\pi$ or any CM period in place of ϖ.*

(ii) *For every CM period Ω, one of e^{π^2}, $e^{\pi\Omega}$, $e^{\Omega/\pi}$ is transcendental, and one of e^{π^2}, $e^{\pi^2/\Omega}$, $e^{1/\Omega}$ is transcendental. More generally, $\pi^2, \pi\Omega, \Omega/\pi$ are not all in $\widetilde{\mathcal{L}}$, and neither are $\pi^2, \pi^2/\Omega, 1/\Omega$.*

(iii) (CM analogue of [36, Cor. 1.3]) *Let $\Lambda_1 \in \overline{\mathbb{Q}}^\times\{\Omega^{\pm 1}, (\Omega/\pi)^{\pm 1}\}$ and let $\Lambda_2, \Lambda_3 \in \widetilde{\mathcal{L}}$ be linearly independent over $\overline{\mathbb{Q}}$. Then $\Lambda_1\Lambda_2$ or $\Lambda_1\Lambda_3$ is not in $\widetilde{\mathcal{L}}$. (Λ_1 need not lie in $\widetilde{\mathcal{L}}$: $\Lambda_1 \notin \widetilde{\mathcal{L}}$ is proved for $\overline{\mathbb{Q}}^\times\Omega$ and $\overline{\mathbb{Q}}^\times\pi/\Omega$, and open for the other two classes.)*

(iv) (Mixed forms, only at the stated slopes.) *For $s \in \overline{\mathbb{Q}}^\times\Omega/\pi$ and $y_1, y_2 \in \widetilde{\mathcal{L}}$ with $2\pi i, y_1, y_2$ $\overline{\mathbb{Q}}$-independent, sy_1, sy_2 are not both in $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathcal{L}_E$; for $s \in \overline{\mathbb{Q}}^\times/\Omega$, not both in $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathrm{Fib}$. E.g. $\varpi \log 2/\pi$ and $\varpi \log 3/\pi$ are not both in $\overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L} + \overline{\mathbb{Q}}\mathcal{L}_{E_i}$. These do not pass to $1/s$, since inversion swaps the rows.*

Status: [Proved modulo published theorems and Lemma A.1]. (For $\Omega \in \overline{\mathbb{Q}}^{\times}\varpi$, items (i)–(iii) follow in one line from Theorem 1.5 and Theorem 4.2, whose status is stated there, and Baker's theorem; these one-line deductions are not formalised.)

Proof. (i) Theorem 1.5 with $(y_1, y_2) = (\log 2, \log 3)$ and $(1+\log 2, \log 3)$ at $t = \varpi$ ($\overline{\mathbb{Q}}$-independent by Baker), and $(1, \log 2)$ at $t = \varpi/\pi$ and at $t = 1/\varpi = \eta(\varpi)/\pi$. (ii) Theorem 4.2 with $(y_1, y_2) = (1, \pi^2)$ and $s = \Omega/\pi$, resp. $s = 1/\Omega \in \overline{\mathbb{Q}}^{\times}\eta/\pi$; $2\pi i, 1, \pi^2$ are $\overline{\mathbb{Q}}$-independent since π is transcendental. If $\pi^2 \notin \widetilde{\mathcal{L}}$ the statements are trivial. (iii) is Theorem 1.5 with $t = \Lambda_1$, $y = (\Lambda_2, \Lambda_3)$. (iv) is Theorem 4.2; for the example, $2\pi i, \log 2, \log 3$ are $\overline{\mathbb{Q}}$-independent by Baker. □

Remark 4.6. (1) The first triple in (ii) and the statement for $e^{\varpi/\pi}$, $2^{\varpi/\pi}$ in (i) are also Corollary 7.7(i), (ii), which §7.4 derives from the count only (proof sketch); the second triple in (ii) is not covered by Corollary 7.7 or by the elliptic results of §3. (2) Rescaling by x_1: if x_1, x_2 and y_1, y_2 are $\overline{\mathbb{Q}}$-independent pairs with $x_2/x_1 \in \overline{\mathbb{Q}}^{\times}\{\Omega^{\pm 1}, (\Omega/\pi)^{\pm 1}\}$, then one of $x_i y_j$ is not in $\widetilde{\mathcal{L}}$; this is a strong five exponentials statement with the fifth number x_1/x_2 replaced by the CM-ratio hypothesis. (3) The sharp form (no algebraic β_{ij} with $x_i y_j - \beta_{ij} \in \mathcal{L}$ for all four pairs) follows and is strictly weaker. (4) The general 2×3 minors statement of [36, Thm. 1.2] at CM data is out of reach of this count: the weight per coordinate is too large (internal note, proof sketch); (iii) is the only specialisation we obtain. (5) [Proof sketch, two internal referees] The same method gives strong *elliptic* four exponentials on a CM curve E at $t \in \overline{\mathbb{Q}}^{\times}(\Omega'/\Omega)^{\pm 1}$ and $\overline{\mathbb{Q}}^{\times}(\pi/(\Omega\Omega'))^{\pm 1}$, Ω' a period of a CM curve not isogenous to E, with $\widetilde{\mathcal{L}}_E = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}_E$ in place of $\widetilde{\mathcal{L}}$; it is classical by conjugation for non-conjugation-stable spans (via the elliptic strong six exponentials theorem [26, Cor. 1] on $\mathbb{G}_{\mathrm{a}} \times E$, since $\widetilde{\mathcal{L}}_E$ is conjugation-stable for CM E and $\bar{t} \in \overline{\mathbb{Q}}^{\times} t$), and we do not write it out.

Remark 4.7 (Strong six does not give Theorem 1.5)**.** Each inversion pair of slope classes contains one class with $t \notin \widetilde{\mathcal{L}}$ proved ($\overline{\mathbb{Q}}^{\times}\Omega$, $\overline{\mathbb{Q}}^{\times}\pi/\Omega$) and one where membership is open ($\overline{\mathbb{Q}}^{\times}\Omega/\pi$, $\overline{\mathbb{Q}}^{\times}/\Omega$). Theorem 1.5 is not known to follow from the strong six exponentials theorem; it would follow partially (via [36, Cor. 1.3], for pairs with $1, y_1, y_2$ $\overline{\mathbb{Q}}$-independent) only if Ω/π or $1/\Omega$ lay in $\widetilde{\mathcal{L}}$, which is open and expected to be false. Apart from the conjugation overlap described in §8.1, we found no classical few-line proof.

5. Two CM curves: Theorems M, MX and MX′

5.1. **Theorem M: first proof (the count).** Let $G = E \times \mathbb{G}_{\mathrm{m}} \times \mathbb{G}_{\mathrm{a}}$, $\Psi(e, t, a) = (e, t + a)$, $V = \mathbb{C}(\Omega, i\pi)$, $W = 0$, and $K \supseteq k$ containing the algebraic coefficients involved. Then $\alpha_K(G) = 1 + 1 + 0 = 2$; $\Psi(\Omega_K(G)) = K\Omega \times K \cdot 2\pi i$ since $K\Lambda_E = K\Omega$ for CM; so $\kappa = 1$. Condition (i): V has a nonzero rational point iff $\Omega/\pi \in \overline{\mathbb{Q}}$, which is false (Masser [20]; cf. [39, Thm. 26]; also Schneider). Theorem 2.4 gives $\dim_K(V \cap \Psi(\Lambda_K)) \leq 1$, and the period $(\Omega, i\pi)$ lies in it. If $i\pi u = \Omega\ell$ then $(u, \ell) \in V \cap \Psi(\Lambda_K)$, so $(u, \ell) = s(\Omega, i\pi)$ with $s \in K$. If $u \in \mathcal{L}_E$, Masser's theorem ([39, Thm. 27]: elements of $\mathcal{L}_E$ that are k-independent are $\overline{\mathbb{Q}}$-independent) gives $s \in k$. □

Controls. If E has no CM, $\alpha_K(E) = 2$ and the count ties (no conclusion). Replacing E by $\mathbb{G}_{\mathrm{m}}$ ($\Omega \to 2\pi i$) makes V rational, so (i) fails — as it must, since $u = 2\ell$ solves $i\pi u = 2\pi i\ell$. For e^{π^2} itself ($G = \mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{m}}^2$, $V = \mathbb{C}(1, i\pi)$) one finds $\kappa = 0$ and an exact tie.

5.2. **Theorem M: second proof for $u \in \mathcal{L}_E$ (Laurent's identity and Wüstholz).** We treat $u \in \mathcal{L}_E$, which covers every item of Corollary 1.10; a version for $u \in \overline{\mathbb{Q}}\mathcal{L}_E$ would need several points Q_j and is not written out here. Let $\Lambda_E = \Omega\,\mathbb{Z}[\alpha]$ (a sublattice of finite index suffices), and write Masser's relation [20] as $\eta(\alpha\Omega) = \bar{\alpha}\eta(\Omega) + c\,\Omega$ with $c \in \overline{\mathbb{Q}}$. Legendre's

relation gives $(\alpha-\bar\alpha)\eta(\Omega)-c\Omega=2\pi i/\Omega$, and $f(z)=\zeta(\alpha z)-\bar\alpha\zeta(z)-cz$ is an elliptic function defined over $\overline{\mathbb{Q}}$. With the third-kind period $\lambda(u,\Omega)=\Omega\zeta(u)-\eta(\Omega)u$ one obtains, for $u\in\mathcal{L}_E$ not a pole of f,

$$\frac{2\pi i\,u}{\Omega}=\bar\alpha\,\lambda(u,\Omega)-\lambda(\alpha u,\Omega)+\Omega f(u),\qquad f(u)\in\overline{\mathbb{Q}}. \tag{4}$$

On E_i this reads $2\eta(\varpi)u=i\lambda(iu,\varpi)-\lambda(u,\varpi)$ (cf. [19, §9, p. 160], where the general CM form is derived; the display above is its case $\mu=i$). Hence $i\pi u=\Omega\ell$ with $\ell\in\widetilde{\mathcal{L}}$ is a $\overline{\mathbb{Q}}$-linear relation among 1, logarithms of algebraic numbers, Ω, $\lambda(u,\Omega)$ and $\lambda(\alpha u,\Omega)$. These are coordinates of a point of $\Lambda(\mathcal{G})$ for $\mathcal{G}=G_Q\times G_{\alpha Q}\times\mathbb{G}_{\mathrm{m}}^r\times\mathbb{G}_{\mathrm{a}}$, where G_Q is the extension of E by $\mathbb{G}_{\mathrm{m}}$ attached to $Q=\exp_E(u)$ (Serre [27]) — add a factor $E^\natural$ when algebraic constants enter. By the analytic subgroup theorem [45], the smallest algebraic subgroup B whose tangent space contains this point lies in the hyperplane defined by the relation. B surjects onto the diagonal of $E\times E$, and over it the toric part carries the extension classes $[a+b\alpha]Q$ for characters (a,b); these vanish only for $a=b=0$ when Q is not torsion. So B contains both fibre tori, on which the relation has nonzero coefficients — a contradiction. Hence Q is torsion, which is the trivial case. □

5.3. **The corollaries of Theorem M.** (M1): $u=c\varpi/\pi$, $\ell=ic$ gives $s=c/\pi\notin\overline{\mathbb{Q}}$. (M2): $\ell=i\log\alpha$ gives $\log\alpha\in\overline{\mathbb{Q}}\pi$, so $\alpha=(-1)^{-is}$ with s algebraic, a root of unity by Gel'fond–Schneider. (M3): if $e^{\pi^2}\in\overline{\mathbb{Q}}$ then $\ell=ic\pi^2\in\widetilde{\mathcal{L}}$ and $u=c\pi\Omega$ give $s=c\pi\notin\overline{\mathbb{Q}}$. (M3′): $u=\varpi\log\alpha$, $\ell=i\pi\log\alpha$ give $\log\alpha\in\overline{\mathbb{Q}}^\times$, contradicting Hermite–Lindemann; (M3) is the case $\alpha=-1$, using $\wp(iz)=-\wp(z)$. (Mη): $\eta(\varpi)=\pi/\varpi$. On a general CM curve $\eta(\Omega)=\kappa_1\Omega+\kappa_2\pi/\Omega$ with $\kappa_2\neq0$, and the statement becomes $(\eta-\kappa_1\Omega)u\notin\widetilde{\mathcal{L}}$.

5.4. **Proof of Theorem 1.6.** The coordinates (z,w) on $E^\natural$, η^*, ζ^* and (3) are as in §2.3. Let $G=E_i^\natural\times E_\rho^\natural\times\mathbb{G}_{\mathrm{a}}^2$, let $K\supseteq\mathbb{Q}(i,\sqrt{-3},\beta)$ be a number field, $n=[K:\mathbb{Q}]$ (so $4\mid n$), and let

$$\Psi(x_i,x_\rho,a_1,a_2)=\big(z(x_i)+z(x_\rho)+a_1,\ w(x_i)+w(x_\rho)+a_2\big),\qquad V=\mathbb{C}\,w_1,$$

with $w_1=(\varpi,\beta\eta_\rho)$, $\eta_i=\eta(\varpi)=\pi/\varpi$, $\eta_\rho=\eta(\Omega_\rho)=2\pi/(\sqrt{3}\,\Omega_\rho)$, and $w_2=(\Omega_\rho,\beta\frac{2}{\sqrt{3}}\eta_i)$. By (3), $w_1,w_2\in\Psi(\Omega_K(G))$ and

$$\Psi(\Omega_K(G))=\mathcal{P}:=(K\varpi+K\Omega_\rho)\times(K\eta_i+K\eta_\rho);$$

$w_1\parallel w_2$ because $\varpi\cdot\frac{2}{\sqrt{3}}\eta_i=\frac{2\pi}{\sqrt{3}}=\Omega_\rho\eta_\rho$, and w_1,w_2 are K-independent since $\Omega_\rho/\varpi\notin\overline{\mathbb{Q}}$ (periods of non-isogenous curves are $\overline{\mathbb{Q}}$-independent [45]).

Proposition 5.1 (Proposition D)**.** *$V\cap\Psi(\Lambda_K(G))=Kw_1+Kw_2$.*

Status: in its single-point form (one $v\in V\cap\Psi(\Lambda(G))$, with conclusion $v\in\overline{\mathbb{Q}}w_1+\overline{\mathbb{Q}}w_2$; this is the form used for Theorem 1.6 and Corollary 7.5(ii)): [Proved modulo Wüstholz's theorem in the forms $\varpi\Omega_\rho/\pi\notin\overline{\mathbb{Q}}$, $\Omega_\rho/\varpi\notin\overline{\mathbb{Q}}$, and the algebraicity of $g_3(\Omega_\rho\mathbb{Z}[\rho])$; formally verified otherwise]. The K-span form as stated: [Proved modulo published theorems and Lemma A.1].

Granting Proposition 5.1: the point (X,Y,a,t) lies in $\Lambda(G)$ and the hypothesis says that its image lies on V, so $z(X)+z(Y)+a=x\varpi+x'\Omega_\rho$ with $x,x'\in K$. By the linear independence of 1 and elliptic logarithms on non-isogenous curves (Wüstholz [45]), $a=0$, $z(X)=x\varpi$, $z(Y)=x'\Omega_\rho$, and by Masser's theorem [20] $x\in\mathbb{Q}(i)$, $x'\in\mathbb{Q}(\rho)$. □

Proof of Proposition 5.1. Suppose $v\in V\cap\Psi(\Lambda_K(G))$ with $v\notin Kw_1+Kw_2$.

Step 1: an elementary lemma. For $L\in\{0,\ \mathbb{C}(1,0),\ \mathbb{C}(0,1)\}$ we have $V\cap(L+\mathcal{P})=Kw_1+Kw_2$. Indeed, if $sw_1\in L+\mathcal{P}$ with $L\neq\mathbb{C}(1,0)$ then $s\varpi\in K\varpi+K\Omega_\rho$, so $s\in K+K\Omega_\rho/\varpi$ and $(\Omega_\rho/\varpi)w_1=w_2$; if $L=\mathbb{C}(1,0)$ then $s\beta\eta_\rho\in K\eta_i+K\eta_\rho$, so $s\in K+K\eta_i/\eta_\rho$ and $(\eta_i/\eta_\rho)w_1=(\pi/\eta_\rho,\beta\eta_i)=\frac{\sqrt{3}}{2}w_2$.

Step 2: the group. Fix a basis $\theta_1,\dots,\theta_n$ of $K/\mathbb{Q}$, let $p_\theta\colon T_{G^n}\to T_G$, $(x_j)\mapsto\sum\theta_j x_j$, and $\Psi_\theta=\Psi\circ p_\theta$. Write the coordinates of T_{G^n} as $z_i,w_i,z_\rho,w_\rho,a_1,a_2\in\mathbb{C}^n$ and let $U\subset G^n$ be the vector subgroup $\{z_i=z_\rho=0,\ \theta^{\mathsf T}a_1=0,\ \theta^{\mathsf T}(w_i+w_\rho+a_2)=0\}$, of dimension $4n-2$, defined over $\overline{\mathbb{Q}}$, with $T_U\subset\ker\Psi_\theta$. Then $G''=G^n/U=\mathbb{G}_{\mathrm a}\times G_\ell$, where G_ℓ is the extension of $A=E_i^n\times E_\rho^n$ by $\mathbb{G}_{\mathrm a}$ obtained from the universal vector extension of A by pushing out along $(w_i,w_\rho)\mapsto\theta^{\mathsf T}w_i+\theta^{\mathsf T}w_\rho$. In the coordinates $s_1=\theta^{\mathsf T}a_1$, z_i, z_ρ, $s_2=\theta^{\mathsf T}(w_i+w_\rho+a_2)$ the map induced by Ψ_θ is

$$\Psi''(s_1,z_i,z_\rho,s_2)=\big(\theta^{\mathsf T}z_i+\theta^{\mathsf T}z_\rho+s_1,\ s_2\big).$$

We apply [33, Thm. 4.1] (Theorem 2.1) to G'' with $d=2n+2$, $d_0=1$ ($\pi_0=s_1$), $d_1=0$, $G_2=G_\ell$, $d_2=2n+1$; $V''=\Psi''^{-1}(V)$, a hyperplane, so $N=2n+1$; $W''=\ker\Psi''$, rational of dimension $2n$; and Y'' generated by the images of $3n$ elements of $\Lambda(G^n)$ whose Ψ_θ-images are $D\theta_h w_1$, $D\theta_h w_2$ (taken in $\Omega(G^n)$, D a common denominator) and $\theta_h v$, $h=1,\dots,n$. Since U is a vector group, $\Omega(G'')$ is the image of $\Omega(G^n)$, so $\Psi''(\Omega(G''))\subset\mathcal P$. By Step 1 with $L=0$, $\kappa=\mathrm{rk}(Y''\cap\Omega(G''))=2n$, and $\Gamma=\exp Y''$ has rank n. The inequality of Theorem 2.1 becomes

$$\lambda+2\delta_2\le(\delta-\tau)(2n+2),\tag{5}$$

and, with $r=\dim\Psi''(T_{G'})$, one has $\tau=\delta+r-2$; so an obstruction must have $r\le1$.

Step 3: the subgroups of G''. Abelian subvarieties of A are $B=B_i\times B_\rho$ (E_i,E_ρ are non-isogenous), and $B_i\subset E_i^n$ corresponds to a $\mathbb{Q}(i)$-subspace $S_i\subset\mathbb{Q}(i)^n$ with $T_{B_i}=\mathbb{C}\,\sigma(S_i)$, $\dim B_i=\dim S_i$; likewise for E_ρ. With ψ,ϕ as in Lemma 2.3 (for $k=\mathbb{Q}(i)$ and for $k=\mathbb{Q}(\rho)$), $\ker\psi\cap\ker\phi=0$ and each kernel has k-dimension $n/2$. Because $[a]$ acts on $T(E^\natural)$ by $\mathrm{diag}(a,\bar a)$ and $\mathrm{Hom}(E^\natural,\mathbb{G}_{\mathrm a})=0$ (the universal vector extension is anti-affine; [22]), the image in G_ℓ of the universal extension of B is a splitting $s(B)\cong B$ (with tangent space $T_B\times0$) if $\phi(S_i)=\phi(S_\rho)=0$, and is $\pi^{-1}(B)$, a non-split extension, otherwise (a splitting would give a nonzero homomorphism from a quotient of $(E_i^\natural)^{b_i}\times(E_\rho^\natural)^{b_\rho}$ to $\mathbb{G}_{\mathrm a}$). A connected subgroup over B either contains the fibre or is a splitting, and a splitting is unique since $\mathrm{Hom}(B,\mathbb{G}_{\mathrm a})=0$. It follows that the connected algebraic subgroups of $G''=\mathbb{G}_{\mathrm a}\times G_\ell$ are exactly

type	G'	condition	$\Psi''(T_{G'})$
A	$\mathbb{G}_{\mathrm a}\times\pi^{-1}(B)$	—	$\mathbb{C}^2$
B	$\mathbb{G}_{\mathrm a}\times s(B)$	$\phi(S)=0$	$\mathbb{C}(1,0)$
C	$0\times\pi^{-1}(B)$	—	$\mathbb{C}(0,1)$ if $\psi(S)=0$, else $\mathbb{C}^2$
D	$0\times s(B)$	$\phi(S)=0$	0 if $B=0$, else $\mathbb{C}(1,0)$
E	$\{(\mu\,\mathrm{pr}_F(g),g):g\in s(B)\times F\}$, $\mu\neq0$	$\phi(S)=0$	$\mathbb{C}(\mu,1)$ if $B=0$, else $\mathbb{C}^2$

(with $\phi(S)=0$ meaning $\phi(S_i)=\phi(S_\rho)=0$, and similarly for ψ). In type E, $F\cong\mathbb{G}_{\mathrm a}$ is the fibre of $G_\ell\to A$ (coordinate s_2), $\pi^{-1}(B)=s(B)\times F$, and pr_F is the projection to F; so G' is the graph of $\mu\,\mathrm{pr}_F$ into the first factor $\mathbb{G}_{\mathrm a}$ (coordinate s_1), and for $B=0$ it is $\{s_1=\mu s_2\}$. In every case allowed by the conditions, $b=\dim B\le n$.

Step 4: no obstruction. If $\Psi''(T_{G'})\in\{0,\mathbb{C}(1,0),\mathbb{C}(0,1)\}$ then $\lambda=n$: if $\exp(\sum c_h\tilde y_h)\in G'$ for the lifts $\tilde y_h$ of $\theta_h v$, then $xv\in L+\mathcal P$ with $x=\sum c_h\theta_h\in K$, so $x=0$ by Step 1. With $\delta_0=0$ for types A, B, E and $\delta_0=1$ for C, D, inequality (5) reads: $5n+2\le4n+4$ for $G'=0$; $5n+2-2b\le2n+2$ for types B and D ($B\neq0$); $5n-2b\le2n+2$ for type C with $r=1$; and $\lambda+4n+2\le2n+2$ for type E with $B=0$. Since $b\le n$, none of these holds for $n\ge3$, and types with $r=2$ fail $\delta>\tau$. So no connected $G'\neq G''$ satisfies the conclusion of Theorem 2.1, a contradiction. □

Remark 5.2 (The count)**.** Theorem 2.4 predicts Proposition 5.1: with $\kappa=2$ and $\alpha_K(G)=2(2+2/n)$ (Lemma 2.6) it gives $\dim_K(V\cap\Psi(\Lambda_K(G)))\le2+4/n$, which is <3 only for $n>4$; with the weight $\alpha(G'')/n=(4n+2)/n$ one gets $2+2/n<3$ for $n\ge3$, which is the threshold of Step 4.

A second derivation of Proposition 5.1 applies Theorem 2.5 to (G'', V'', W'', Y''): $d = 2n+2$, $d_0 = 1$, $d_1 = 0$, $d_2 = 2n+1$, $t_W = \dim W'' = 2n$, $m = 3n$, $\kappa = 2n$, and condition (5.4) of [33] reads $3n > (4n+2-2n)\cdot 1$, i.e. $3n > 2n+2$ (for $K = \mathbb{Q}(i,\sqrt{3})$, $n = 4$: $12 > 18 - \kappa = 10$). It remains to see that no connected $G' \neq 0$ has $T_{G'} \subset V''$. First, $T_{G'} \not\subset \ker \Psi''$. If G' meets the kernel $\mathbb{G}_{\mathrm{a}} \times F$ of $G'' \to A$ in a subgroup of positive dimension, $T_{G'}$ contains a vector $(c,0,0,d) \neq 0$, whose image is (c,d). Otherwise G' is finite over its image B in A, and $B \neq 0$. Then G' is an abelian variety, so $G' \subset 0 \times G_\ell$ ($\mathrm{Hom}(B, \mathbb{G}_{\mathrm{a}}) = 0$) is a splitting $s(B)$; by Step 3 this forces $\phi(S) = 0$, and then $\psi \neq 0$ on S (Lemma 2.3), so $\Psi''(T_{G'}) = \mathbb{C}(1,0)$. Since $T_{G'}$ is $\overline{\mathbb{Q}}$-rational, it contains a $\overline{\mathbb{Q}}$-rational x with $\Psi''(x) \neq 0$. If $T_{G'} \subset V''$, then $\Psi''(x) \in V$ is a nonzero $\overline{\mathbb{Q}}$-rational point of $V = \mathbb{C}w_1$, so $\varpi/(\beta\eta_\rho) \in \overline{\mathbb{Q}}$. This contradicts $\varpi/\eta_\rho \notin \overline{\mathbb{Q}}$, which is Wüstholz's theorem [45] (a period of E_i and a quasi-period of E_ρ are $\overline{\mathbb{Q}}$-linearly independent). The same exclusion can be read off the table in Step 3. Unlike Step 4, this derivation uses $\varpi/\eta_\rho \notin \overline{\mathbb{Q}}$, and it rests on Theorem 2.5 as published.

Corollary 1.7 is the case $X = (u, \zeta_i(u))$, $Y = 0$ (the logarithm of the origin), resp. $X = 0$; the value 0 of the ratio is excluded separately, since $1, u, \zeta(u)$ are $\overline{\mathbb{Q}}$-linearly independent for non-torsion u on a CM curve [18]; it also follows from [45] applied on $E^\natural \times \mathbb{G}_{\mathrm{a}}$. Note that the base and fibre coordinates of a single logarithm cannot be chosen independently: $w(X) - \zeta^*(z(X)) \in \overline{\mathbb{Q}}$.

Remark 5.3 (Other pairs of curves)**.** Let E, E' be non-isogenous CM curves with lattices $\omega\mathbb{Z}[\alpha]$, $\omega'\mathbb{Z}[\alpha']$ (up to finite index), so that $(\alpha - \bar\alpha)\,\omega\,\eta^*(\omega) = 2\pi i = (\alpha' - \bar\alpha')\,\omega'\eta^*(\omega')$. With $V = \mathbb{C}(\omega, \beta\eta^*(\omega'))$ the vectors $w_1 = (\omega, \beta\eta^*(\omega'))$ and $w_2 = \big(\omega', \beta\frac{\alpha-\bar\alpha}{\alpha'-\bar\alpha'}\eta^*(\omega)\big)$ are parallel, and the same proof gives: if $X \in \Lambda(E^\natural)$, $Y \in \Lambda(E'^\natural)$ and $(z(X)+z(Y)+a)\beta\eta^*(\omega') = (w(X)+w(Y)+t)\omega$, then $a = 0$ and $z(X) \in k\omega$, $z(Y) \in k'\omega'$.

Remark 5.4. By the explicit formulas for ϖ and Ω_ρ, $\varpi\Omega_\rho/\pi \in \overline{\mathbb{Q}}^\times \cdot \Gamma(1/4)^2\Gamma(1/3)^3/\pi^{5/2}$, and $\varpi/\eta_\rho = \frac{\sqrt{3}}{2\pi}\varpi\Omega_\rho$. If $\varpi/\eta_\rho = c \in \overline{\mathbb{Q}}$, then $X = Y = 0$, $a = 1$, $t = \beta/c$ satisfy the hypothesis of Theorem 1.6 with $a \neq 0$; so Corollary 1.8 is the case $X = Y = 0$. It also follows directly from Wüstholz's theorem [45] (a period of E_i and a quasi-period of E_ρ are $\overline{\mathbb{Q}}$-linearly independent).

5.5. **Controls for Theorem MX.** (1) If either curve has no CM there is no relation making two periods parallel, so κ drops to n in Step 2 and $G' = 0$ satisfies the inequality of Theorem 2.1 ($5n+2 \le 6n+4$): no conclusion. The same happens if only one Legendre relation is used. If the fibre action were $\mathrm{diag}(a,a)$ instead of $\mathrm{diag}(a,\bar a)$, subgroups of type D with image 0 would satisfy it; and for a rational line $V = \mathbb{C}(\mu, 1)$ with true data on it ($\kappa = 0$) the type E line does. So the conjugate character, the second Legendre relation and the irrationality of V are all used. (2) With the same curve twice, the antidiagonal of $E^\natural \times E^\natural$ lies in $\ker\Psi$ and the count collapses to the one-curve cell, which only gives $(\zeta(u)+t)/(u+a) \neq \pi/\varpi^2$ — a weak form of Chudnovsky's theorem [8]. (3) Data on the period plane satisfy the relation only trivially.

5.6. **A twisted variant.** Exchanging the roles of base and fibre on E_ρ gives a second statement of the same kind. Let $G = E_i^\natural \times E_\rho^\natural \times \mathbb{G}_{\mathrm{a}}^2$ and

$$\Psi'(X_i, X_\rho, a_1, a_2) = \big(z(X_i)+w(X_\rho)+a_1,\ w(X_i)+z(X_\rho)+a_2\big), \qquad V' = \mathbb{C}\,w_1', \quad w_1' = (\varpi, \beta\Omega_\rho).$$

With $w_2' = (\eta_\rho, \frac{2\beta}{\sqrt{3}}\eta_i)$ one has $w_2' = (\eta_\rho/\varpi)\,w_1'$ by the two Legendre relations, and for $K \supseteq \mathbb{Q}(i, \sqrt{-3}, \beta)$ both lie in $\Psi'(\Omega_K(G)) = (K\varpi + K\eta_\rho) \times (K\eta_i + K\Omega_\rho)$. The vectors w_1', w_2' are K-independent because $\eta_\rho/\varpi \notin \overline{\mathbb{Q}}$ (Corollary 1.8).

Proposition 5.5 (Proposition D′)**.** *Let $K \supseteq \mathbb{Q}(i, \sqrt{-3}, \beta)$ be a number field of degree $n \ge 8$. Then $V' \cap \Psi'(\Lambda_K(G)) = Kw_1' + Kw_2'$.*

Status: in its single-point form (one $v \in V' \cap \Psi'(\Lambda(G))$, with conclusion $v \in \overline{\mathbb{Q}}w_1' + \overline{\mathbb{Q}}w_2'$, every $\beta \in \overline{\mathbb{Q}}^\times$; this is the form used for Corollary 7.5(i)): [Proved modulo Wüstholz's theorem in the forms $\varpi\Omega_\rho/\pi \notin \overline{\mathbb{Q}}$, $\Omega_\rho/\varpi \notin \overline{\mathbb{Q}}$, and the algebraicity of $g_3(\Omega_\rho\mathbb{Z}[\rho])$; formally verified otherwise]. The K-span form as stated: [Proved modulo published theorems and Lemma A.1]. The formal verification rests on `lean_lsg`'s proof of Theorem 2.1 for the group G'' below, a route separate from the one checked against Lemma A.1 (§9.3, Appendix A).

Proof. With θ, p_θ and $\Psi'_\theta = \Psi' \circ p_\theta$ as before, a canonical lift $B^\natural$ of an abelian subvariety $B_{S_i} \times B_{S_\rho}$ has tangent space in $\ker \Psi'_\theta$ only if $\psi(S_\bullet) = \phi(S_\bullet) = 0$, i.e. $B = 0$ (Lemma 2.3). So the largest connected subgroup U' with $T_{U'} \subset \ker \Psi'_\theta$ is the vector subgroup $\{\theta^{\mathsf{T}}(w_\rho + a_1) = 0,\ \theta^{\mathsf{T}}(w_i + a_2) = 0\}$, and $G'' = G^n/U'$ is the pushout of $(E_i^n \times E_\rho^n)^\natural$ along $(w_i, w_\rho) \mapsto (\theta^{\mathsf{T}} w_i, \theta^{\mathsf{T}} w_\rho)$, an extension of $E_i^n \times E_\rho^n$ by $\mathbb{G}_{\mathrm{a}}^2$ of dimension $2n+2$. It is a quotient of an anti-affine group, so $\mathrm{Hom}(G'', \mathbb{G}_{\mathrm{a}}) = \mathrm{Hom}(G'', \mathbb{G}_{\mathrm{m}}) = 0$: $d_0 = d_1 = 0$ (maximal), $d_2 = d = 2n+2$. Let $\bar\Psi' \colon T_{G''} \to \mathbb{C}^2$ be the map induced by Ψ'_θ, $\bar V' = \bar\Psi'^{-1}(V')$ (a hyperplane) and $W'' = \ker \bar\Psi' \subset \bar V'$, $\overline{\mathbb{Q}}$-rational of dimension $t_W = 2n$. Suppose $v \in V' \cap \Psi'(\Lambda_K(G))$ with $v \notin Kw_1' + Kw_2'$, and let $Y'' \subset \bar V'$ be generated by the images of lifts in $\Lambda(G^n)$ of $D\theta_h w_1'$, $D\theta_h w_2'$ (in $\Omega(G^n)$, D a common denominator) and $\theta_h v$, $h = 1, \dots, n$. Since w_1', w_2', v are K-independent, their $\bar\Psi'$-images are $\mathbb{Q}$-independent, so $m = \mathrm{rk}\, Y'' = 3n$; the $2n$ period lifts give $\kappa \geq 2n$. The κ-refined Theorem 1.1 of [33, §5d] (Theorem 2.5) applies to $(G'', \bar V', W'', Y'')$: condition (5.4), $m > (d_1 + 2d_2 - \kappa)(d - 1 - t_W)$, holds since $(4n+4-\kappa)\cdot 1 \leq 2n+4 < 3n$ for $n > 4$, hence for $n \geq 8$ ($4 \mid n$). So $\bar V'$ contains $T_{G'}$ for some connected algebraic subgroup $G' \neq 0$ defined over $\overline{\mathbb{Q}}$. Then $\bar\Psi'(T_{G'})$ is a $\overline{\mathbb{Q}}$-rational subspace of V', which has no nonzero $\overline{\mathbb{Q}}$-rational point ($\Omega_\rho/\varpi \notin \overline{\mathbb{Q}}$); so $T_{G'} \subset \ker \bar\Psi'$, the preimage of G' in G^n is a connected subgroup with tangent space in $\ker \Psi'_\theta$, and $G' = 0$ by the maximality of U', a contradiction. (A full case check against Theorem 2.1, with the analogue of Step 1 of the proof of Theorem 1.6, should give the same threshold; we have not written it out.) □

Theorem 5.6 (Theorem MX′)**.** *Let $X \in \Lambda(E_i^\natural)$, $Y \in \Lambda(E_\rho^\natural)$, $a, b \in \overline{\mathbb{Q}}$ and $\beta \in \overline{\mathbb{Q}}^\times$. If*

$$\big(z(X) + w(Y) + a\big) \cdot \beta\, \Omega_\rho = \big(w(X) + z(Y) + b\big) \cdot \varpi,$$

then $z(X) = z(Y) = 0$ and both sides vanish.

Proof (sketch). By Proposition 5.5, $z(X) + w(Y) + a = x\varpi + x'\eta_\rho$ and $w(X) + z(Y) + b = x\beta\Omega_\rho + x'\frac{2\beta}{\sqrt{3}}\eta_i$ with $x, x' \in K$. The analytic subgroup theorem gives $z(X) = p\varpi$, $z(Y) = q\Omega_\rho$ with $p \in \mathbb{Q}(i)$, $q \in \mathbb{Q}(\rho)$; then $\zeta_i(p\varpi) \in \bar p\,\eta_i + \overline{\mathbb{Q}}$ and $\zeta_\rho(q\Omega_\rho) \in \bar q\,\eta_\rho + \overline{\mathbb{Q}}$, and the $\overline{\mathbb{Q}}$-independence of $1, \varpi, \eta_\rho$ and of $1, \eta_i, \Omega_\rho$ gives $x = p$, $x' = \bar q$, $q = p\beta$, $\bar p = \frac{2\beta}{\sqrt{3}}\bar q$, and vanishing constant terms. If $p \neq 0$ this gives $\sqrt{3}|p|^2 = 2|q|^2$ with $|p|^2, |q|^2 \in \mathbb{Q}_{>0}$, which is impossible. □

Corollary 5.7. *For every nonzero $u \in \mathcal{L}_{E_i}$ (off the lattice) and $a, b \in \overline{\mathbb{Q}}$: $(\zeta_i(u) + b)/(u + a) \notin \overline{\mathbb{Q}}^\times \cdot \Omega_\rho/\varpi = \overline{\mathbb{Q}}^\times \cdot \Gamma(1/3)^3/(\sqrt{\pi}\,\Gamma(1/4)^2)$.*

Together with Corollary 1.7, the ratio $(\zeta_i(u) + t)/(u + a)$ avoids both $\overline{\mathbb{Q}}\,\pi/(\varpi\Omega_\rho)$ and $\overline{\mathbb{Q}}^\times \Omega_\rho/\varpi$. *Status:* Theorem 5.6 and Corollary 5.7: [Proof sketch]. The step from Proposition 5.5 (whose status is stated there) to Theorem 5.6 rests on Masser's and Wüstholz's theorems and is sketched, not written out in full; it is not formalised. Corollary 5.7 is the case $X = (u, \zeta_i(u))$, $Y = 0$.

5.7. **The abelian-surface analogue.** A simple CM abelian surface A has two eigen-pairs of periods with $P_1P_4,\ P_2P_3 \in \pi\,\overline{\mathbb{Q}}^\times$ (Riemann's bilinear relations). For $A = \mathrm{Jac}(y^2 = x^5 + 1)$, $P_a = (1 - \zeta_5^a)B(\frac{a}{5}, \frac{1}{2} - \frac{a}{5})/5$, and the reflection formula gives these relations exactly. With $G = A^\natural \times \mathbb{G}_{\mathrm{a}}^2$, $\Psi = (z_1 + z_2 + a_1,\ w_3 + w_4 + a_2)$ and $V = \mathbb{C}(P_1, \beta P_3)$, the same argument (with $\alpha_K(A^\natural) = 4 + 4/n$) shows: for a non-torsion point of $A(\overline{\mathbb{Q}})$ with logarithm v, $(z_1(v) + z_2(v) +$

$a)\beta P_3 \neq (w_3(v)+w_4(v)+t)P_1$. We state this only as a proof sketch; the steps not written out are: the computation $\alpha_K(A^\natural) = 4+4/n$ (no abelian part in H, since the four coordinates carry four distinct characters of $\mathbb{Q}(\zeta_5)$); condition (i) via Wüstholz on $A^\natural$, which uses that $A^\natural$ has no proper algebraic subgroup surjecting onto A; the description of $\Lambda(A^\natural)$; and the final deduction that v is torsion. This sketch uses Theorem 2.4. The direct argument of §5.4 should carry over with the same numbers ($G'' = \mathbb{G}_\mathrm{a} \times G_\ell$ of dimension $2n+2$, abelian subvarieties given by $\mathbb{Q}(\zeta_5)$-subspaces and four conditions from the four embeddings), but we have not written it out.

5.8. **Single-partner limits for e^{π^2}.** The three exponentials conjecture asks that $\lambda_1\lambda_2 = \gamma\lambda_3$ with $\lambda_j \in \mathcal{L}$, $\gamma \in \overline{\mathbb{Q}}^\times$ force $\lambda_3 = 0$; $e^{\pi^2} \notin \overline{\mathbb{Q}}$ is the instance $\lambda_1 = \lambda_2 = i\pi$. In the count above, a configuration of rank-one shape gives a theorem only if κ exceeds what the true data need. For lines with a single coordinate per factor, $\kappa \leq 1$, and one CM factor supplies exactly one unit of weight. On the period side, Laurent's identity (4) and its abelian analogue show that these configurations are linearisable: multiplying by π turns them into linear relations among 1-periods, decided by the analytic subgroup theorem.

To get $\kappa = 2$ on a line one needs two independent periods with parallel images, i.e. a true *homogeneous* degree-two relation $p_a p_d = p_b p_c$ among periods. The CM Legendre relations $(\alpha - \bar{\alpha})\,\omega\,\eta^*(\omega) = 2\pi i$ of two non-isogenous CM curves provide one; so do two eigen-pairs of a CM abelian variety. The multiplicative group cannot take part: its only period is $2\pi i$, and the corresponding relation would pair $2\pi i$ with 1, which is not a period (for $E^\natural \times \mathbb{G}_\mathrm{m}$ with $\mathbb{G}_\mathrm{m}$ added to either coordinate, parallel periods force a $\overline{\mathbb{Q}}$-linear relation among $1, \omega, \eta^*$, so $\kappa \leq 1$). Under the hypothesis $e^{\pi^2} \in \overline{\mathbb{Q}}$ the number π^2 is a logarithm, not a period, so it adds data, not κ. This is why Theorem 1.6 exists while the same count cannot reach e^{π^2}. The $\mathbb{G}_\mathrm{m}$-analogue of Theorem 1.6 (replace Ω_ρ by $i\pi$ and the E_i-data by logarithms) is $i\pi \cdot \log\beta = \gamma \log\alpha$, an open case of the conjecture.

With two partners the mechanism does reach e^{π^2}: Corollary 7.1 (one of e^{π^2}, $e^{\pi\Omega}$, $e^{\pi^2/\Omega}$) uses π^2 and π^2/Ω as data and the period $(2\pi i, 2ic\Omega)$ for κ. In the configurations we enumerated (lines in $\mathbb{C}^2$, at most three factors per coordinate; not refereed), no single-partner disjunction "one of e^{π^2}, X" appears beyond (M3) and the known strong-six partners $e^{c/\pi}$, $e^{c\pi^3}$ (from the strong six exponentials theorem [24, Cor. 2] with $x = (1, \pi)$ and $y = (1/\pi, 1, \pi)$, resp. $y = (1, \pi, \pi^2)$).

[Heuristic] Within the monomial model (not refereed), a slack count explains this. On a line $V = \mathbb{C}(1, r)$, each factor contributes monomials: its period, plus the constant 1 for $\mathbb{G}_\mathrm{a}$ and for the fibre. So $\mathbb{G}_\mathrm{a}$ gives $\{1\}$, $\mathbb{G}_\mathrm{m}$ gives $\{\pi\}$, the base of E gives $\{\Omega\}$, and the fibre of $E^\natural$ gives $\{\pi/\Omega, 1\}$. Their weights in the count are 0, 1, 1, $1+2/n$, and the true relations are homogeneous for the grading $\deg\pi = 2$, deg(1-period) $= 1$. A line count is then a matching between the monomials of the two coordinates. The hypothesis $e^{\pi^2} \in \overline{\mathbb{Q}}$ adds the monomial π^2 (degree 4) to the $\mathbb{G}_\mathrm{m}$-factors, and a partner hypothesis adds one monomial m_X. A single-partner disjunction needs the partner to sit opposite π^2, so $m_X = \pi^2 r^{\pm 1}$, or the e^{π^2}-edges alone to fill the slack. Running through the cases leaves exactly $m_X = \pi \cdot p$ with p a period of the factor carrying the partner, i.e. $\wp(c\pi\Omega)$ (the case (M3)), together with the strong-six partners $e^{c/\pi}$ and $e^{c\pi^3}$. An exhaustive enumeration in a bounded range (at most four factors per coordinate) confirms this. On lines of type MX the slope has degree 0, so a $\mathbb{G}_\mathrm{m}$-factor can never be matched; e^{π^2} can enter such a count only together with a second hypothesis.

6. First-kind period relations: Theorem X

6.1. **Set-up.** A *first-kind slot* is an eigen-slot of the first kind (§2.2), written $j = (A_j, \sigma_j)$ with $A_j/\overline{\mathbb{Q}}$ a simple CM abelian variety (CM field F_j, primitive type Φ_j) and $\sigma_j \in \Phi_j$; z_j is the corresponding eigen-coordinate on $\mathrm{Lie}(A_j)$ and $P_j = z_j(\gamma_j) \neq 0$ for a fixed $\gamma_j \in \Omega(A_j)$, so

$z_j(\Omega(A_j)) \subset \sigma_j(F_j)P_j$. As in §2.2, two slots are equal iff the abelian varieties are isogenous and the embeddings correspond; distinct slots of one A have distinct embeddings. By (W), periods of distinct slots are $\overline{\mathbb{Q}}$-linearly independent. Let a, b, c, d be four pairwise distinct first-kind slots (distinct in the sense just defined, i.e. up to isogeny) with a *first-kind relation*

$$P_a P_b = \mu\, P_c P_d, \qquad \mu \in \overline{\mathbb{Q}}^{\times}. \tag{6}$$

Let $G = \prod_A A \times \mathbb{G}_{\mathrm{a}}^2$, the product over the isogeny classes among A_a, A_b, A_c, A_d, each taken once (so the factors A are pairwise non-isogenous), let $\beta \in \overline{\mathbb{Q}}^{\times}$ and

$$\Psi = \big(z_a + z_d + q_1,\ z_c + z_b + q_2\big)\colon T_G \to \mathbb{C}^2, \qquad V = \mathbb{C}w, \quad w = (P_a, \beta P_c),$$

where q_1, q_2 are the $\mathbb{G}_{\mathrm{a}}$-coordinates. Then $w' := (P_d, \beta P_b/\mu) = (P_d/P_a)\, w$ by (6). Let K be a number field containing the Galois closures of all F_j, μ and β, and $n = [K : \mathbb{Q}]$. By (Ch) and $z_j(\Omega(A_j)) \subset \sigma_j(F_j)P_j$,

$$\Psi(\Omega_K(G)) = \mathcal{P} := (KP_a + KP_d) \times (KP_c + KP_b),$$

so $w, w' \in \Psi(\Omega_K(G))$, and they are K-independent since $P_d/P_a \notin \overline{\mathbb{Q}}$.

Proposition 6.1. $V \cap \Psi(\Lambda_K(G)) = Kw + Kw'$.

Theorem 6.2 (Theorem X)**.** *In the set-up above, let $v_j \in \overline{\mathbb{Q}}\Lambda(A_j)$ for $j = a, b, c, d$ and $q_1, q_2 \in \overline{\mathbb{Q}}$ satisfy*

$$\big(z_a(v_a) + z_d(v_d) + q_1\big) \cdot \beta P_c = \big(z_c(v_c) + z_b(v_b) + q_2\big) \cdot P_a. \tag{7}$$

Then $q_1 = q_2 = 0$ and $z_j(v_j) \in \overline{\mathbb{Q}}P_j$ for each j. If $v_j \in \Lambda(A_j)$, then $\exp_{A_j}(v_j)$ is a torsion point.

Status: Proposition 6.1 and Theorem 6.2: [Proved modulo published theorems and Lemma A.1] (three internal referees, one with a blind classification of the subgroups; independent exact enumerations; minimum excess n). The deduction of Theorem 6.2 from the proposition is the standard analytic-subgroup step, as in §5.4 for Theorem 1.6, written out below.
When several of the slots lie on the same A (as in §6.4), the v_j may be logarithms of different points: by (Ch), a single element of $\Lambda_K(A)$ has slot coordinates $z_j(v_j)$ for any prescribed v_j.

6.2. **Proof of Proposition 6.1.** Suppose $v \in V \cap \Psi(\Lambda_K(G))$, $v \notin Kw + Kw'$. Let θ, p_θ and $\Psi_\theta = \Psi \circ p_\theta$ be as before.

The group. For each isogeny class A let Σ_A be the set of slots on A used by Ψ. By (P) and (S1), the largest connected subgroup with tangent space in $\ker \Psi_\theta$ is $U = \prod_A B_{\ker(\psi_\sigma)_{\sigma\in\Sigma_A}} \times \{\theta^{\mathsf{T}} q_1 = \theta^{\mathsf{T}} q_2 = 0\}$, and $G'' = G^n/U = \mathbb{G}_{\mathrm{a}}^2 \times A''$ with $\dim A'' = \sum_A |\Sigma_A|\, n/2 = 2n$ by Lemma 2.2. So $d = 2n + 2$, $d_0 = 2$, $d_1 = 0$, $d_2 = 2n$. The induced $\Psi''\colon T_{G''} \to \mathbb{C}^2$ is onto, $V'' = \Psi''^{-1}(V)$ is a hyperplane ($N = 2n + 1$), and $W'' = \ker \Psi'' \subset V''$ is $\overline{\mathbb{Q}}$-rational of dimension $2n$.

The data. Let Y'' be generated by the images of lifts of $D\theta_h w$, $D\theta_h w'$ (in $\Omega(G^n)$) and $\theta_h v$, $h = 1, \dots, n$. Since U is connected, $\Omega(G'')$ is the image of $\Omega(G^n)$, so $\Psi''(\Omega(G'')) \subset \mathcal{P}$, and $V \cap \mathcal{P} = Kw + Kw'$ (Lemma 6.4 with $L = 0$). Hence $\operatorname{rk} Y'' = 3n$, $\kappa = 2n$ exactly, and $\Gamma = \exp Y''$ has rank n. Since $d_1 + 2d_2 - \kappa = 2n$ and $d - N = 1$, (1) reads

$$\lambda + 2\delta_2 \le (2 - r') \cdot 2n. \tag{8}$$

Lemma 6.3 (Lemma S′)**.** *Every connected algebraic subgroup of G'' is $L \times B$ with L a $\overline{\mathbb{Q}}$-subspace of $\mathbb{G}_{\mathrm{a}}^2$ and B an abelian subvariety of A''. $\Psi''(T_B)$ is spanned by the target directions of the slots on which B is not killed: it is 0 iff $B = 0$; it is $\subset \mathbb{C}(1, 0)$ iff the slots c, b are killed, and then $\dim B \le n$; similarly for $\mathbb{C}(0, 1)$ (slots a, d killed); otherwise it is $\mathbb{C}^2$.*

Proof. The product decomposition is (P). By (S1) the slot forms $\theta^{\mathsf{T}} z_j$ on $T(B_S)$ involve independent vectors u_σ, and each slot feeds exactly one target coordinate, so $\Psi''(T_B)$ is a coordinate subspace; it is 0 only for $B = 0$ by maximality of U. If only the slots of one target survive, $S/\ker$ embeds into K^2 via the two surviving ψ_σ, so $\dim B \le 2 \cdot n/2 = n$ (Lemma 2.2). □

Lemma 6.4 (Lemma L′)**.** *For $L \in \{0,\ \mathbb{C}(1,0),\ \mathbb{C}(0,1)\}$ one has $V \cap (L + \mathcal{P}) = Kw + Kw'$.*

Proof. If $L \neq \mathbb{C}(1,0)$ and $\xi w \in L + \mathcal{P}$, then $\xi P_a \in KP_a + KP_d$, so $\xi = x + yP_d/P_a$ $(x, y \in K)$ and $\xi w = xw + yw'$. If $L = \mathbb{C}(1,0)$, then $\xi\beta P_c \in KP_c + KP_b$, so $\xi = x + yP_b/P_c = x + y\mu P_d/P_a$ and $\xi w = xw + y\mu w'$. □

The case check. Let $G' = L \times B$ with $r' = \dim \Psi''(T_{G'}) \le 1$, $e = \dim B$. Then $\delta_2 = 2n - e$. If $\Psi''(T_{G'}) \in \{0, \mathbb{C}(1,0), \mathbb{C}(0,1)\}$, then $\lambda = n$: if $\exp(\sum c_h \tilde{y}_h) \in G'$ for the lifts $\tilde{y}_h$ of $\theta_h v$, then $xv \in \Psi''(T_{G'}) + \mathcal{P}$ with $x = \sum c_h \theta_h \in K$, so $x = 0$ by Lemma 6.4.

G'	r'	λ	(8) reads	holds?
0	0	n	$5n \le 4n$	never
$B \neq 0$, image an axis ($e \le n$), $L \in \{0, \text{same axis}\}$	1	n	$5n - 2e \le 2n$	never ($e \le n$)
L an axis, $B = 0$	1	n	$5n \le 2n$	never
L a non-axis line, $B = 0$	1	≥ 0	$\lambda + 4n \le 2n$	never
$r' = 2$			$\delta > \tau$ fails	

So no connected $G' \neq G''$ satisfies the conclusion of Theorem 2.1, for any n; the minimum excess (left side minus right side) is n, attained at $G' = 0$ and at axis subgroups with $e = n$. Condition (2) reads $4n + 2 \ge 2n + 4$, and $d_0 = 2 \ge 1$ is covered by the final paragraph of Lemma A.1. This contradiction proves Proposition 6.1. □

Remark 6.5 (Second derivation)**.** Waldschmidt's [33, §5d] (κ-refined Theorem 1.1, hyperplane case) applies to (G'', V'', W'', Y'') directly: condition (5.4) there reads $m = 3n > (d_1 + 2d_2 - \kappa)(d - 1 - t) = 2n$, and V'' contains no nonzero Lie algebra of an algebraic subgroup, by Lemma 6.3 and $P_c/P_a \notin \overline{\mathbb{Q}}$. Here $m = \operatorname{rk} Y''$; p. 394 of [33] prints "$\kappa = \ell - m$", which must be read $m - \ell$ ($\ell = \operatorname{rk}\Gamma$, cf. pp. 392–393), as noted after Theorem 2.5.

6.3. **Proof of Theorem 6.2 from Proposition 6.1.** Enlarge K to contain q_1, q_2 and the coefficients of the v_j, so that $v_j \in K\Lambda(A_j)$; the proposition holds for every such K. When several slots lie on one A, (Ch) gives a single element of $\Lambda_K(A)$ with the prescribed slot coordinates $z_j(v_j)$. So hypothesis (7) says that the point $\Psi(v_a, \dots, v_d, q_1, q_2)$ lies in $V \cap \Psi(\Lambda_K(G))$, and Proposition 6.1 gives $x, y \in K$ with

$$z_a(v_a - x\gamma_a) + z_d(v_d - y\gamma_d) + q_1 = 0,$$

and similarly in the second coordinate. *Slot splitting.* Write $v_a - x\gamma_a$ and $v_d - y\gamma_d$ as K-combinations of finitely many $u_1, \dots, u_m \in \Lambda(A)$ (for each isogeny class A involved). The point $\xi = (1, (u_i)_i) \in \operatorname{Lie}(\mathbb{G}_{\mathrm{a}} \times \prod_A A^m)$ has $\exp \xi$ algebraic, and the displayed relation is a $\overline{\mathbb{Q}}$-linear form ℓ vanishing at ξ. By Wüstholz's theorem [45], the smallest algebraic subgroup H whose Lie algebra contains ξ has $T_H \subset \ker \ell$. By (P) and (S1), $H = L \times \prod_A B_{S_A}$, and $L = \mathbb{G}_{\mathrm{a}}$ because the $\mathbb{G}_{\mathrm{a}}$-coordinate of ξ is 1; so $(1, 0) \in T_H$ and $q_1 = 0$. On $T(B_S)$ the slot forms involve independent vectors u_σ (S1), and the slots a, d are distinct; so each slot form vanishes on T_H separately, and $z_a(v_a) = xP_a$, $z_d(v_d) = yP_d$. Likewise $q_2 = 0$, $z_c(v_c) \in KP_c$ and $z_b(v_b) \in KP_b$. *Torsion.* Let $v_j \in \Lambda(A_j)$ with $z_j(v_j) = x_j P_j$. If $x_j = 0$, Wüstholz on the simple A_j gives $v_j = 0$. Otherwise the smallest algebraic subgroup of A_j^2 whose Lie algebra contains (v_j, γ_j) is a proper abelian subvariety B_S (its tangent space lies in $\ker(z_j \oplus -x_j z_j)$) that projects onto the second factor. So $\dim_F S = 1$ and B_S is the graph of some $\alpha \in F_j = \operatorname{End}^0(A_j)$; hence $v_j \in F_j\gamma_j \subset \mathbb{Q}\,\Omega(A_j)$ and $\exp(v_j)$ is torsion. □

6.4. **The level-seven instance.** Let $C\colon y^7 = x(1-x)$. It has genus 3 (branch points $0, 1, \infty$), and dx/y^k is holomorphic exactly for $k = 4, 5, 6$ (orders $6-k$ at 0 and 1, $2k-8$ at ∞). The automorphism $y \mapsto \zeta y$ ($\zeta = e^{2\pi i/7}$) multiplies $dx/y^6, dx/y^5, dx/y^4$ by ζ, ζ^2, ζ^3, so $A = \mathrm{Jac}(C)$ has CM by $\mathbb{Q}(\zeta_7)$ with type $\{1,2,3\}$, which has trivial stabiliser in $(\mathbb{Z}/7)^\times$; so A is simple. Let z_1, z_2, z_3 be the eigen-coordinates dual to $dx/y^6, dx/y^5, dx/y^4$. On the closed cycle γ = (lift of $[0,1]$ to the sheet $y>0$) minus (its image under $y \mapsto \zeta y$),

$$\int_\gamma \frac{dx}{y^{7-j}} = (1-\zeta^j)\,B(\tfrac{j}{7},\tfrac{j}{7}) =: P_j \qquad (j = 1,2,3).$$

Let $E = E_{-7}$ be an elliptic curve over $\overline{\mathbb{Q}}$ with CM by $\mathbb{Q}(\sqrt{-7})$; by Chowla–Selberg [Lit] [15] it has a period $\Omega \in \overline{\mathbb{Q}}^\times B(\frac17,\frac27)$. The Beta cocycle $B(x,y)B(x+y,z) = B(y,z)B(x,y+z)$ at $(\frac17,\frac17,\frac27)$ and one reflection give, exactly,

$$B(\tfrac17,\tfrac17)B(\tfrac27,\tfrac27) = B(\tfrac17,\tfrac27)B(\tfrac17,\tfrac37), \qquad B(\tfrac17,\tfrac37) = \mu\, B(\tfrac37,\tfrac37),$$

$$\mu = \frac{\Gamma(\frac17)\Gamma(\frac67)}{\Gamma(\frac37)\Gamma(\frac47)} = \frac{\sin(3\pi/7)}{\sin(\pi/7)} = 1 + 2\cos\tfrac{2\pi}{7},$$

so $P_1P_2 \in \overline{\mathbb{Q}}^\times P_3\Omega$, a relation (6) with $a = z_1$, $b = z_2$, $c = z_3$, $d = E$. (In Koblitz–Ogus labels: $\{1,2,3\}+\{1,4,5\} = \{1,3,5\}+\{1,2,4\}$; the label $\{1,2,4\}$ is induced from $\mathbb{Q}(\sqrt{-7})$.) No use of Deligne's theorem is needed at this level.

Theorem 6.6 (Theorem X7)**.** *Let $v_1, v_2, v_3 \in \Lambda(A)$, $u \in \mathcal{L}_E$, $q_1, q_2 \in \overline{\mathbb{Q}}$ and $\beta \in \overline{\mathbb{Q}}^\times$. If*

$$\big(z_1(v_1) + u + q_1\big)\cdot \beta P_3 = \big(z_3(v_3) + z_2(v_2) + q_2\big)\cdot P_1,$$

then $q_1 = q_2 = 0$ and $\exp(v_1), \exp(v_2), \exp(v_3)$, $\exp_E(u)$ are torsion points.

This is Theorem 6.2 (with Ω normalised so that $P_1P_2 = \mu' P_3\Omega$ and $\mu' \in K$); here n is a multiple of 6 and the margin is n.
Status: [Proved modulo published theorems and Lemma A.1].

Corollary 6.7. (a) *For every non-torsion point $\exp(v) \in A(\overline{\mathbb{Q}})$: $z_2(v)/z_1(v) \notin \overline{\mathbb{Q}}\cdot P_3/P_1 = \overline{\mathbb{Q}}\cdot B(\frac37,\frac37)/B(\frac17,\frac17)$. In particular Corollary 1.11(a) holds.*

(b) *For every non-torsion $u \in \mathcal{L}_E$ and every $v \in \Lambda(A)$: $z_3(v)/u \notin \overline{\mathbb{Q}}^\times \cdot P_3/P_1$.*

(c) *P_1P_2, P_1P_3, P_1, P_3 are $\overline{\mathbb{Q}}$-linearly independent; equivalently, Corollary 1.11(b).*

Status: [Proved modulo published theorems and Lemma A.1] ((a) and (c) contain Corollary 1.11). That (c) is not a consequence of the analytic subgroup theorem is [Heuristic] (Remark 6.8).

Proof. (a) Data $v_1 = v_2 = v$, $v_3 = 0$, $u = 0$, $q_1 = q_2 = 0$, $\beta = \gamma$: if $z_2(v) = \gamma\, z_1(v) P_3/P_1$ then the hypothesis of Theorem 6.6 holds, so $\exp(v)$ is torsion. The values 0 and ∞ of the ratio do not occur for $v \neq 0$: $z_j(v) = 0$ forces $v = 0$ by Wüstholz on the simple A. For $O, Q \in C(\overline{\mathbb{Q}})$, z_j of the Abel–Jacobi logarithm of $Q - O$ is $\int_O^Q$ of the dual differential. (b) Data $v_3 = v$, u, $\beta = \gamma \in \overline{\mathbb{Q}}^\times$, all others 0. (c) Substituting (6) into $\alpha_1P_1P_2 + \alpha_2P_1P_3 + \alpha_3P_1 + \alpha_4P_3 = 0$ gives $(\alpha_2P_1 + \alpha_1\mu'\Omega + \alpha_4)\cdot P_3 = (-\alpha_3)\cdot P_1$. So, with $\beta = 1$ and $K \ni \alpha_i$, $v = (\alpha_2P_1 + \alpha_1\mu'\Omega + \alpha_4,\ -\alpha_3) \in V \cap \Psi(\Lambda_K(G))$, and Proposition 6.1 gives $v = (xP_1 + y\Omega,\ xP_3 + yP_2/\mu')$ with $x, y \in K$. The $\overline{\mathbb{Q}}$-independence of $1, P_2, P_3$ (W) gives $x = y = \alpha_3 = 0$, and then that of $1, P_1, \Omega$ gives $\alpha_1 = \alpha_2 = \alpha_4 = 0$. Dividing the four numbers by P_1 gives $P_2, P_3, 1, P_3/P_1$, which are $B(\frac27,\frac27), B(\frac37,\frac37), 1, B(\frac37,\frac37)/B(\frac17,\frac17)$ up to factors in $\mathbb{Q}(\zeta_7)^\times$. The Gamma form follows from $B(\frac17,\frac17) = \Gamma(\frac17)^2/\Gamma(\frac27)$ (the definition) and, by the reflection formula, $B(\frac27,\frac27) = \Gamma(\frac27)^2\Gamma(\frac37)\sin(\frac{3\pi}{7})/\pi$ and $B(\frac37,\frac37) = \Gamma(\frac37)^2\Gamma(\frac17)\sin(\frac{\pi}{7})/\pi$, after multiplying by π. □

No hidden symmetry. C is hyperelliptic: $(2x-1)^2 = 1-4y^7$, and $x \mapsto 1-x$ is the hyperelliptic involution, acting as $[-1]$ on A and multiplying z_1, z_2, z_3 by -1 together. $\mathrm{Aut}(C) \cong \mathbb{Z}/14$ and $\mathrm{End}^0(A) = \mathbb{Q}(\zeta_7)$ acts diagonally, $z_j([x]v) = \sigma_j(x) z_j(v)$; so no symmetry links z_1, z_2, z_3 of one point in a way that could produce P_3/P_1 in (a).

6.5. Remarks: linearisation, other relations, and e^{π^2}.

Remark 6.8 (Not linearisations)**.** [Heuristic] By the abelian analogue of Laurent's identity (internal note, refereed internally, not reproduced), each $\ell_j := \pi z_j(v)/P_j$ is a $\overline{\mathbb{Q}}$-combination of 1-periods; it linearises only products $P_{\bar\sigma} z_\sigma$ of a period with the logarithm of the conjugate slot on the same variety. Dividing (7) by $P_a P_c$ and using (6) gives

$$\ell_c - \beta\ell_a + \frac{P_d}{P_a}(\mu\ell_b - \beta\ell_d) + \pi\Big(\frac{q_2}{P_c} - \frac{\beta q_1}{P_a}\Big) = 0,$$

where the ratio P_d/P_a of periods of two different slots multiplies a combination of 1-periods. Unless $\mu\ell_b = \beta\ell_d$ this is not a $\overline{\mathbb{Q}}$-linear relation among 1-periods, so the analytic subgroup theorem does not apply as in §5.2. For (c): $P_3/P_1 \in \overline{\mathbb{Q}}^\times P_3 P_{\bar 1}/\pi$ has Koblitz–Ogus function $\mathbf{1}_{\{1,3,5\}} - \mathbf{1}_{\{1,2,3\}}$, which takes three values, while every slot period and π have functions with at most two values; under the Rohrlich–Lang (or period) conjecture, P_3/P_1 is not in the $\overline{\mathbb{Q}}$-span of 1-periods, and Wüstholz's theorem does not decide (c). Chudnovsky's theorem gives the algebraic independence of π and $\Omega_{-7} \sim \Gamma(\frac17)\Gamma(\frac27)/\Gamma(\frac37)$, which does not suffice, since (a)–(c) involve P_1, P_2, P_3. By the period conjecture (the Mumford–Tate group of A is a 4-dimensional torus and Ω adds no transcendence degree), Theorem 6.6 is a consequence of the period conjecture.

Remark 6.9 (Where first-kind relations occur)**.** *Elliptic slots.* Among CM elliptic slots (any number of non-isogenous curves) the relations of Deligne's theorem are only the Legendre relations [Proved]: the indicators are $(1\pm\chi_k)/2$ with χ_k odd quadratic characters, which are linearly independent. (That there are no other relations is the Rohrlich–Lang conjecture, [Heuristic].) So for products of CM elliptic curves, Theorems 1.6 and 5.6 (two Legendre relations on a line) are the only constructions of this kind.

Labels. [Proved] (elementary combinatorics). For CM types of $\mathbb{Q}(\zeta_N)$ containing 1, a first-kind partner of (Φ_a, Φ_b) exists iff Φ_a, Φ_b differ on at least two $\pm$-pairs: with $\Delta = \Phi_a \triangle \Phi_b$, the partners are $(\Phi_a \cap \Phi_b) \cup X$ and $(\Phi_a \cap \Phi_b) \cup (\Delta \setminus X)$, X one element from each $\pm$-pair in Δ, $X \notin \{\Phi_a \setminus \Phi_b, \Phi_b \setminus \Phi_a\}$. Hence first-kind relations among labels exist at every level N with $\varphi(N)/2 \geq 3$. The passage from labels to periods is Deligne's theorem [Lit] (§2.2 for Beta values; in general [10]).

Examples. The smallest levels are $N = 7$ and $N = 9$, with one relation each: at $N = 7$ the instance above; at $N = 9$ one with slots of a simple CM 3-fold with CM by $\mathbb{Q}(\zeta_9)$ and of E_ρ, $B(\frac19,\frac19)B(\frac19,\frac49) = 3^{1/3}\frac{\sin(2\pi/9)}{\sin(\pi/9)}B(\frac19,\frac29)B(\frac13,\frac13)$ (labels [Proved]; the constant [Numerical], 300 digits). An exact enumeration of label relations (four pairwise distinct first-kind labels) gives 100, 720, 12, 12, 12, 720, 12 relations at $N = 11, 13, 15, 16, 20, 21, 24$. At $N = 24$ there is one whose slots are two CM elliptic curves (CM by $\mathbb{Q}(\sqrt{-6})$ and $\mathbb{Q}(i)$) and two slots of a simple CM 4-fold: $B(\frac{1}{24},\frac{1}{24})B(\frac{1}{24},\frac14) \in \overline{\mathbb{Q}}^\times B(\frac{1}{24},\frac{1}{12})B(\frac{1}{24},\frac18)$ (labels checked exactly; the constant [Numerical], PSLQ).

Remark 6.10 (Several pairs)**.** [Proof sketch, not refereed] With all CM types of $\mathbb{Q}(\zeta_N)$, slope classes containing $p = 2^{g-2}$ parallel first-kind pairs occur ($g \geq 3$); the case check of Proposition 6.1 with p slots in each target has margin n for every p, and the line again carries no datum beyond the periods ("free room" 0), so nothing stronger in kind than Theorem 6.2 results.

Remark 6.11 (e^{π^2})**.** [Heuristic] The multiplicative group cannot enter a parallel pair: its only period is $2\pi i$, and a relation $p_a p_d = p_b p_c$ involving it would pair $2\pi i$ with 1, which is not a period (this is argued in §5.8). So we do not expect Theorem 6.2 to reach e^{π^2}.

7. Corollaries

7.1. **Corollaries MG1 and MG3.**

Corollary 7.1 (MG1)**.** *For every CM curve E, every period Ω and all $a, b, c \in \overline{\mathbb{Q}}^\times$, at least one of $e^{a\pi^2}$, $e^{b\pi\Omega}$, $e^{c\pi^2/\Omega}$ is transcendental. On E_i: one of e^{π^2}, $e^{\pi\varpi}$, $e^{\pi\eta(\varpi)}$ is transcendental.*

Status: [Proved modulo Wüstholz's theorem in the form $\varpi, \eta(\varpi) \notin \widetilde{\mathcal{L}}$; formally verified otherwise] for $\Omega \in \overline{\mathbb{Q}}^\times \varpi$ (every $a, b, c \in \overline{\mathbb{Q}}^\times$); [Proved modulo published theorems and Lemma A.1] for every other CM period.

Proof. Take $s = \Omega/\pi$ (up to $\overline{\mathbb{Q}}^\times$), $x_1 = \pi^2$, $x_2 = \pi^2/\Omega$: if all three exponentials are algebraic then $x_1, x_2 \in \overline{\mathbb{Q}}\mathcal{L}$ and $sx_1 = \pi\Omega$, $sx_2 = \pi$ lie in $\overline{\mathbb{Q}}\mathcal{L}$. The numbers $2\pi i, \pi^2, \pi^2/\Omega$ are $\overline{\mathbb{Q}}$-independent. A relation without the π^2/Ω term contradicts the transcendence of π. Otherwise $\pi/\Omega = a_1 + b_1\pi$ with $a_1, b_1 \in \overline{\mathbb{Q}}$. If $b_1 = 0$ then $\Omega/\pi \in \overline{\mathbb{Q}}$, which is false. If $b_1 \neq 0$ then $\pi\Omega = (\pi - a_1\Omega)/b_1$; since $\pi\Omega \in \overline{\mathbb{Q}}\mathcal{L}$ (as $e^{b\pi\Omega} \in \overline{\mathbb{Q}}$) and $\pi \in \overline{\mathbb{Q}}\mathcal{L}$, this gives $a_1\Omega \in \overline{\mathbb{Q}}\mathcal{L}$, so $a_1 = 0$ because $\Omega \notin \overline{\mathbb{Q}}\mathcal{L}$ (proof of Corollary 3.2); then $\Omega = 1/b_1 \in \overline{\mathbb{Q}}$, which is false. (Chudnovsky's algebraic independence of π and Ω [8] would give this at once, but is not needed.) On E_i, $\pi/\varpi = \eta(\varpi)$. □

For a general CM curve, $\eta(\Omega) = \kappa_1\Omega + \kappa_2\pi/\Omega$ with $\kappa_1, \kappa_2 \in \overline{\mathbb{Q}}$ and $\kappa_2 \neq 0$ (Legendre); $\kappa_1 = 0$ when $j(E) \in \{0, 1728\}$. Since $\pi\Omega, \pi\eta \in \overline{\mathbb{Q}}\mathcal{L}$ imply $\pi^2/\Omega = (\pi\eta - \kappa_1\pi\Omega)/\kappa_2 \in \overline{\mathbb{Q}}\mathcal{L}$, the statement "one of e^{π^2}, $e^{\pi\Omega}$, $e^{\pi\eta}$ is transcendental" holds for every CM period as well.

Corollary 7.2 (MG3)**.** *If e^{π^2} and $e^{c\pi\Omega}$ ($c \in \overline{\mathbb{Q}}^\times$) are algebraic, then $\alpha^{\Omega/\pi} \notin \overline{\mathbb{Q}}$, for every determination, for every $\alpha \in \overline{\mathbb{Q}}^\times$ such that $2\pi i, \pi^2, \log\alpha$ are linearly independent over $\mathbb{Q}$.*

Status: [Proved modulo Baker's theorem; formally verified otherwise] for $\Omega = \varpi$ (the exponent ϖ/π); [Proved modulo published theorems and Lemma A.1] for every other CM period, including the other elements of $\overline{\mathbb{Q}}^\times\varpi$ (the constant cannot be absorbed into $\log\alpha$).

Proof. With $s = \Omega/\pi$, take $x_1 = \pi^2$ (then $sx_1 = \pi\Omega \in \overline{\mathbb{Q}}\mathcal{L}$) and $x_2 = \log\alpha$ (then $sx_2 \in \mathcal{L}$ if $\alpha^{\Omega/\pi} \in \overline{\mathbb{Q}}$). These are $\mathbb{Q}$-independent logarithms together with $2\pi i$, hence K-independent, and Theorem 3.1 gives a contradiction. The hypothesis is needed: for $c = 1$ and $\alpha = e^{\pi^2}$ ($\log\alpha = \pi^2$) one has $\alpha^{\Omega/\pi} = e^{\pi\Omega} \in \overline{\mathbb{Q}}$. (Corollary 7.2 is also the case $\beta = e^{\pi^2}$ of the second sentence of Corollary 3.2.) □

7.2. **Corollary A2.**

Corollary 7.3. *Let $P = P_\sigma$, $\sigma \in \Phi$, be a first-kind eigen-period and $a, b, c \in \overline{\mathbb{Q}}^\times$. At least one of $e^{a\pi^2}$, $e^{b\pi P}$, $e^{c\pi^2/P}$ is transcendental. Since $\pi^2/P \in \overline{\mathbb{Q}}^\times \pi P_{\bar\sigma}$, this is "one of e^{π^2}, $e^{\pi P_\sigma}$, $e^{\pi P_{\bar\sigma}}$", up to algebraic factors in the exponents.*

Status: [Proved modulo published theorems and Lemma A.1].

Proof. Suppose all three are algebraic. Take $s = P/\pi$ (Theorem 3.7 with its constant equal to 1), $x_1 = \pi^2$, $x_2 = \pi^2/P$: then $x_1, x_2 \in \overline{\mathbb{Q}}\mathcal{L}$, $sx_1 = \pi P \in \overline{\mathbb{Q}}\mathcal{L}$, $sx_2 = \pi \in \overline{\mathbb{Q}}\mathcal{L}$. It remains to show that $2\pi i, \pi^2, \pi^2/P$ are $\overline{\mathbb{Q}}$-independent. A relation without the π^2/P term contradicts the transcendence of π. Otherwise $\pi/P = a_1 + b_1\pi$ with $a_1, b_1 \in \overline{\mathbb{Q}}$. If $b_1 = 0$ then $P/\pi \in \overline{\mathbb{Q}}$, contradicting (W). If $b_1 \neq 0$ then $P(a' + \pi) = c''\pi$ with $a', c'' \in \overline{\mathbb{Q}}$, so $\pi P = c''\pi - a'P$. Since

$\pi P \in \overline{\mathbb{Q}}\mathcal{L}$ and $\pi \in \overline{\mathbb{Q}}\mathcal{L}$, this gives $a'P \in \overline{\mathbb{Q}}\mathcal{L}$, so $a' = 0$ by (W), and then $P = c'' \in \overline{\mathbb{Q}}$, again contradicting (W). □

Remark 7.4. In Corollary 7.1 (one of e^{π^2}, $e^{\pi\Omega}$, $e^{\pi^2/\Omega}$ for a CM elliptic period Ω), the K-independence of $2\pi i, \pi^2, \pi^2/\Omega$ follows at once from Chudnovsky's algebraic independence of π and Ω [8], which is not available for abelian periods. The argument of Corollary 7.3, with $g = 1$, replaces it by the analytic subgroup theorem under the hypothesis $e^{\pi\Omega} \in \overline{\mathbb{Q}}$; this is the proof given for Corollary 7.1 above.

7.3. **Two degree-two period statements.** Propositions 5.1 and 5.5 also apply to data that are periods plus algebraic constants; their single-point forms suffice. This gives statements about products of periods of the two curves.

Corollary 7.5. (i) *The numbers* $\pi,\ \varpi,\ \Omega_\rho,\ \varpi\Omega_\rho$ *are linearly independent over* $\overline{\mathbb{Q}}$*; equivalently, so are* $\pi, \eta_i, \eta_\rho, \eta_i\eta_\rho$.

(ii) *The numbers* $1,\ \varpi,\ \Omega_\rho,\ \varpi\Omega_\rho/\pi$ *are linearly independent over* $\overline{\mathbb{Q}}$*; equivalently, so are* $\pi, \varpi, \eta_\rho, \varpi\eta_\rho$.

Status: (i) and (ii): [Proved modulo Wüstholz's theorem in the form that $1, \pi, \varpi, \eta_i, \Omega_\rho, \eta_\rho$ are $\overline{\mathbb{Q}}$-linearly independent, and the algebraicity of $g_3(\Omega_\rho\mathbb{Z}[\rho])$; formally verified otherwise]. Part (i) is proved from Proposition 5.5, part (ii) from Proposition 5.1, in the paper and in Lean alike; neither uses Theorem 5.6.

Proof. The equivalences follow by multiplying a relation by $\eta_i\eta_\rho/\pi$, resp. by η_ρ, and using $\varpi\eta_i = \pi$, $\Omega_\rho\eta_\rho = 2\pi/\sqrt{3}$. We use that $1, \pi, \varpi, \eta_i, \Omega_\rho, \eta_\rho$ are linearly independent over $\overline{\mathbb{Q}}$ (analytic subgroup theorem [45] on $E_i^\natural \times E_\rho^\natural \times \mathbb{G}_{\mathrm{m}} \times \mathbb{G}_{\mathrm{a}}$).

(i) A relation with zero coefficient on $\varpi\Omega_\rho$ is excluded by that independence. A relation $c_1\varpi + c_2\Omega_\rho + c_3\varpi\Omega_\rho = 0$ with zero coefficient on π is excluded too: multiplying it by $\pi/(\varpi\Omega_\rho)$ and using $\pi/\varpi = \eta_i$, $\pi/\Omega_\rho = \frac{\sqrt{3}}{2}\eta_\rho$ gives $c_3\pi + c_2\eta_i + \frac{\sqrt{3}}{2}c_1\eta_\rho = 0$, so $c_1 = c_2 = c_3 = 0$. So suppose $\pi = A\varpi\Omega_\rho + B\varpi + C\Omega_\rho$ with $A \neq 0$. In the set-up of §5.6 take $\beta = A$, $X = (\varpi, \eta_i) \in \Lambda(E_i^\natural)$, $Y = 0$, $a = C/A$, $b = -B$. The point $v = \Psi'(X, Y, a, b) = (\varpi + C/A,\ \eta_i - B)$ of $\Psi'(\Lambda(G))$ lies on $V' = \mathbb{C}(\varpi, A\Omega_\rho)$: $(\varpi + C/A)\,A\Omega_\rho = \pi - B\varpi = (\eta_i - B)\varpi$. By Proposition 5.5, $v = x(\varpi, A\Omega_\rho) + x'(\eta_\rho, \frac{2A}{\sqrt{3}}\eta_i)$ with $x, x' \in \overline{\mathbb{Q}}$. The first coordinate, $\varpi + C/A = x\varpi + x'\eta_\rho$, and the independence of $1, \varpi, \eta_\rho$ give $C = 0$, $x = 1$, $x' = 0$. Then $\eta_i - B = A\Omega_\rho$, which contradicts the independence of $1, \eta_i, \Omega_\rho$.

(ii) Suppose $\varpi\Omega_\rho/\pi = a + b\varpi + c\Omega_\rho$ (a relation with zero left-hand coefficient is excluded by the independence above). If $c = 0$, dividing by ϖ and using $\pi/\varpi = \eta_i$, $\Omega_\rho = 2\pi/(\sqrt{3}\eta_\rho)$ gives $\Omega_\rho = a\eta_i + b\pi$. If $b = 0$, multiplying by η_ρ gives $\frac{2}{\sqrt{3}}\varpi = a\eta_\rho + \frac{2c}{\sqrt{3}}\pi$. Both contradict the independence above, so $b, c \neq 0$. Put $t = -1/c$, $\beta = -\sqrt{3}\,b/(2c)$, $a_1 = a/b$, and take $X = (\varpi, \eta_i + t) \in \Lambda(E_i^\natural)$, $Y = 0$. Then $v = (\varpi + a_1, \eta_i + t) \in \Psi(\Lambda(G))$ lies on $V = \mathbb{C}(\varpi, \beta\eta_\rho)$: expanding $(\varpi + a_1)\beta\eta_\rho = (\eta_i + t)\varpi$ with $\varpi\eta_i = \pi$ and $\eta_\rho = 2\pi/(\sqrt{3}\Omega_\rho)$ gives exactly the assumed relation. Proposition 5.1 gives $v = x(\varpi, \beta\eta_\rho) + x'(\Omega_\rho, \frac{2\beta}{\sqrt{3}}\eta_i)$ with $x, x' \in \overline{\mathbb{Q}}$. The first coordinate and the independence of $1, \varpi, \Omega_\rho$ give $a_1 = 0$, $x = 1$, $x' = 0$; then $\eta_i + t = \beta\eta_\rho$, contradicting the independence of $1, \eta_i, \eta_\rho$. □

[Heuristic] Both statements are genuinely of degree two: rewriting with the Legendre relations always leaves a product of periods of the two different curves, so under the period conjecture they are not consequences of the analytic subgroup theorem. They are predicted by the period conjecture (by Chowla–Selberg [15] they concern π, $\Gamma(1/4)$, $\Gamma(1/3)$, whose algebraic independence is open), and they do not follow from known algebraic independence results. They have

the shape of a nonvanishing 2×2 determinant of periods, like the theorem of Barré-Sirieix, Diaz, Gramain and Philibert [3] (see also [39]). We found them nowhere in print; they are among the claims flagged for an expert check in the introduction and in §9.2. Neither statement contains the constant 1 together with $\varpi\Omega_\rho$, so they do not give $\varpi\Omega_\rho \notin \overline{\mathbb{Q}}$, a case of a conjecture of Waldschmidt that we do not claim (§8.2).

7.4. **The affine version and strong four exponentials.** Adding one factor $\mathbb{G}_{\mathrm{a}}$ to each coordinate of Ψ (so $G = \mathbb{G}_{\mathrm{a}}^2 \times \mathbb{G}_{\mathrm{m}}^2 \times E$ with coordinates (a, a', x, y, z) and $\Psi(a, a', x, y, z) = (x + a,\ y + z + a')$) gives the following statement. In the quotient $G'' = \mathbb{G}_{\mathrm{a}}^2 \times \mathbb{G}_{\mathrm{m}}^{2n} \times A''$ one has $d_0 = 2$, which does not enter the left-hand side of Theorem 2.1. The only new subgroups are $\mathbb{G}_{\mathrm{a}}$-lines $\{(\mu a, a)\}$ with $\mu \notin K$, which are excluded with margin n. So the count is the same as for Theorem 3.1; Lemma A.1 applies since $5n + 2 \geq 3n + 4$. This was checked at proof-sketch level by both internal referees of Theorem 3.1 and by a third; the subgroup list is not written out here. The base change in K now uses Baker's inhomogeneous theorem.

Theorem 7.6 (Affine MG; proof sketch)**.** *With E, Ω, c, $s = c\Omega/\pi$ and K as in Theorem 3.1,*

$$\dim_K\{x \in K + K\mathcal{L} :\ s\,x \in K + K\mathcal{L} + K\mathcal{L}_E\} \leq 2.$$

Corollary 7.7 (proof sketch)**.**

(i) *For $a, b, c \in \overline{\mathbb{Q}}^\times$, one of $e^{a\pi^2}$, $e^{b\pi\Omega}$, $e^{c\Omega/\pi}$ is transcendental.*

(ii) *For $\alpha \in \overline{\mathbb{Q}}^\times$ not a root of unity, $e^{\Omega/\pi}$ and $\alpha^{\Omega/\pi}$ (any determination) are not both algebraic.*

(iii) *(Strong four exponentials at ratio $t \in \overline{\mathbb{Q}}^\times\Omega/\pi$.) If y_1, y_2 are linearly independent over $\overline{\mathbb{Q}}$, then y_1, y_2, ty_1, ty_2 are not all in $\widetilde{\mathcal{L}} = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}$. By $y \mapsto ty$ the same holds at $t \in \overline{\mathbb{Q}}^\times\pi/\Omega$. (The strong statement is invariant under $t \mapsto ct$, $c \in \overline{\mathbb{Q}}^\times$, and $\overline{\mathbb{Q}}^\times\Omega/\pi$ contains a real number. So this is a statement at real t, where the $\widetilde{\mathcal{L}}$-version of the conjugation argument* [40, Thm. 4.1] *applies when $y_1, y_2, \bar{y}_2$ are $\overline{\mathbb{Q}}$-independent. The new content is the remaining y.)*

Sketch. (i) Take $x_1 = 1$ ($s \cdot 1 \in \overline{\mathbb{Q}}\mathcal{L}$ if $e^{c\Omega/\pi}$ is algebraic, with s adjusted by a constant) and $x_2 = \pi^2$. With $2\pi i$ these are K-independent, because $1, \pi, \pi^2$ are $\overline{\mathbb{Q}}$-independent (Lindemann). (ii) Take $x_1 = 1$ and $x_2 = \log\alpha$; then $1, 2\pi i, \log\alpha$ are K-independent by Baker's theorem. (iii) Take K containing all the coefficients. Then $y_1, y_2 \in T_K := \{x \in K+K\mathcal{L} :\ sx \in K+K\mathcal{L}+K\mathcal{L}_E\}$, which has dimension ≤ 2 and contains $2\pi i$. Since y_1, y_2 are K-independent, $2\pi i \in Ky_1 + Ky_2$. Then $2ic\Omega = s \cdot 2\pi i \in \widetilde{\mathcal{L}}$, which contradicts the analytic subgroup theorem ($\Omega \notin \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}$). □

The subgroup list for the count of Theorem 7.6 is written out in §4 in the closely related form of Theorem 4.2(a) (with $\overline{\mathbb{Q}}$-independent data and K chosen to contain all coefficients, so that no base change in K is needed). Accordingly Corollary 7.7(i) is contained in the first triple of Corollary 4.5(ii) ($\pi^2, \pi\Omega, \Omega/\pi$ are not all in $\widetilde{\mathcal{L}}$), (ii) follows from Theorem 1.5 with $(y_1, y_2) = (1, \log\alpha)$ at $t = \Omega/\pi$, and (iii) is Theorem 1.5 at $t \in \overline{\mathbb{Q}}^\times(\Omega/\pi)^{\pm 1}$; these are proved in §4, with the labels stated there. The K-dimension form of Theorem 7.6 itself remains a proof sketch.

8. Overlap with known results, and novelty

This section compares our results with the literature. Theorem, corollary, conjecture, slide and page numbers cited below from [6, 35, 38, 39, 40] have not been rechecked against the printed texts (§9.2); the arguments use only the statements as quoted.

8.1. **Complex conjugation.** Waldschmidt's [40, Cor. 2.7] (after Diaz [11]) says: if $x_1, x_2 \in \mathbb{R} \cup i\mathbb{R}$ are linearly independent over $\mathbb{Q}$ and $y_1, y_2, \bar{y}_2$ are linearly independent over $\mathbb{Q}$, then one of the four numbers $e^{x_i y_j}$ is transcendental. Its $\widetilde{\mathcal{L}}$-version [40, Thm. 4.1], a consequence of Roy's strong six exponentials theorem and complex conjugation, says: if $x_1, x_2 \in \mathbb{R} \cup i\mathbb{R}$ are $\overline{\mathbb{Q}}$-linearly independent and $y_1, y_2, \bar{y}_2$ are $\overline{\mathbb{Q}}$-linearly independent, then one of the numbers $x_i y_j$ is not in $\widetilde{\mathcal{L}}$.

Every slope class contains a real number. For a CM elliptic period, $\bar{E}$ has CM by the same field, so it is isogenous to E over $\overline{\mathbb{Q}}$ and $\bar{\Omega} \in \overline{\mathbb{Q}}^\times \Omega$; and π is real. For an eigen-period of a CM abelian variety A, let A^c be the complex conjugate of A, with the conjugate F-action; it has CM type $\bar{\Phi}$. Composing that action with complex conjugation of F gives type Φ, so A^c with the twisted action is F-isogenous to A (two CM abelian varieties of the same type (F, Φ) are isogenous; a standard fact, see [28]). The conjugate $\bar{\omega}_\sigma$ of ω_σ is a σ-eigen class for the twisted action. Its pull-back along such an isogeny is a nonzero σ-eigen class of A, so it lies in $\overline{\mathbb{Q}}^\times \omega_\sigma$, and comparing periods gives $\bar{P}_\sigma \in \overline{\mathbb{Q}}^\times P_\sigma$. Hence $e^{2i \arg P_\sigma} \in \overline{\mathbb{Q}}$, and the same holds for non-simple A by the reduction in §2.2. So every t in Theorems 1.1, 1.2 and 1.5 is γr with $\gamma \in \overline{\mathbb{Q}}^\times$ and $r \in \mathbb{R}$.

Four exponentials. If $\alpha^t, \beta^t \in \overline{\mathbb{Q}}$, then $y = (\log\alpha, \log\beta)$ satisfies $y_j,\ ry_j = \gamma^{-1} t y_j \in \widetilde{\mathcal{L}}$. The $\widetilde{\mathcal{L}}$-version [40, Thm. 4.1] at the real point r then excludes every pair with $\log\alpha, \log\beta, \overline{\log\beta}$ linearly independent (over $\overline{\mathbb{Q}}$, which for logarithms is the same as over $\mathbb{Q}$ by Baker's theorem), and, with the roles of α and β exchanged, every pair with $\log\beta, \log\alpha, \overline{\log\alpha}$ linearly independent, whatever the argument of t. For the remaining pairs both $\overline{\log\alpha}$ and $\overline{\log\beta}$ lie in $\mathbb{Q}\log\alpha + \mathbb{Q}\log\beta$, so they have a conjugation-stable $\mathbb{Q}$-span of logarithms, which has a basis in $\mathbb{R} \cup i\mathbb{R}$ (for $(\alpha, \bar{\alpha})$, for instance, $\log|\alpha|^2$ and $\log(\alpha/\bar{\alpha})$). For them, [40, Cor. 2.7] with the roles exchanged (x that basis, $y = (1, t, \bar{t})$) applies as soon as $t \notin \mathbb{R} \cup i\mathbb{R}$: then $1, t, \bar{t}$ are $\mathbb{Q}$-independent, because t is transcendental and $\bar{\gamma}/\gamma \notin \mathbb{Q}$ (e.g. $t = e^{i\pi/5}\varpi/\pi$). So Theorems 1.1 and 1.2 (and the four exponentials statements of Corollaries 3.2 and 3.5) are already known at every $t \notin \mathbb{R} \cup i\mathbb{R}$, and at every t for pairs whose logarithms span a space that is not conjugation-stable (e.g. $(2, 1+2i)$). What remains new is real or purely imaginary t — as for ϖ, ϖ/π, Ω_ρ, and the Beta values $B(a/N, b/N)^{\pm 1}$ and $(B(a/N, b/N)/\pi)^{\pm 1}$ themselves — with a conjugation-stable span of logarithms, such as $(2, 3)$ or $(5, (1+2i)/(1-2i))$; Corollaries 7.1 and 7.3; and the dimension statements themselves.

Strong four exponentials. The strong statement at t depends only on the class $\overline{\mathbb{Q}}^\times t$. Hence Theorem 1.5 is *already known whenever $\overline{\mathbb{Q}}y_1 + \overline{\mathbb{Q}}y_2$ is not stable under complex conjugation*, whatever the argument of t. What is new is the case of a conjugation-stable span (equivalently, a span with a real basis), e.g. $(\log 2, \log 3)$, $(1 + \log 2, \log 3)$, $(1, \log 2)$, $(1, \pi^2)$: this contains all our examples. (Unlike for four exponentials, the second orientation does not help here: since $\bar{t} \in \overline{\mathbb{Q}}^\times t$, the numbers $1, t, \bar{t}$ are never $\overline{\mathbb{Q}}$-independent.) The same applies to Corollary 7.7(iii): $\overline{\mathbb{Q}}^\times \Omega/\pi$ contains a real number, so it is a statement at real t, where [40, Thm. 4.1] applies when $y_1, y_2, \bar{y}_2$ are $\overline{\mathbb{Q}}$-independent. The mixed forms SMG, SMG$^\natural$ (second row in $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathcal{L}_E$, resp. $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathrm{Fib}$) are not reached by conjugation, since no *mixed* analogue of the strong six exponentials theorem (with rows in $\widetilde{\mathcal{L}}$ and $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathcal{L}_E$, resp. $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathrm{Fib}$) is in print; the purely elliptic strong six exponentials theorem on $\mathbb{G}_\mathrm{a} \times E$ is [26, Cor. 1].

8.2. **Waldschmidt's "Variations" papers and slides, and Roy.** The closest published theorem to Theorem 1.1 is Waldschmidt's [38, Thm. 4.2] ("Variations on the six exponentials theorem", Hyderabad 2003), a mixed statement on $\mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m} \times E$ without CM whose second row consists of elliptic logarithms; it needs seven columns (one of six numbers $\ell_1 u_j - \ell_j u_1$ is transcendental) in general, and five columns (one of four) in its period-column case. CM and the mixing map bring the count down to two data with ordinary logarithms. The nearest published

statement to Theorem 1.6 we know is [38, Thm. 4.3]: a bilinear statement for two elliptic curves (non-isogenous, or equal with k-independent first column), but for 2×9 matrices of elliptic logarithms, without quasi-periods and without a 2×2 case. The case $W = 0$ of Theorem 2.5 is [38, Thm. 5.1], and the strong four exponentials conjecture is [38, Conj. 1.5]. Waldschmidt's "Further variations" [36] concerns 2×2 minors with entries in $\widetilde{\mathcal{L}}$ only (no elliptic functions); Corollary 4.5(iii) is a CM analogue of its Cor. 1.3, and its general 2×3 minors statement [36, Thm. 1.2] at CM data is out of reach of our count (Remark 4.6(4)). In lecture slides [35] (we cite them from our notes and have not rechecked the slide), Waldschmidt conjectures that $\omega_1\omega_2^* - \omega_1^*\omega_2$ is transcendental for non-isogenous curves. For E_i, E_ρ this would give $\varpi\Omega_\rho \notin \overline{\mathbb{Q}}$, since $\varpi \cdot \rho\Omega_\rho - \Omega_\rho \cdot i\varpi = (\rho - i)\varpi\Omega_\rho$; neither statement of Corollary 7.5 gives it (§7.3), and we do not claim it.

The six exponentials theorem (through [40, Cor. 2.7]) and Roy's strong six exponentials theorem [24, Cor. 2] (through [40, Thm. 4.1]) are the sources of the conjugation overlap above; we do not know Theorem 1.5 to follow from it (Remark 4.7). Theorem 1.2 is not a restatement of the Roy–Waldschmidt theorem: [26, Théorème (iii)] as stated uses $\alpha_K(A) = g$ for a CM abelian variety and gives no conclusion for $g \geq 2$; the inequality inside its proof (step (b), with the quotient G^n/H) gives Theorem 3.7 at once (Remark 3.8). We present Theorems 3.1, 3.4, 1.6, 3.7, 3.9, 4.2 and 6.2 as applications of a published theorem to explicit groups, not as a new method.

8.3. **Theorem M and the two-curve results.** With $\ell \in \mathcal{L}$, Theorem 1.9 is the CM case, with one row of Waldschmidt's matrix a period of $E \times \mathbb{G}_{\mathrm{m}}$, of his Conjecture 42 in [39] (four exponentials for $E \times \mathbb{G}_{\mathrm{m}}$). Its special case $\ell \in \overline{\mathbb{Q}} + \overline{\mathbb{Q}} \cdot \pi i$, $u \in \mathcal{L}_E$, follows in one line from Laurent's [19, Cor. 3, p. 146]: for any CM lattice, a period $\omega \neq 0$ and $u \notin \mathbb{Q}\Lambda_E$ with $\wp(u) \in \overline{\mathbb{Q}}$, the numbers $1, \omega, \eta(\omega), 2\pi i, \omega H(u), 2\pi i u/\omega$ are $\overline{\mathbb{Q}}$-linearly independent, where $H = \zeta - \gamma z$ and $\gamma = 0$ when $g_2 g_3 = 0$. (M1) is one line from it ($2\pi i u/\Omega = 2ic$), and so is (Mη) modulo $\overline{\mathbb{Q}} + \overline{\mathbb{Q}}\varpi + \overline{\mathbb{Q}}\eta(\varpi) + \overline{\mathbb{Q}}\pi i$. The remaining items of Corollary 1.10 (which only use $u \in \mathcal{L}_E$) follow quickly from Laurent's identity and the analytic subgroup theorem (§5.2); we found none of them stated in print (Abe [1] claimed (M1); the preprint was withdrawn). The theorem of Masser that $1, \omega, \eta, 2\pi i, \log \sigma$ are $\overline{\mathbb{Q}}$-linearly independent on a CM curve, for $\sigma \in \overline{\mathbb{Q}}^\times$ not a root of unity (we cite it as [21], an attribution we have not rechecked), is linear and gives none of these; nor do the algebraic independence results of Brownawell–Kubota [7], which have six-exponentials strength, or the third-kind results surveyed in [41] (Wüstholz's linear independence of $1, \omega, \eta, \lambda(u_i, \omega)$).

Theorem 1.1, Corollary 1.3, Theorem 1.6 and Corollary 1.7 we found nowhere in the literature (apart from what follows from [40] as shown in §8.1); the four exponentials problem is listed as open in [6, p. 27] and [42], and [42], which we read in full, mentions no case at t of these kinds. Laurent [19] never mixes two curves; Corollary 1.7 is the cross-curve analogue of his Cor. 3, with $\eta(\Omega_\rho)$ in place of $\eta(\varpi)$. For Corollary 1.8 see the remark after it (§1); for Corollary 7.5, the remark after its proof (§7.3). Algebraic independence of $\pi, \Gamma(1/4), \Gamma(1/3)$ is open, so neither Corollary 1.7 nor Corollary 7.5 follows from known algebraic independence results.

8.4. **CM abelian varieties and first-kind relations.** Apart from what follows from [40] as shown in §8.1 (all pairs at every $t \notin \mathbb{R} \cup i\mathbb{R}$), we found no t of the kind of Theorem 1.2 (an abelian eigen-period, a Beta value, or such a value divided by π, up to $\overline{\mathbb{Q}}^\times$) at which the four exponentials conjecture was stated in the literature for all pairs, and no real t of this kind at which it was known for a single pair with conjugation-stable span of logarithms, such as $(2, 3)$. We looked in Waldschmidt's survey [42] (read in full), in [38, 36, 40, 31, 37] and elsewhere. Nor did we find any use of a first-kind (Deligne–Koblitz–Ogus) relation as the period term of a subgroup theorem.

Corollary 1.11(b) lies in territory that Gao–Ullmo–Yafaev [14] and Gao–Ullmo [13] describe as open: they write that almost nothing is known about algebraic relations among holomorphic CM periods beyond the linear ones (apart from cases reducible to the linear case, such as $\theta\theta'/\pi \notin \overline{\mathbb{Q}}$ [14, Cor. 3.1]), and ask whether all quadratic relations are elementary [14, Question 1.3]; the relation used in Theorem 6.2 is an elementary quadratic relation in their sense. Corollary 1.11(b) is therefore, with Corollary 7.5, among the claims flagged for an expert check in the introduction and in §9.2. Wolfart–Wüstholz [44] determine the $\overline{\mathbb{Q}}$-linear relations among Beta values; Corollary 1.11(b) involves the quotient $B(\frac{3}{7},\frac{3}{7})/B(\frac{1}{7},\frac{1}{7})$, which is not known to be a Beta value up to $\overline{\mathbb{Q}}^{\times}$, so it is not covered (its Koblitz–Ogus function takes three values, so under the Rohrlich–Lang conjecture it is not one; §6.5). Tosi [29] proves algebraic independence results of Gel'fond–Chudnovsky type for abelian periods and their exponentials; they do not give Theorem 1.2 or Corollary 1.4. Grinspan and Vasil'ev (two of $\pi, \Gamma(1/5), \Gamma(2/5)$ are algebraically independent, see [37, §3]) do not imply Theorem 1.2: they concern π and Gamma values only and contain no statement about α^t for algebraic α.

8.5. **Other related work.** The recent preprint [9] of G. V. and D. V. Chudnovsky, with accompanying Lean files, concerns, according to its title, irrationality exponents of real CM elliptic periods and of powers of the lemniscate period ϖ. We have not checked that paper or its formalisation. Irrationality exponents are not four exponentials statements. Sources we could not consult are listed in §9.2.

9. Consistency checks, status, and formal verification

9.1. **Consistency checks.** The main results are consequences of a published theorem (completed by Lemma A.1 in the case used) applied to explicit groups, so an error would most likely sit in the bookkeeping. We checked that the same bookkeeping does not prove false statements, and that it gives no conclusion in the control configurations below, where it should not.

- *Rational lines.* If V contains a $\overline{\mathbb{Q}}$-rational point, the statement is false (e.g. $\mathbb{G}_{\mathrm{m}}$ in place of E, $s = 2$, $x = \log 2$). In each configuration we checked (MG, MG$^\natural$, MX) the count finds an obstruction, typically a subtorus or a $\mathbb{G}_{\mathrm{a}}$-line; for MG and MG$^\natural$ it occurs at equality in the inequality of Theorem 2.1.
- *No CM, wrong field, one Legendre relation, wrong coordinates.* An obstruction appears (the trivial subgroup, or a type-D subgroup) in each of these cases: a non-CM curve; $k \not\subset K$; only one Legendre relation on an MX-type line; for MX, $\mathrm{diag}(a, a)$ in place of $\mathrm{diag}(a, \bar{a})$ on $T(E^\natural)$; for MG$^\natural$, a non-eigen fibre coordinate. So no conclusion is claimed in these cases. (For MG$^\natural$ the conjugation itself is not load-bearing; what matters is that Ψ reads an eigen-coordinate.)
- *Pure periods.* Data consisting of periods alone are true, so they test the period term by itself. In all the configurations we built (semi-abelian G with V a hyperplane containing a corank-one sublattice of periods, $\mathbb{G}_{\mathrm{m}}^{d_1} \times E^g$, the groups of §§3.1, 5.4), an obstruction exists.
- *Sharpness of the period term.* For the graph of complex multiplication in $E \times E$ ($\kappa = 2$), the diagonal in $\mathbb{G}_{\mathrm{m}}^2$ ($\kappa = 1$), and the Weil-restricted graph of complex multiplication with maximal $\kappa = n$, the obstruction occurs exactly at equality. A period term larger by one would prove false statements there. The applications in §§3.1, 5.4 use $\kappa = \mathrm{rk}(V'' \cap \ker \exp)$ exactly. This is calibration evidence, not a proof of [33, Prop. 6.1].
- $\mathbb{G}_{\mathrm{m}}$ *alone* [Proved]. For $G = \mathbb{G}_{\mathrm{m}}^r$ the period space $\Psi(\Omega_K)$ is $2\pi i$ times a K-rational subspace. So a line $V = \mathbb{C}(1, s)$ meets it only if $s \in K$, where the statement is false and

the method is silent. With a logarithm in place of the elliptic period ($\kappa = 0$), the same count reproduces the six exponentials theorem (a calibration).

- *Special exponents.* Values of t that make one power algebraic ($2^t = 3$, $2^t = 1$) are not known to lie outside $\overline{\mathbb{Q}}^\times \Omega/\pi$. They lie inside only under degree-two relations such as $\pi \log 3 \in \overline{\mathbb{Q}}\,\Omega \log 2$ or $\Omega \log 2 \in \overline{\mathbb{Q}}\pi^2$, which are conjecturally false. If they did lie inside, the resulting consequences of Corollary 3.2 would be conjecturally true and would contradict no known result.
- *The results of §§3.3, 4, 6.* All case checks were rerun by exact integer enumeration, each by at least two scripts that share no code; all controls below obstruct, as they must.
- *Theorems MG-A, MG$^\natural$-A.* Exact tables for $g = 1, \dots, 6$ and all n divisible by $2g$ up to 60: margins n and $n-2$, the latter obstructing only at $n = 2$. One referee built U, A'', Q'' concretely as subspaces ($g = 2$: $K = \mathbb{Q}(\zeta_{20})$, $\mathbb{Q}(\zeta_{40})$; $g = 3$: $K = \mathbb{Q}(\zeta_{28})$). Controls: κ deleted; one datum; a non-eigen first-kind coordinate $z_{\phi_1} + \mu z_{\phi_2}$ (abelian part of dimension n); a non-CM A; a fictitious rational V ($P/\pi \in \overline{\mathbb{Q}}$); $\mathbb{G}_{\mathrm{m}}$ in place of A; for MG$^\natural$-A, κ deleted, a non-eigen fibre coordinate, and a rational V.
- *Theorem X.* Minimum excess n for $n \le 120$. Controls: no relation or wrong pairing ($\kappa = n$); dropping slot d (equality obstruction: the fourth slot is load-bearing); a non-CM slot, with or without $\kappa = 2$ imposed; κ deleted with three data; λ forced to 0 (Lemma 6.4 deleted: equality obstructions, so the lemma is load-bearing); a slot-weight sweep (equality obstruction at weight $5n/8$ per slot, so the weight $n/2$ of Lemma 2.2, which needs $K \supseteq$ the Galois closures, is load-bearing). The same code reproduces a known linearisable slot-reading cell with no obstruction, and obstructs its non-CM twin.
- *Theorem SMG, SMG$^\natural$.* No obstruction for $n = 2, \dots, 40$ (resp. $4, \dots, 40$), margins n and $n-2$. Controls: non-CM curve; κ deleted (so the CM period is load-bearing; for SMG the margin is exactly κ); one datum only; $s \in K$ (equality obstruction, as it must, since the strong statement is false at $t = 2$); SMG$^\natural$ at $n = 2$.
- *Falsity hunts.* [Numerical] For Theorem 1.5: 64 targets $t \cdot y$ ($\Omega \in \{\varpi, \Omega_\rho\}$, four slope classes, eight y's) against real $\widetilde{\mathcal{L}}$-bases at 210 digits, PSLQ and LLL: no relation. No corollary is implied by Nesterenko's theorem (which controls only $e^{\pi\sqrt{d}\,r}$) or by Chudnovsky's. For Corollary 6.7, all computations use the real numbers $B_j := B(\frac{j}{7}, \frac{j}{7})$ in place of P_j (they differ by the algebraic factors $1 - \zeta^j$; a $\mathbb{Q}(\zeta_7)$-linear relation among real numbers gives a $\mathbb{Q}(\mu)$-linear one). With $Q = (x_0, (x_0(1-x_0))^{1/7})$, $O = (0,0)$ and $x_0 = 1/3, 1/5, 2/7$ (220 digits), $(z_2/z_1)/(B_3/B_1)$ satisfies no integer polynomial of degree ≤ 12 with coefficients $\le 10^{10}$. At $x_0 = 1/2$ (a torsion point) $z_j = B_j/2$ by the symmetry $x \mapsto 1 - x$, and the pipeline returns $(z_2/z_1)/(B_2/B_1) = 1$ to working precision (with the P_j of §6.4 this ratio is $1/(1+\zeta)$). PSLQ on $\{1, B_2, B_3, B_3/B_1\} \otimes \{1, \mu, \mu^2\}$ finds no relation with coefficients $\le 10^{14}$, while the positive control $\{B_1 B_2, B_3 B(\frac{1}{7}, \frac{2}{7}), B_1, B_3\}$ finds (6) at once.
- *Period-conjecture plausibility.* [Heuristic] Under the period conjecture, $y, ty \in \widetilde{\mathcal{L}}$ already forces $y = 0$ for t in a CM slope class (every monomial on one side contains Ω), so Theorem 1.5 is strictly weaker than predicted; and Theorem 6.6 is a consequence of the period conjecture. A counterexample would have to refute the period conjecture.

9.2. **Dependencies and status.** The table lists the main results with their labels ((a), (b), (c) as in the conventions of §1; "(b): X" names the classical input) and what they depend on.

result	label	depends on
Thm. 1.1; Cor. 3.5; Cor. 3.2, first sentence	(a) at ϖ; (b): g_3 at Ω_ρ; (c) otherwise	Thms. 3.1, 3.4; Wüstholz or Remarks 3.3, 3.6
Cor. 3.2, second sentence	(c)	Thm. 3.1, Baker
Thms. 3.1, 3.4 (K-forms)	(a) at ϖ for the K stated there; (c) otherwise	Thm. 2.1, Lemma A.1, Lemma 2.3, (F1)–(F4) for $\mathrm{MG}^\natural$, Baker, transcendence of Ω and Ω/π
Thm. 3.7 (MG-A)	(c)	Thm. 2.1, Lemma A.1, (S1), (P), (W), Baker
Thm. 3.9 ($\mathrm{MG}^\natural$-A)	(c), given (F1)–(F4)	as above, plus (F1)–(F4)
Thm. 1.2 ($g \geq 2$), Cor. 7.3	(c)	Thms. 3.7, 3.9; (W); Beta dictionary (Lit)
Thm. 1.2 ($g = 1$)	as Thm. 1.1; (c) for $t \in \overline{\mathbb{Q}}^\times/\Omega \cup \overline{\mathbb{Q}}^\times\pi/\Omega$	Thm. 1.1, $t \leftrightarrow 1/t$
Thm. 4.2 (SMG, $\mathrm{SMG}^\natural$)	(a) at ϖ with rows in $\widetilde{\mathcal{L}}$; (b): g_3 at Ω_ρ; (c) otherwise	Thm. 2.1, Lemma A.1 (exponent $b > 1$, $d_0 \leq 2$), (P), Lemma 2.3
Thm. 1.5	(b): Wüstholz ($\Omega, \eta \notin \widetilde{\mathcal{L}}$) at ϖ, also g_3 at Ω_ρ; (c) otherwise	Thm. 4.2, Lemma 4.1
Cor. 4.5	(c)	Thms. 1.5, 4.2, Baker
Thm. 1.9	(a) at ϖ; (b): Masser for $s \in k$ at ϖ; (b): g_3 at Ω_ρ, (c) for $s \in k$ there; (c) otherwise	Thm. 2.4 (hyperplane case: Thm. 2.5); for $u \in \mathcal{L}_E$ also Laurent's identity and Wüstholz
Cor. 1.10	by item (status line)	Thm. 1.9; Gel'fond–Schneider (M2), Hermite–Lindemann (M3′), Masser (Mη)
Prop. 5.1 (single point), Thm. 1.6	(b): Wüstholz, Masser (for Thm. 1.6), g_3; Prop. 5.1 as stated (K-span): (c)	Thm. 2.1, Lemma A.1, Lemma 2.3, (F1)–(F4) for $g = 1$, Wüstholz, Masser
Cor. 1.7	(b): as Thm. 1.6, and Laurent (on E_i); (c) on E_ρ	Thm. 1.6, [18]
Cor. 1.8	(c)	Wüstholz
Prop. 5.5 (single point)	(b): Wüstholz ($\varpi\Omega_\rho/\pi$, $\Omega_\rho/\varpi \notin \overline{\mathbb{Q}}$), g_3; as stated (K-span): (c)	Thm. 2.5 as published ([33, §5d]), Wüstholz, Lemma 2.3, (F1)–(F4) for $g = 1$; in Lean: `lean_lsg`'s Thm. 2.1 for this group
Thm. 5.6, Cor. 5.7	Proof sketch (from Prop. 5.5)	Prop. 5.5, Masser, Wüstholz
Prop. 6.1	(c)	Thm. 2.1, Lemma A.1, (S1), (P), (Ch)
Thm. 6.2, Thm. 6.6	(c)	Prop. 6.1; Wüstholz, (P), (S1), (Ch)
Cor. 6.7	(c)	Thm. 6.6, Prop. 6.1, (W); Beta cocycle, reflection, Chowla–Selberg
Cor. 7.1	(b): Wüstholz ($\varpi, \eta(\varpi) \notin \widetilde{\mathcal{L}}$) at ϖ; (c) otherwise	Thm. 3.1, Wüstholz ($\Omega \notin \overline{\mathbb{Q}}\mathcal{L}$), Masser ($\Omega/\pi \notin \overline{\mathbb{Q}}$)
Cor. 7.2	(b): Baker for $\Omega = \varpi$; (c) otherwise	Thm. 3.1, Baker
Cor. 7.5(i)	(b): Wüstholz (six-independence), g_3	Prop. 5.5, Wüstholz

result	label	depends on
Cor. 7.5(ii)	(b): Wüstholz (six-independence), g_3	Prop. 5.1, Wüstholz
Thms. 7.6, 3.13, Cors. 3.14, 7.7, §5.7	Proof sketch	as indicated there
Rem. 6.10, Rem. 4.6(5)	Proof sketch	as indicated there
non-linearisability (Rems. 3.11, 6.8, §7.3); new instances (Rem. 3.10); §5.8	Heuristic	period conjecture, Rohrlich–Lang
(F1)–(F4)	Proved (given [22])	Hodge sequence, Poincaré reducibility, universality

In more detail, the main results rest on the following.

- Theorem 2.1 [33, Thm. 4.1]. Its proof [33, §§6–7] uses:
  - the auxiliary function with periods and the κ-saving [32, Prop. 2.1, 2.4];
  - multiplicities along W (Lemma A.1; a proof sketch, checked by two independent internal referees, not peer-reviewed; formalised for the groups listed at the end of Appendix A). Its published inputs are [32, Prop. 2.1, Lemma 2.2], Serre's embedding and height bound [27], Bertrand's [30, App. I, Lemme 7 and its Corollaire, with Lemme 1] (cited in [33, p. 396] as "Lemma 7"), and the lower bound of [32, p. 640];
  - Philippon's zero estimate [23];
  - a complete system of addition laws of one fixed bidegree for the projective embedding of G_2. This exists for any quasi-projective commutative group variety by an elementary argument (Remark A.2); for abelian varieties see also Lange–Ruppert [17]. (Serre's appendix [27] gives the compactification with its G-action, but not addition laws; cf. also [12], which we have not checked.)
- For Proposition 5.5 (and so for the sketch of Theorem 5.6) and the second derivations (Remarks 3.3, 3.6, 5.2, 6.5): Theorem 2.5, as published in [33, §5d].
- Legendre's relation; Wüstholz's analytic subgroup theorem [45]; Masser [20]; Baker; the transcendence of Ω and of Ω/π (Schneider, Masser [20]).
- Standard facts on CM abelian varieties and universal vector extensions: (S1), (F1)–(F4) (§2.2); the Beta dictionary, and the Beta cocycle and Chowla–Selberg [15] at level 7.

For $u \in \mathcal{L}_E$ (all of Corollary 1.10), Theorem 1.9 follows from Laurent's identity and the analytic subgroup theorem; for $u \in \overline{\mathbb{Q}}\mathcal{L}_E$ it rests on Statement R (in the hyperplane case, which is Theorem 2.5 in substance). Proposition 6.1 and Theorem 4.2(a) use $d_0 = 2$; Theorem 2.1 allows any d_0; [33] itself uses $d_0 = 2$ (Cor. 2.1) and $d_0 = n$ (Cor. 3.3, second proof); and Lemma A.1 is uniform in d_0 (by a two-sentence step that is part of trust point T), its hypotheses (2) and $d_2 \geq 1$ being checked in each application (§2.1).

What most needs an expert read, in order: Corollaries 6.7(c) and 7.5, which say more about $\Gamma(\frac{1}{7}), \Gamma(\frac{2}{7}), \Gamma(\frac{3}{7}), \pi$, resp. $\pi, \Gamma(1/4), \Gamma(1/3)$, than any known algebraic-independence result (the first lies where [14, 13] say little is known; Corollary 7.5(i) and (ii) rest on Propositions 5.5 and 5.1, which are formally verified in their single-point forms modulo named inputs, §9.3, so what needs the expert read is their mathematical content and the group models); the use of a first-kind relation as the period term (Proposition 6.1); Lemma A.1 (the one step of [33, Prop. 6.1] that is only indicated in print); and the subgroup classifications in §§3, 4, 5 and 6. The claims that MX, the degree-two statements and Corollary 1.11 are not linearisations, and the discussion in §5.8, are heuristic (they use the period conjecture).

Sources we could not consult, any of which might contain these statements: Bertrand [4] (the source of the K-structure used in [26]); Waldschmidt [34] (the general form of Theorem 2.1); Baker–Wüstholz [2, §6.2]; Bertrand–Laurent [5]; Faltings–Wüstholz [12]; Roy [25]; the full text of Waldschmidt's *Diophantine approximation on linear algebraic groups*, Ch. 11–13; Tosi's published version of [29]; Bertrand's papers on 1-motives; §§2–7 of [13]; MathSciNet/zbMATH citation lists of [26], [33], [44]. Bibliographic data not checked against the printed sources: the theorem, corollary, conjecture, slide and page numbers cited from [35, 38, 39, 40] and [6, p. 27] (and the date of [35]); the metadata of [10] and [43]; the page numbers of [16]; [28] as the precise source of (S1) and of the isogeny criterion used in §8.1; and the attribution to [21] in §8.3. Numerical values were computed with mpmath (50–60 digits unless stated otherwise); they illustrate and prove nothing. The results were found with substantial use of AI tools; all checks reported here are internal to the project and none is peer review.

9.3. **Formal verification.** A Lean 4 certificate, `lean_mstar` (Lean and Mathlib v4.35.0), checks deductions of this paper from explicitly stated hypotheses. Every declaration depends only on Lean's standard axioms (`propext`, `Classical.choice`, `Quot.sound`); there is no `sorry`, no `axiom` declaration and no `native_decide`. Each use of the subgroup theorem is a hypothesis about an explicit *group model* (the tangent space of G'' with its $\overline{\mathbb{Q}}$-structure, the logarithms and periods used, and the family of connected algebraic subgroups); named classical inputs enter as further explicit hypotheses, never as axioms.

The subgroup theorem. The companion project `lean_lsg` proves Theorem 2.1 under (2), including the multiplicity step of Lemma A.1 and the κ-saving of [32, Prop. 2.1], for $\mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times E_1 \times \cdots \times E_m$ (E_j elliptic curves over $\overline{\mathbb{Q}}$), and for vector extensions of such groups in the form its §5d consequence needs (vector part of dimension $r \leq 1$ with $V \cap T(\text{fibre}) = 0$, which the §5d deduction guarantees in this range; and, for $d_0 = 0$, vector part of dimension $r \leq 2$, which covers the group of Proposition 5.5); it also uses Roy's theorem for linear groups. The internal comparison of Lemma A.1 with `lean_lsg`'s proofs covered the groups $\mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times E_1 \times \cdots \times E_m$ and the vector extensions with $r \leq 1$; it did *not* cover the $r = 2$ route used for Proposition 5.5 (a separate Lean proof of Theorem 2.1 for that group, Appendix A), whose formal verification therefore rests on the Lean kernel check and on the internal review of its group model only. From these it proves the hyperplane statement Theorem 2.5 for $\mathbb{G}_{\mathrm{m}}^{d_1} \times E$, for the standard model of $\mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times E^m$ (one curve repeated, every period pair with algebraic invariants), and for the vector-extension groups below. The Lean proofs therefore follow the route through [33, §5d] (Theorem 2.5; Remarks 3.3, 3.6, 5.2; for Theorem 1.9, the reduction of §2.4 written out and §5d applied to a hyperplane), not the direct application of Theorem 2.1 to the quotient groups in our main proofs; the labels refer to the statements, not to the text of those proofs. Since `lean_lsg` derives the §5d statements from its own Theorem 4.1, the caveat after Theorem 2.5 (the second application of Theorem 4.1 inside [33, §5d]) does not arise for these groups.

The bridges. The groups $\mathbb{G}_{\mathrm{m}}^{2n} \times A''$, $\mathbb{G}_{\mathrm{a}}^{2} \times \mathbb{G}_{\mathrm{m}}^{2n} \times A''$ and $A'' \times \mathbb{G}_{\mathrm{m}}^{n} \times \mathbb{G}_{\mathrm{a}}$ of Theorems 3.1, 4.2(a) and 1.9 are reduced to the standard model by the isomorphism $A'' = E^n/B_{\ker\psi} \cong E^{n/2}$, and the models with elliptic logarithms among the data by the arithmetic of elliptic logarithms: the addition and duplication formulas make $\mathcal{L}_E$ a group, and $\wp(iz) = -\wp(z)$, $\wp(\rho z) = \rho\wp(z)$ make it stable under complex multiplication for $\varpi\mathbb{Z}[i]$ and $\Omega_\rho\mathbb{Z}[\rho]$. All of this is proved in Lean. The vector-extension groups of Theorems 3.4, 4.2(b) and 1.6 (vector part of dimension 1) and of Proposition 5.5 (vector part of dimension 2) are covered by `lean_lsg`'s theorems for vector extensions. So the hyperplane statements (instances of Theorem 2.5) that the Lean proofs use are proved in Lean for these groups, for the group of Proposition 5.5 by the separate route of Appendix A. (Trust point T, the direct use of Theorem 2.1 in our main proofs, is not what

Lean checks.) Also proved in Lean: $g_2(\varpi\mathbb{Z}[i]) = 4$, $g_3(\varpi\mathbb{Z}[i]) = 0$, $g_2(\Omega_\rho\mathbb{Z}[\rho]) = 0$, Schneider's theorem ($\varpi$ and ϖ/π are transcendental), and the transcendence of π.

Formally verified results (labels (a), (b)). With no hypothesis beyond Lean's axioms (label (a)): Theorem 1.1, both parts, for $\Omega \in \overline{\mathbb{Q}}^\times\varpi$ (every algebraic $c \neq 0$; in particular $2^\varpi, 3^\varpi$ and $2^{\varpi/\pi}, 3^{\varpi/\pi}$ are not both algebraic); the K-forms of Theorems 3.1 and 3.4 at ϖ, for the K stated in their status lines; Theorems SMG and SMG$^\natural$ at ϖ with second rows in $\widetilde{\mathcal{L}}$; Theorem 1.9 for $\Omega \in \overline{\mathbb{Q}}^\times\varpi$ (first conclusion) with (M1) and (M3). Formally verified modulo named classical inputs (label (b)): Theorem 1.5 at ϖ (Wüstholz's theorem in the form $\Omega, \eta \notin \widetilde{\mathcal{L}}$, Lemma 4.1); Corollary 7.1 on E_i (the same form of Wüstholz's theorem); Corollary 7.2 for the exponent ϖ/π, i.e. $\Omega = \varpi$ (Baker's theorem); (M3′) (the Hermite–Lindemann theorem); $s \in \mathbb{Q}(i)$ in Theorem 1.9 at ϖ and (Mη) (Masser's theorem); Proposition 5.1 in its single-point form (one $v \in V \cap \Psi(\Lambda(G))$, $v \in \overline{\mathbb{Q}}w_1 + \overline{\mathbb{Q}}w_2$) and Theorem 1.6 (Wüstholz's theorem in the forms $\varpi\Omega_\rho/\pi \notin \overline{\mathbb{Q}}$, $\Omega_\rho/\varpi \notin \overline{\mathbb{Q}}$ and, for Theorem 1.6, the linear independence of 1 and elliptic logarithms of E_i, E_ρ, and Masser's theorem); Corollary 1.7 on E_i (also Laurent's theorem [18]); Corollary 7.5(ii) (Wüstholz's theorem: $1, \pi, \varpi, \eta_i, \Omega_\rho, \eta_\rho$ are $\overline{\mathbb{Q}}$-linearly independent). Every statement at Ω_ρ, and Proposition 5.1, Theorem 1.6, Corollaries 1.7 and 7.5(ii), additionally assume that $g_3(\Omega_\rho\mathbb{Z}[\rho])$ is algebraic ($g_3 = 4$ is classical but not formalised).

Also formally verified modulo named classical inputs, by the $r = 2$ route of `lean_lsg` described above: Proposition 5.5 in its single-point form (one $v \in V' \cap \Psi'(\Lambda(G))$, $v \in \overline{\mathbb{Q}}w_1' + \overline{\mathbb{Q}}w_2'$, every $\beta \in \overline{\mathbb{Q}}^\times$; Wüstholz's theorem in the forms $\varpi\Omega_\rho/\pi \notin \overline{\mathbb{Q}}$, $\Omega_\rho/\varpi \notin \overline{\mathbb{Q}}$) and Corollary 7.5(i) (Wüstholz's theorem: $1, \pi, \varpi, \eta_i, \Omega_\rho, \eta_\rho$ are $\overline{\mathbb{Q}}$-linearly independent; including the case of a zero coefficient on π), both also assuming that $g_3(\Omega_\rho\mathbb{Z}[\rho])$ is algebraic. The Lean proof of Corollary 7.5(i) goes through Proposition 5.5 directly, as does ours. The step from Proposition 5.5 to Theorem 5.6 (Masser's and Wüstholz's theorems) is not written out in Lean.

Results with label (c) checked in Lean conditionally on Theorem 2.5 for CM abelian varieties of dimension 2 and 3 and their vector extensions (these instances are not formalised): the K-forms of Theorems 3.7 and 3.9 for the stated instances (the first-kind eigen-periods $(1-\zeta_N)B(\frac{1}{N}, \frac{1}{N})$ of $y^N = x(1-x)$, $N = 5, 7$, resp. the second-kind eigen-period of $y^5 = x(1-x)$), for every number field $K \supseteq \mathbb{Q}(\zeta_N, i, c)$; Theorem 1.2 at the classes $t \in \overline{\mathbb{Q}}^\times B(\frac{1}{5}, \frac{1}{5})$, $\overline{\mathbb{Q}}^\times B(\frac{1}{5}, \frac{1}{5})/\pi$, $\overline{\mathbb{Q}}^\times \pi/B(\frac{1}{5}, \frac{1}{5})$ and $\overline{\mathbb{Q}}^\times B(\frac{1}{7}, \frac{1}{7})/\pi$ (including $t = \Gamma(\frac{1}{5})^2/\Gamma(\frac{2}{5})$ and $t = \Gamma(\frac{1}{5})^2/(\pi\Gamma(\frac{2}{5}))$ with $(\alpha, \beta) = (2, 3)$), but not at $\overline{\mathbb{Q}}^\times/B(\frac{1}{5}, \frac{1}{5})$; Corollary 7.3 for $B(\frac{1}{5}, \frac{1}{5})$ (with Wüstholz's theorem in the form $P \notin \widetilde{\mathcal{L}}$); Proposition 6.1 and Theorem 6.2 in general (for any four pairwise distinct first-kind slots of simple CM abelian varieties with a relation (6) taken as hypothesis, as in the paper: Proposition 6.1 for $[K : \mathbb{Q}] \geq 5$, and Theorem 6.2 without its torsion clause; (Ch) is proved in Lean; conditional on the abelian-variety instance of Theorem 2.5 for the block groups); Theorem 6.6 at level seven, including the torsion clause (the torsion step a named input); and Corollary 6.7(a)–(c) ((a), (b) for Abel–Jacobi integrals along paths avoiding the branch points). The level-seven relation $P_1P_2 = \mu' P_3\Omega$ and $B(\frac{1}{5}, \frac{1}{5})B(\frac{4}{5}, \frac{4}{5}) \in \pi\overline{\mathbb{Q}}^\times$ are proved in Lean from the Beta cocycle and Euler's reflection formula.

Not formalised at all: Theorem 1.2 for other eigen-periods (each needs its own period data), and the torsion clause of Theorem 6.2 in general; Theorem 1.1 for CM periods outside $\overline{\mathbb{Q}}^\times\varpi \cup \overline{\mathbb{Q}}^\times\Omega_\rho$, and its $t \leftrightarrow 1/t$ extension to $\overline{\mathbb{Q}}^\times/\Omega$ and $\overline{\mathbb{Q}}^\times\pi/\Omega$; the passage to arbitrary $K \supseteq k$ by Baker's theorem; the second sentence of Corollary 3.2; (M2); $s \in k$ in Theorem 1.9 at Ω_ρ; the mixed second rows of SMG and SMG$^\natural$; Corollary 7.2 for $\Omega \neq \varpi$; Propositions 5.1 and 5.5 in their K-span forms; the step from Proposition 5.5 to Theorem 5.6; Corollaries 4.5 and 1.8 (the latter

is a named hypothesis); the proof sketches; Lemmas 2.3 and 2.2; and Lemma A.1 for general G_2.

What is not machine-checked. The group models are hand-written. They were refereed (by internal AI referees), not machine-derived from algebraic geometry: that a model's logarithms, periods and subgroup family are true of the group is checked by these reviews, not by Lean. The same holds for the facts that the Beta expressions are periods of the stated cycles (Weil, Rohrlich) and for Chowla–Selberg, which are checked numerically. No human expert has reviewed the definitions or the proofs. Details, with the Lean name of each statement: the certificate's `TRUST.md` (§§1a, 2, 5, 8).

Appendix A. Multiplicities along W in the auxiliary function

We use the notation of [33, §6]. $G = \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times G_2$ over a number field K_0 with $d_2 \geq 1$ and G_2 embedded in $\mathbb{P}_M$ by Serre's embedding (coordinates $\psi_0, \dots, \psi_M$, entire of order ≤ 2; $\xi = \psi/\psi_0$). $V \subset T_G$ has dimension $N < d$, $W \subset V$ is K_0-rational of dimension t, $y_1, \dots, y_m \in V$ with $\exp_G(y_j) \in G(K_0)$, $Y(S) = \{\sum h_j y_j : 0 \leq h_j \leq S\}$, $0 \leq \kappa \leq \mathrm{rk}_{\mathbb{Z}}(V \cap \ker \exp_G)$, and $\Delta^{d-N} = C\, S^{d_1+2d_2-\kappa}(\log S)^{bd_0}$, $T(\log S)^a = D_0(\log S)^b = D_1 S = D_2 S^2 = \Delta$.

Lemma A.1. *Assume $2N \geq d_1 + 2d_0 + \kappa$, $d_2 \geq 1$, $a, b \geq 1$. For ν large and $S \geq S_0(\nu)$ there is a polynomial $P_S \in K_0[U, V, W]$ of degree $\leq D_0$ in U, $\leq D_1$ in V, homogeneous of degree $\leq D_2$ in W, not vanishing identically on G, which vanishes with multiplicity $\geq T$ along W at every point of $\exp_G(Y(S))$. Hence* [33, Prop. 6.1] *holds under these assumptions.*

Sketch. Step 1 is [32, Prop. 2.1 and p. 640] unchanged. With $r = \nu^2 S$, $R = \nu^3 S$, it gives a nonzero $Q_S \in \mathbb{Z}[U, V, W]$ of degrees $\leq D_i/\nu$ and coefficients $\leq e^{\Delta/\nu}$ with $|Q_S(\lambda, \mu, \psi)(z)| \leq e^{-\nu^6\Delta}$ for $z \in V$, $|z| \leq \nu^2 S$. The hypothesis $2N \geq d_1 + 2d_0 + \kappa$ is the condition $U \geq r^2 \log(R/r)$ of Prop. 2.1 (printed with $-\kappa$ on p. 641, a misprint). No condition along W enters the linear system, so the saving r^κ in the hypothesis of [32, Prop. 2.1, p. 633] is untouched. We also use the lower bound $\max_i |\psi_i(t+z)| \geq e^{-c(\nu S)^2}$ for $t \in Y(\nu S) \cap \mathcal{T}$ and $|z| \leq \rho$, where $Y = \mathcal{T} + F$ with F finite ([32, p. 640], [30, Lemme 1.2.2]; the ψ_i have no common zero [27, Prop. 8]). In Step 2 it bounds the scalar $\mu_j(z)$ in $E^{(j)}(\xi(\gamma), \xi(z)) = \mu_j(z)\psi(t+z)$. (The polynomial variables U, V, W below are not the subspaces V, W.)

Step 2 (derivatives). Fix $t \in Y(\nu S) \cap \mathcal{T}$, $\gamma = \exp_G(t)$, and an addition law $E^{(j)}$ of the embedding with $E^{(j)}(\xi(\gamma), \xi(0)) \neq 0$. For $z = \sum s_i w_i \in W$ small put $F_t(s) = Q_S(\lambda(\gamma) + \lambda(z), \mu(\gamma)\mu(z), E^{(j)}(\xi(\gamma), \xi(z)))$. By homogeneity, F_t is a unit times $Q_S(\lambda, \mu, \psi)(t+z)/\psi_0(t+z)^{D^\circ}$, so it has the same order along W. Its Taylor coefficients θ' of order $< T$ satisfy $|\theta'| \leq e^{-\nu^6\Delta + c\nu\Delta}$ (Cauchy). They are polynomials, with no denominator depending on γ, in the coefficients of Q_S, the coordinates of γ, and the Taylor coefficients of λ, μ, ξ along W at the origin. The latter are K_0-rational with house and denominator $\leq c^{|\tau|+1}(|\tau|!)^c$: by Bertrand's Corollaire to his Lemme 7 [30, App. I, p. 177] the coordinates of the exponential map have finite arithmetic functional order, every point being regular, and his Lemme 1 [30, App. I, pp. 166–167] turns this into such bounds for derivatives ([33, p. 396] cites this as "Lemma 7"). Since $|\tau| < T$ and $T \log T \leq c\Delta$ (as $a \geq 1$ and $\log \Delta \leq c \log S$), Serre's height bounds for γ then give $h(\theta') \leq c\nu\Delta$ with c independent of ν. Liouville's inequality forces $\theta' = 0$ once ν^5 exceeds a constant. (Dividing by a power of $E_0^{(j)}(\xi(\gamma), \xi(0))$ instead would cost a height $T(\nu S)^2$, which is too large.)

Step 3 (translation). Write $Y(S) \subset (Y(\nu S) \cap \mathcal{T}) + F'$ with F' finite. For $f \in F'$ take the product over the addition laws $E^{(j)}$ with $E^{(j)}(\cdot, \xi(-\exp f)) \not\equiv 0$ of Q_S translated by $-\exp f$; then take the product over $f \in F'$. Translations commute with invariant derivations, and each factor is a unit times the translate on a dense open set, so the product vanishes with multiplicity

$\geq T$ along W on $\exp_G(Y(S))$. It is not identically zero on G, and its degrees are $\leq D_i$ once ν exceeds (number of laws)·(bidegree)·$|F'|$. No norm to $\mathbb{Q}$ is taken, since [33, Prop. 6.1] allows $\overline{\mathbb{Q}}$-coefficients.

The exponent $b \geq 1$ changes only the number of unknowns, which is compensated exactly by the factor $(\log S)^{bd_0}$ of Δ^{d-N}, and lowers the height term $(D_0/\nu)\log S$. In the proof of [33, Thm. 4.1], $b > 1$ is used only in the case $\delta = \tau$, which requires $d_0 \geq 1$ (Theorems 1.6 and 4.2, Proposition 6.1). □

Remark A.2 (Addition laws)**.** Let $G_2 \subset \mathbb{P}_M$ be locally closed over K_0, and $X = G_2 \times G_2 \subset \mathbb{P}_M \times \mathbb{P}_M$. The local ring of X at a point p is a quotient of that of $\mathbb{P}_M \times \mathbb{P}_M$, whose elements are ratios of bihomogeneous forms of equal bidegree. Clearing denominators, near p the group law $X \to G_2 \subset \mathbb{P}_M$ is given by bihomogeneous forms $(A_0, \dots, A_M)$ of one bidegree, not all zero at p. By quasi-compactness finitely many such systems suffice, and multiplying each by suitable monomials $x_i^a y_l^b$ gives a complete system of addition laws of one bidegree (β, β), with coefficients in a finite extension of K_0. (This replaces K_0 by a finite extension in Step 2, which only changes $[K_0 : \mathbb{Q}]$ in the Liouville inequality.) Step 2 uses these laws only on $G_2(K_0) \times \exp(\text{a neighbourhood of 0 in } W)$, and Step 3 only with a fixed second argument.

What is formalised. The companion Lean project `lean_lsg` (§9.3) proves Lemma A.1, in a form at least as strong for these groups (with $a = 1$, integer $b \geq 1$, some constant C, and a Weierstrass–Segre model in place of Serre's embedding), for $G = \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times E_1 \times \cdots \times E_m$ (`LSG.W88.prop61_b`), including the d_0-part of the last paragraph of the proof; and for push-outs of $(E_1 \times \cdots \times E_m)^\natural$ along an algebraic linear map, under the additional hypothesis $V \cap T(\text{fibre}) = 0$ (`LSG.AuxV.prop61V_b`). For the group of Proposition 5.5 ($d_0 = 0$, vector part of dimension 2 meeting V) `lean_lsg` proves Theorem 2.1 by a separate route (its `Theorem41Vec2`); that route has not been compared with Lemma A.1 line by line, and the internal comparison of Lemma A.1 with `lean_lsg` did not cover it. Lemma A.1 as stated, for general G_2, is not formalised.

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